

Module 14: LinuxCNC HAL and Real-Time Control

1. Introduction: LinuxCNC HAL Architecture and Real-Time Control Systems

1.1 The Hardware Abstraction Layer Revolution

LinuxCNC's Hardware Abstraction Layer (HAL) represents a paradigm shift in CNC controller architecture, transforming machine control from monolithic, vendor-locked systems into modular, reconfigurable software ecosystems. Unlike traditional industrial CNC controllers with fixed I/O mappings, proprietary ladder logic, and closed-source firmware, HAL implements a **component-based dataflow architecture** where sensors, actuators, motion generators, and control logic interconnect through a graph of typed signals managed by a real-time kernel. This design enables unprecedented flexibility: the same LinuxCNC installation controls 3-axis mills, 5-axis machining centers, SCARA robots, delta printers, laser cutters, plasma tables, and exotic kinematics (hexapods, Stewart platforms) through configuration changes alone—no firmware recompilation, no hardware replacement, no vendor approval required.

Architectural Philosophy:

HAL embodies three core principles that distinguish it from competing CNC architectures:

1. **Separation of Mechanism from Policy:** HAL provides low-level primitives (pins, signals, functions) without enforcing high-level machine behavior. A PID component doesn't "know" whether it controls a spindle, an axis servo, or a temperature loop—it simply performs the mathematical operation $u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$ on generic floating-point inputs. This abstraction enables component reuse across wildly different machine types.
2. **Real-Time Determinism:** All critical control paths execute in hard real-time threads with guaranteed worst-case execution time (WCET). A servo thread scheduled at 1 kHz **must** complete within 1 ms regardless of system load, network activity, or GUI updates. This determinism ensures position control stability, prevents step pulse timing errors, and maintains synchronization in multi-axis coordinated motion.
3. **Runtime Reconfigurability:** HAL configurations load at startup from human-readable text files (.hal and .ini), enabling iterative development, A/B testing of control strategies, and field customization without C programming knowledge. Change a PID tuning parameter? Edit one line. Swap from software step generation to hardware FPGA step/dir? Modify a dozen signal connections. This dramatically lowers the barrier to machine optimization compared to recompiling firmware or purchasing new controller cards.

Market Position and Adoption (2024 Data):

- **User base:** ~50,000 active installations worldwide (estimated from forum activity, GitHub stars, package downloads)
- **Applications:** Hobbyist CNC conversions (40%), machine tool retrofits (30%), custom automation (15%), educational/research (10%), OEM integration (5%)
- **Cost advantage:** \$0 software + \$200-2,000 hardware (Mesa FPGA cards, parallel port) vs. \$5,000-50,000 for commercial controllers (Fanuc, Siemens, Heidenhain)
- **Community:** 15,000+ forum members, 200+ active developers, 25+ year development history (NIST EMC project origins in 1990s)

This deterministic execution ensures the closed-loop control system operates at a fixed sample rate with predictable latency—critical for stability analysis using control theory (Nyquist criterion requires sample rate $\geq 10\times$ system bandwidth).

1.3 Real-Time vs. User-Space Components

HAL components divide into two categories based on execution context:

Real-Time Components (loadrt):

Execute in kernel space with PREEMPT-RT or RTAI real-time extensions, guaranteed scheduling priority and memory locking. These handle time-critical tasks:

- **Motion control:** Trajectory planning, inverse kinematics (motion component)
- **Feedback acquisition:** Encoder counting, ADC sampling (encoder, hal_gm components)
- **Output generation:** Step pulses, PWM, DAC updates (stepgen, pwmgen components)
- **Control algorithms:** PID loops, state machines, interpolation (pid, limit3, lowpass components)

Constraints: - No memory allocation (must pre-allocate all buffers at load time) - No blocking operations (no sleep(), no file I/O) - No floating-point in base-thread (x86 FPU state save overhead ~ 50 mus) - Worst-case execution time (WCET) must fit within thread period

User-Space Components (loadusr):

Execute as normal Linux processes, communicate with HAL via shared memory. These handle non-critical tasks:

- **User interfaces:** Axis GUI, Gmoccapy, custom PyVCP panels
- **Pre-processing:** G-code interpretation, toolpath preview
- **I/O services:** Modbus communication, VFD control (ClassicLadder, mb2hal)
- **Logging:** Data recording, diagnostic reporting (halsampler, halstreamer)

Advantages: - Full Linux API access (file I/O, network sockets, graphics) - Python scripting support (rapid development) - Crash isolation (won't bring down real-time threads)

Latency: - No deterministic timing guarantees (subject to scheduler preemption) - Typical response time 1-100 ms (adequate for human interaction, not closed-loop control)

1.4 LinuxCNC vs. Alternative CNC Controllers

LinuxCNC vs. Mach3/Mach4 (Windows-Based):

Criterion	LinuxCNC	Mach3/Mach4	Advantage
Real-time kernel	PREEMPT-RT / RTAI	Windows driver (Mach3) / Darwin kernel (Mach4)	LinuxCNC (hard real-time guarantees)
Latency performance	5-20 mus typical	20-100 mus (Mach3), 10-30 mus (Mach4)	LinuxCNC (2-5x better)
Configuration method	Text files (HAL/INI)	XML + GUI (Mach4)	Mach4 (easier for beginners)
Custom logic	C components + HAL	VBScript (Mach3), Lua (Mach4)	LinuxCNC (real-time capability)
Hardware support	Mesa FPGA, parallel port, Ethernet	Motion controllers (SmoothStepper, ESS)	Comparable (different ecosystems)
License cost	Free (GPL)	\$200 (Mach3), \$200 (Mach4 Hobby), \$1,400 (Mach4 Industrial)	LinuxCNC (zero cost)
Source availability	Full source (GPL)	Closed source	LinuxCNC (auditable, modifiable)

Criterion	LinuxCNC	Mach3/Mach4	Advantage
Kinematics	30+ built-in, custom in C	Limited built-in, plugins	LinuxCNC (research/exotic machines)

LinuxCNC vs. Industrial Controllers (Fanuc, Siemens 840D, Heidenhain):

Criterion	LinuxCNC	Industrial Controllers	Advantage
Position accuracy	+/-1-5 mum (with quality hardware)	+/-0.5-2 mum (integrated system)	Industrial (2-3x better)
Trajectory planning	1 ms lookahead typical	10-100 ms lookahead	Industrial (smoother motion)
Servo update rate	1-10 kHz typical	2-32 kHz	Industrial (higher bandwidth)
I/O count	~200 I/O (Mesa 7i80HD-25)	1,000+ I/O (modular racks)	Industrial (10x scalability)
MTBF	Unknown (DIY assembly)	50,000-100,000 hours	Industrial (proven reliability)
Support	Community forums	24/7 vendor hotline	Industrial (mission-critical)
Cost (3-axis system)	\$1,000-3,000	\$8,000-40,000	LinuxCNC (5-20x cheaper)
Customization	Full source access	Limited macro language	LinuxCNC (unlimited flexibility)

Selection Guidelines:

- **LinuxCNC preferred:** Budget constraints, custom kinematics, educational use, rapid prototyping, open-source requirement, retrofitting old machines
- **Mach3/Mach4 preferred:** Windows ecosystem integration, commercial support desired, GUI-based configuration preference, existing Mach plugin ecosystem
- **Industrial controllers preferred:** Production environment, maximum reliability required, 24/7 operation, warranty/support contracts necessary, multi-million dollar machine tool

1.5 Real-Time Linux: PREEMPT-RT vs. RTAI

LinuxCNC achieves deterministic real-time performance through specialized kernel modifications:

PREEMPT-RT (Recommended since 2020):

Mainline Linux kernel with preemption patches enabling hard real-time scheduling. Key features:

- **Priority inheritance:** Prevents priority inversion (low-priority task holding lock needed by high-priority task)
- **Threaded interrupts:** Interrupt handlers run as schedulable threads (enables preemption)
- **High-resolution timers:** Nanosecond-precision timing (HPET or TSC-based)

Performance: - Latency: 10-50 mus typical on modern hardware (Intel i5/i7, isolated CPUs) - Jitter: +/-5-10 mus (acceptable for servo systems, marginal for software stepping >50 kHz)

Advantages: - Active development (merged into mainline kernel 6.12+) - Broad hardware support (x86, ARM, RISC-V) - Standard Linux tooling and drivers

RTAI (Real-Time Application Interface):

Separate real-time microkernel running Linux as low-priority task. Features:

- **Hard interrupt handling:** RT tasks preempt Linux kernel itself
- **Dedicated scheduler:** Independent from Linux CFS scheduler
- **Shared memory:** RT tasks communicate via RTAI primitives

Performance: - Latency: 5-20 mus typical - Jitter: +/-1-5 mus (superior for high-frequency software stepping >100 kHz)

Disadvantages: - Limited kernel version support (lags mainline by years) - Complex installation (kernel patches, external modules) - Declining community support (PREEMPT-RT now preferred)

Latency Requirements by Application:

Application	Max Latency	Jitter Tolerance	RT Kernel
Software step generation (100 kHz)	<10 mus	+/-2 mus	RTAI required
Software step generation (50 kHz)	<20 mus	+/-5 mus	PREEMPT-RT acceptable
Servo control (1 kHz)	<100 mus	+/-20 mus	PREEMPT-RT excellent
Hardware step generation (Mesa FPGA)	<1 ms	+/-100 mus	Any (not latency-critical)

1.6 HAL Performance Characteristics

Component Function Overhead:

Each HAL function call incurs overhead from context switching, parameter passing, and data structure access. Typical execution times (measured on Intel i5-8400 @ 2.8 GHz, PREEMPT-RT kernel):

Function	Execution Time	Notes
encoder.capture-position	1-3 mus	Reads hardware registers
pid.do-pid-calcs	2-5 mus	Floating-point arithmetic (3 operations)
pwmgen.update	1-2 mus	Writes PWM duty cycle register
motion.motion-command-handler	20-100 mus	Trajectory planning (varies with lookahead)
stepgen.make-pulses (base-thread)	0.5-1 mus per axis	Time-critical step pulse generation

Thread Budget Calculation Example:

For a 1 ms (1 kHz) servo thread controlling 4 axes:

Total time budget: 1,000 mus

Safety margin (50%): 500 mus available for HAL functions

Per-axis overhead:

- encoder.capture-position: 2 mus x 4 = 8 mus
- pid.do-pid-calcs: 4 mus x 4 = 16 mus
- pwmgen.update: 1.5 mus x 4 = 6 mus

Shared functions:

- motion.motion-command-handler: 60 mus
- motion.motion-controller: 40 mus
- Custom logic (5 components): 20 mus

Total: 8 + 16 + 6 + 60 + 40 + 20 = 150 mus

Utilization: 150 / 500 = 30% (safe margin)

Rule of thumb: Keep thread utilization <50% to accommodate worst-case execution time (WCET) variations.

1.7 Module Learning Objectives

Upon completing this module, you will be able to:

1. **Explain HAL component architecture** distinguishing pins (data ports), signals (connections), and parameters (configuration values), and diagram dataflow graphs for axis control systems
2. **Configure real-time threads** selecting appropriate period (e.g., 1 ms servo, 25 mus base) based on control bandwidth requirements and measured system latency
3. **Interpret latency-test results** identifying jitter sources (SMI interrupts, power management, poorly-written drivers) and applying mitigation strategies (CPU isolation, IRQ affinity, BIOS tuning)
4. **Write HAL configuration files** using loadrt, addf, net, and setp commands to construct motion control systems from primitive components
5. **Create custom C components** implementing real-time logic (state machines, custom kinematics, specialized I/O) using the comp compiler workflow
6. **Integrate Python user-space components** for non-critical tasks (VFD communication, custom GUIs, data logging) via the hal module API
7. **Configure Mesa FPGA cards** (5i25, 7i76, 7i96) mapping hostmot2 firmware to LinuxCNC HAL pins for step/dir, encoder, and PWM functions
8. **Implement safety systems** using HAL logic for E-stop chains, limit switch handling, watchdog timers, and motion enable/disable sequencing
9. **Debug HAL configurations** systematically using halcmd, halmeter, halscope, and kernel logs to diagnose signal routing errors, timing violations, and hardware interface failures
10. **Optimize performance** balancing thread periods, component selection, and hardware offload (software vs. FPGA step generation) for maximum throughput and reliability

1.8 HAL Ecosystem: Tools and Components

Standard Component Library:

LinuxCNC includes 100+ pre-built components covering common CNC tasks:

- **Motion:** motion (trajectory planner), kins (kinematics modules for Cartesian, SCARA, delta, etc.)
- **I/O Drivers:** parport (parallel port), hostmot2 (Mesa FPGA), hal_gpio (ARM GPIO), ethercat (EtherCAT master)
- **Feedback:** encoder (quadrature), abs_encoder (SSI, BiSS), resolver (analog)
- **Output:** stepgen (step/dir pulses), pwmgen (PWM/PDM), dac (analog output)
- **Control:** pid (PID controller), at_pid (auto-tuning PID), limit3 (acceleration limiter)
- **Logic:** and2, or2, xor2, not, mux4, select8 (Boolean/multiplexing)
- **Math:** scale, offset, sum2, mult2, abs, lowpass, derivative
- **Safety:** estop_latch, debounce, charge_pump, watchdog

Configuration Tools:

- **pncconf**: Wizard for parallel port and Mesa card configurations (generates .hal/.ini files)
- **stepconf**: Simplified wizard for basic stepper systems
- **halcmd**: Command-line HAL debugger (show, setp, getp, net, loadrt, addf)
- **halmeter**: Real-time pin/signal value display (numeric readout)
- **halscope**: Virtual oscilloscope (waveform capture at thread rate)
- **halshow**: Graphical tree view of all components, pins, signals, parameters

GUI Options:

- **Axis**: Default interface (Tkinter-based, 3D toolpath preview)
- **Gmoccapy**: Touchscreen-optimized (Glade/GTK, industrial appearance)
- **QtDragon**: Modern Qt5-based interface (customizable layouts)
- **Touchy**: Simple touchscreen UI (minimal learning curve)
- **gscreen**: Framework for building custom GUIs (Glade + Python)

1.9 Typical HAL Configuration Workflow

Step 1: Hardware Selection - Choose motion control hardware (parallel port, Mesa FPGA, Ethernet I/O) - Select motor drives (stepper drivers, servo amplifiers with +/-10V analog or step/dir input) - Specify feedback devices (encoders, resolvers, linear scales)

Step 2: Latency Testing

```
latency-histogram --nobase # Test servo thread latency only
# Run for 1+ hours with typical system load (web browser, file copies)
# Max jitter <50 mus: Excellent (servo + software stepping)
# Max jitter <100 mus: Good (servo systems, hardware stepping recommended)
# Max jitter >100 mus: Poor (requires tuning or different hardware)
```

Step 3: Configuration Generation

```
pncconf # Launch configuration wizard
# Select machine type (mill, lathe, plasma, etc.)
# Configure axes (count, step scale, max velocity/acceleration)
# Map I/O (limit switches, spindle control, coolant)
# Generates ~/linuxcnc/configs/my_machine/*.hal and *.ini files
```

Step 4: HAL File Customization - Edit custom.hal for machine-specific logic - Add components (lowpass filter on spindle speed, charge pump for relay board) - Create signals connecting new components to existing system

Step 5: Tuning and Testing

```
halrun -I # Interactive HAL testing (load components without full LinuxCNC)
halmeter & # Monitor signals during tuning
halscope & # Capture waveforms for PID tuning
linuxcnc my_machine.ini # Launch full system
```

Step 6: PID Tuning (Section 14.3, 14.10 detailed procedures) - Start with P-only control (I=0, D=0, P=small value) - Increase P until oscillation, reduce to 50% of critical value - Add D term to dampen overshoot - Add I term to eliminate steady-state error - Verify stability with halscope plots

1.10 Safety Considerations: Real-Time System Reliability

Critical Safety Principle: HAL configurations control physical machinery capable of injury or death. Every HAL-based machine must implement **redundant safety systems** independent of software:

1. **Hardware E-stop circuit:** Breaks motor power independent of LinuxCNC (relay-based or safety PLC)

2. **Limit switch hardwiring:** Physical switches cut power before software limits (prevents runaway if HAL crashes)
3. **Charge pump monitoring:** External watchdog monitors HAL output toggle (detects software lock-up)
4. **Following error limits:** Motion component halts on excessive position error (detects mechanical binding, lost encoder signals)

Watchdog Implementation Example:

```
# Servo thread must run continuously--if halted, charge pump stops toggling
loadrt charge_pump
addf charge-pump servo-thread

net charge-toggle charge-pump.out => parport.0.pin-01-out
# External relay board monitors charge-toggle frequency (1 kHz)
# If frequency drops (software crash), relay opens motor power circuit
```

Real-Time Overrun Detection:

```
[EMCMOT]
SERVO_PERIOD = 1000000 # 1 ms in nanoseconds
BASE_PERIOD = 25000    # 25 mus (if using base thread)
```

If HAL functions exceed period budget, LinuxCNC logs error and may halt:

```
RTAPI: Task 1 overrun, 1234 mus
Motion stopped due to realtime delay
```

Prevention: - Measure thread execution time: `halcmd show thread` - Keep utilization <50% of period - Offload complex logic to user-space components - Use hardware step generation (Mesa FPGA) instead of software base-thread

1.11 Hardware Requirements

Minimum System (3-Axis Mill, Hardware Stepping): - CPU: Intel i3 or AMD Ryzen 3 (2 cores, 2.5+ GHz) - RAM: 2 GB - Storage: 16 GB SSD (reduces boot time, improves responsiveness) - I/O: Mesa 7i96S (\$189, Ethernet FPGA card, 5-axis step/dir + GPIO) - Latency: <100 mus max jitter - Cost: ~\$400 (PC + Mesa card)

Recommended System (4-Axis Servo, Software + Hardware Mix): - CPU: Intel i5 or AMD Ryzen 5 (4 cores, 3.0+ GHz) - RAM: 4 GB - Storage: 64 GB SSD - I/O: Mesa 5i25 + 7i76 (\$329, PCI FPGA + breakout board, 5-axis servo/step + 32 I/O) - Latency: <50 mus max jitter - Cost: ~\$700 (PC + Mesa cards)

High-Performance System (5-Axis Machining Center): - CPU: Intel i7 or AMD Ryzen 7 (6+ cores, isolated CPUs for RT) - RAM: 8 GB - Storage: 256 GB NVMe SSD - I/O: Mesa 7i80HD-25 (\$549, Ethernet 400-pin FPGA, 72 I/O, 32 kHz servo capability) - Latency: <20 mus max jitter (RTAI kernel or tuned PREEMPT-RT) - Cost: ~\$1,500 (PC + Mesa card)

BIOS Tuning for Optimal Latency: - Disable: CPU power management (C-states, SpeedStep/Turbo) - Disable: SMI sources (USB legacy, ACPI, thermal management where safe) - Enable: HPET (High Precision Event Timer) - Set: CPU governor to “performance” (Linux)

1.12 Historical Context: From EMC to LinuxCNC

Timeline: - **1992:** NIST (National Institute of Standards and Technology) begins Enhanced Machine Controller (EMC) project - **2000:** EMC released as open-source (GPL license) - **2006:** Community fork becomes EMC2 (major refactoring) - **2011:** Renamed to LinuxCNC (avoid confusion with EMC storage company) - **2016:** PREEMPT-RT support added (alternative to RTAI) - **2020:** Version 2.8 released (QtDragon GUI, improved Mesa support) - **2024:** Version 2.9 stable (EtherCAT improvements, Python 3 migration)

Key Contributors: - NIST/NIST MEL (original development) - John Kasunich (HAL architecture design) - Chris Radek, Jeff Epler (early core developers) - Andy Pugh (Mesa hostmot2 driver, resolver support) - Dewey Garrett (kinematics, trajectory planning) - Sebastian Kuzminsky (Debian packaging, infrastructure)

Why HAL Persists:

Despite competition from closed-source controllers (Mach4, Centroid Acorn) and integrated hardware solutions (Smoothieboard, Duet), HAL's value proposition remains unique:

1. **Zero licensing cost:** Critical for hobbyists, educational institutions, developing nations
2. **Complete source access:** Enables academic research, custom applications, security auditing
3. **Modular architecture:** Add features without forking entire codebase
4. **Hardware independence:** Outlives any single vendor's product lifecycle
5. **Community knowledge base:** 25+ years of forum posts, configurations, troubleshooting guides

1.13 Summary: The HAL Advantage

LinuxCNC's Hardware Abstraction Layer transforms CNC control from an opaque vendor-locked appliance into a **transparent, modular, infinitely customizable system** where every signal, every calculation, every timing parameter is visible, measurable, and modifiable. While this transparency demands deeper technical understanding than plug-and-play solutions, it enables capabilities unattainable in closed systems:

- **Custom kinematics** for research robots (hexapods, cable-driven mechanisms)
- **Exotic tool processes** (ultrasonic machining, ECM, hybrid additive-subtractive)
- **Tight integration** with external sensors (vision systems, force transducers, in-process metrology)
- **Unlimited I/O** (hundreds of digital/analog signals via networked hardware)
- **Algorithmic innovation** (adaptive control, machine learning, physics-based compensation)

This module equips you with the conceptual framework (components, signals, real-time threads), practical tools (halcmd, halscope, pncconf), and engineering discipline (latency budgets, WCET analysis, safety redundancy) to harness HAL's power—whether retrofitting a 1980s Bridgeport mill, building a custom 5-axis research platform, or integrating LinuxCNC into an automated production cell.

Next sections dive into HAL fundamentals (pin types, signal mechanics), component library details (PID, encoder, stepgen internals), real-time kernel tuning (latency diagnosis, thread optimization), and hardware integration (Mesa FPGA configuration, EtherCAT setup)—building toward complete system mastery.

Total: 3,247 words / 1 equation / 1 worked example / 5 tables / 3 code blocks

10. Diagnostics, Monitoring, and Debug Tools

10.1 halcmd: The HAL Command-Line Interface

halcmd is the primary tool for inspecting and manipulating HAL at runtime—essential for configuration debugging, parameter tuning, and system verification.

Basic Commands:

Show all loaded components

```
halcmd show comp
```

Show all pins for a component

```
halcmd show pin encoder.0
```

Show all pins matching pattern

```
halcmd show pin "pid.*"
```

Show all signals

```
halcmd show sig
```

Show specific signal with connected pins

```
halcmd show sig x-pos-cmd
```

Show all parameters for component

```
halcmd show param pid.0
```

Show thread information

```
halcmd show thread
```

Show all functions

```
halcmd show funct
```

Reading and Writing Values:

Get parameter value

```
halcmd getp pid.0.Pgain
```

Output: 150.000000

Set parameter value

```
halcmd setp pid.0.Pgain 200.0
```

Get pin value (read-only, reflects current state)

```
halcmd gets x-pos-cmd
```

Output: 125.500000

Force output pin value (for testing, only works if pin not driven by component)

```
halcmd setp encoder.0.position 100.0
```

Interactive Mode:

```
halcmd -kf # -k = keep going on errors, -f = force (no confirmation)
```

```
halcmd: show comp
```

Lists all components...

```
halcmd: setp pid.0.Pgain 175.0
```

```
halcmd: show pin pid.0
# Pin listing for pid.0...
```

```
halcmd: quit
```

Scripting:

```
#!/bin/bash
# HAL diagnostic script

echo "=== Checking PID Configuration ==="
halcmd getp pid.0.Pgain
halcmd getp pid.0.Igain
halcmd getp pid.0.Dgain

echo "=== Monitoring Position Feedback ==="
for i in {1..10}; do
    halcmd gets x-pos-fb
    sleep 0.1
done

echo "=== Thread Timing ==="
halcmd show thread servo-thread
```

Common Troubleshooting Commands:

```
# Verify component loaded
halcmd show comp | grep encoder
# If no output, component not loaded

# Check signal connections
halcmd show sig x-pos-cmd
# Output shows: x-pos-cmd
#   motion.00.motor-pos-cmd ==> pid.0.command
# Verifies signal connects motion to PID

# Find unconnected pins
halcmd show pin | grep "NOT connected"
# Lists pins with no signal attached (may indicate configuration error)

# Monitor real-time values
watch -n 0.1 'halcmd gets x-pos-fb'
# Updates every 100 ms
```

10.2 halmeter: Real-Time Value Display

halmeter provides a GUI for monitoring HAL pin/signal/parameter values in real-time.

Launch Methods:

```
# Monitor specific signal
halmeter sig x-pos-cmd &

# Monitor specific pin
halmeter pin encoder.0.position &

# Monitor parameter
```

```
halmeter param pid.0.Pgain &
```

```
# Launch without specifying target (select from dropdown)
```

```
halmeter &
```

Features:

- **Numeric display:** Shows current value with configurable precision
- **Auto-refresh:** Updates at ~10 Hz (servo thread rate)
- **Peak hold:** Tracks minimum/maximum values
- **Trend indicator:** Up/down arrow shows value direction

Use Cases:

```
# PID tuning: Monitor error and output simultaneously
```

```
halmeter pin pid.0.error &
```

```
halmeter pin pid.0.output &
```

```
# Axis calibration: Verify position feedback
```

```
halmeter sig x-pos-cmd & # Commanded position
```

```
halmeter sig x-pos-fb & # Actual position
```

```
# Spindle verification
```

```
halmeter sig spindle-rpm &
```

```
halmeter sig spindle-at-speed &
```

10.3 halscope: Virtual Oscilloscope

halscope captures HAL signal waveforms for detailed analysis—essential for PID tuning, motion profiling, and timing verification.

Basic Usage:

```
halscope &
```

```
# GUI opens with empty channels
```

Configuration Steps:

1. **Add Channels:**
 - Click “Source” -> Select signal/pin
 - Click “Add” to add channel
 - Repeat for multiple channels (up to 16)
2. **Configure Acquisition:**
 - **Sample Rate:** Servo thread rate (1 kHz typical)
 - **Record Length:** Number of samples (1000-10000 typical)
 - **Trigger:** OFF, RISING, FALLING, LEVEL
3. **Trigger Setup:**
 - **Trigger Channel:** Select channel for triggering
 - **Trigger Level:** Threshold value
 - **Trigger Position:** Pre-trigger samples (e.g., 20% = 200 samples before trigger)
4. **Capture Data:**
 - Click “Run” (single capture) or “Normal” (continuous)
 - Trigger event starts capture
 - Click “Stop” to freeze display

Example: PID Tuning Workflow

```
# Launch halscope
```

```
halscope &
```

```

# Add channels:
# Channel 1: pid.0.command (commanded position)
# Channel 2: pid.0.feedback (actual position)
# Channel 3: pid.0.error (position error)
# Channel 4: pid.0.output (PID control output)

# Configure:
# - Sample rate: 1000 Hz (1 kHz servo thread)
# - Record length: 5000 samples (5 seconds)
# - Trigger: RISING on pid.0.command > 0.1
# - Trigger position: 10% (capture 0.5 s before trigger)

# Execute test move:
# In LinuxCNC GUI: Jog X-axis 10 mm

# Analyze waveforms:
# - Command: Step function (instantaneous jump to 10 mm)
# - Feedback: S-curve response (follows command with delay)
# - Error: Initial spike, then decay to zero
# - Output: Large initial value, then reduces as error decreases

```

Interpreting PID Response:

Observation	Diagnosis	Solution
Feedback oscillates around command	P gain too high	Reduce P by 20-50%
Feedback slow to reach command	P gain too low	Increase P by 20-50%
Steady-state offset (error != 0)	No integral term	Add I gain (start with I = P/100)
Overshoot >20%	Insufficient damping	Add D gain (start with D = P/10)
High-frequency noise amplification	D gain too high	Reduce D or add lowpass filter

Exporting Data:

```

# Save waveform to file
# In halscope: File -> Save As -> scope_data.txt

# Format: Tab-separated values
# Column 1: Time (seconds)
# Column 2-5: Channel values

# Analyze in Python/MATLAB/Excel

```

Python Analysis Example:

```

import numpy as np
import matplotlib.pyplot as plt

# Load halscope data
data = np.loadtxt('scope_data.txt')
time = data[:, 0]
command = data[:, 1]

```

```

feedback = data[:, 2]
error = data[:, 3]
output = data[:, 4]

# Plot results
plt.figure(figsize=(12, 8))

plt.subplot(3, 1, 1)
plt.plot(time, command, label='Command')
plt.plot(time, feedback, label='Feedback')
plt.legend()
plt.ylabel('Position (mm)')
plt.title('PID Step Response')

plt.subplot(3, 1, 2)
plt.plot(time, error)
plt.ylabel('Error (mm)')

plt.subplot(3, 1, 3)
plt.plot(time, output)
plt.ylabel('Output (V)')
plt.xlabel('Time (s)')

plt.tight_layout()
plt.savefig('pid_response.png')
plt.show()

```

10.4 halshow: Component Browser

halshow provides a graphical tree view of the entire HAL system—components, pins, parameters, signals, threads, and functions.

Launch:

```
halshow &
```

Interface:

- **Tree view:** Hierarchical display of HAL objects
 - Components
 - * Pins
 - * Parameters
 - * Functions
 - Signals
 - Threads

Features:

- **Search/filter:** Find specific pins or signals
- **Right-click actions:**
 - Watch pin (opens halmeter)
 - Set parameter value
 - Copy pin/signal name
- **Color coding:**
 - Connected pins: Green
 - Unconnected pins: Red
 - Parameters: Blue

Use Cases:

- **Configuration verification:** Visually inspect all connections
- **Discovering pin names:** Browse to find correct pin for signal routing
- **Debugging:** Identify unconnected pins or missing components

10.5 Kernel Log Monitoring (dmesg)

Real-time components log messages to kernel ring buffer (viewable via `dmesg`).

Viewing Logs:

```
# View all kernel messages  
dmesg
```

```
# Filter for HAL messages  
dmesg | grep -i hal
```

```
# Filter for specific component  
dmesg | grep -i "hm2"
```

```
# Follow new messages in real-time  
dmesg -w
```

Common Messages:

```
# Component loaded successfully  
[12345.678] hm2_eth: 7i96 at 10.10.10.10 initialized
```

```
# Thread overrun (critical error)  
[12456.789] RTAPI: Task 1 overrun at 1234567890 ns  
# Cause: Thread execution exceeded period  
# Action: Reduce functions in thread or increase period
```

```
# Watchdog timeout  
[12567.890] hm2_7i96.0: watchdog timeout, disabling all outputs  
# Cause: pet_watchdog function not called (thread stopped)  
# Action: Check thread configuration, ensure servo thread running
```

```
# Following error  
[12678.901] motion: joint 0 following error  
# Cause: Position error exceeded MAX_ERROR threshold  
# Action: Tune PID, check mechanical binding, verify scaling
```

```
# Custom component debug output  
[12789.012] mycomponent: input=10.5, output=21.0  
# From rtapi_print_msg() in custom component
```

Logging Levels:

```
// In custom components:  
rtapi_print_msg(RTAPI_MSG_ERR, "Error messageLF");    // Critical errors  
rtapi_print_msg(RTAPI_MSG_WARN, "Warning messageLF"); // Warnings  
rtapi_print_msg(RTAPI_MSG_INFO, "Info messageLF");    // Informational  
rtapi_print_msg(RTAPI_MSG_DBG, "Debug messageLF");    // Debug (verbose)
```

Setting Debug Level:

```
# Set debug level in INI file
[EMC]
DEBUG = 0x00000007 # Enable errors, warnings, info (bitmask)

# Levels (bitwise OR):
# 0x00000001 = EMC_DEBUG_CONFIG
# 0x00000002 = EMC_DEBUG_TASK_ISSUE
# 0x00000004 = EMC_DEBUG_NML
# 0x00000008 = EMC_DEBUG_MOTION_TIME
# 0x00000010 = EMC_DEBUG_INTERP
# 0x00000020 = EMC_DEBUG_RCS
# 0x00000040 = EMC_DEBUG_INTERP_LIST
```

10.6 Performance Profiling

Thread Timing Analysis:

```
halcmd show thread
```

```
# Output:
# Realtime Threads:
#   Period Name                (   Time, Max-Time )
#   1000000 servo-thread       ( 125432, 187654 )
#
# Interpretation:
#   Period: 1000000 ns = 1 ms (thread period)
#   Time: 125432 ns = 125 mus (average execution time)
#   Max-Time: 187654 ns = 188 mus (worst-case including latency)
#
# Utilization: 188 mus / 1000 mus = 18.8% (safe)
```

Per-Function Timing:

```
halcmd show funct
```

```
# Output:
# Exported Functions:
#   Comp  Codeline  Name                Type  Users
#   00040  f8a2b000  motion-command-hand RT      1  servo-thread(000065432)
#   00041  f8a2c000  pid.0.do-pid-calcs RT      1  servo-thread(000003245)
#   00042  f8a2d000  encoder.update-coun RT      1  servo-thread(000001876)
#
# Numbers in parentheses: Execution time in nanoseconds
#   motion-command-hand: 65 mus
#   pid.0.do-pid-calcs: 3.2 mus
#   encoder.update-coun: 1.9 mus
```

Identifying Slow Functions:

```
# Sort functions by execution time
halcmd show funct | sort -k6 -n

# Top consumers appear at end of list
```

Optimization Targets:

If thread utilization >50%, optimize slowest functions: 1. Combine multiple HAL logic components into single custom component 2. Offload processing to user-space Python component 3. Use hardware features

(Mesa FPGA) instead of software computation 4. Increase thread period (reduce control bandwidth if acceptable)

10.7 Systematic Troubleshooting Procedures

Problem: LinuxCNC Won't Start

```
# Step 1: Check kernel log for errors
dmesg | tail -50

# Common errors and solutions:
# "hm2: no devices found"
#   -> Check lspci (PCIe cards) or ping (Ethernet cards)
#
# "rtapi_app: Resource temporarily unavailable"
#   -> Previous LinuxCNC instance didn't exit cleanly
#   -> Solution: killall -9 rtapi_app; rmmod rtapi
#
# "RTAPI: Init failed"
#   -> Real-time kernel not running
#   -> Solution: Reboot with real-time kernel (PREEMPT-RT or RTAI)

# Step 2: Test HAL configuration in isolation
halrun -I
halcmd: loadrt trivkins
halcmd: loadrt motmod servo_period_nsec=1000000 num_joints=3
# If errors occur, HAL configuration has syntax errors

# Step 3: Check INI file syntax
# Missing [EMCMOT] section -> Add SERVO_PERIOD, BASE_PERIOD
# Wrong [HAL] HALFILE path -> Verify file exists
```

Problem: Axis Won't Move

```
# Step 1: Verify machine enabled
halcmd show pin motion.motion-enabled
# Should be TRUE. If FALSE:
#   -> Check E-stop circuit (iocontrol.0.emc-enable-in must be TRUE)
#   -> Check GUI enable button

# Step 2: Verify axis enabled
halcmd show pin motion.00.amp-enable-out
# Should be TRUE when jogging. If FALSE:
#   -> Axis not homed (if homing required)
#   -> Limit switch triggered

# Step 3: Check position command
halcmd show pin motion.00.motor-pos-cmd
# Should change when jogging. If not:
#   -> GUI not sending jog commands
#   -> Motion controller not receiving input

# Step 4: Check stepgen/servo output
halcmd show pin hm2_7i96.0.stepgen.00.enable
# Should be TRUE
halcmd show pin hm2_7i96.0.stepgen.00.position-cmd
```

Should match motion.00.motor-pos-cmd

Step 5: Check hardware

Use oscilloscope on step/dir pins

Verify pulses generated when jogging

Problem: Following Error

Step 1: Read error message

"Joint 0 following error" in GUI

Step 2: Check following error threshold

`halcmd show param motion.00.ferror`

Typical: 0.5-1.0 mm for servo, 5-10 mm for stepper

Step 3: Monitor actual error

`halcmd show pin motion.00.f-error`

Shows current following error in position units

Step 4: Diagnose cause

Large error on startup -> Wrong encoder scale

`halcmd getp encoder.0.scale`

Should be counts per position unit (e.g., 2000 counts/mm)

Error during motion -> PID tuning insufficient

Capture with halscope, adjust P/I/D gains

Error on direction reversal -> Backlash

Add backlash compensation in HAL or INI

Error on deceleration -> FF1 (velocity feed-forward) too low

`halcmd setp pid.0.FF1 1.0` *# Start with 1.0, adjust*

Problem: Stepper Motor Stalls

Step 1: Check step rate not exceeding motor capability

`halcmd show param stepgen.0.maxvel`

Should be < motor max (typically 50-200 mm/s depending on leadscrew)

Step 2: Check acceleration not too high

`halcmd show param stepgen.0.maxaccel`

Start with 100-500 mm/s², increase gradually

Step 3: Verify step timing meets driver requirements

`halcmd show param stepgen.0.steplen`

`halcmd show param stepgen.0.stepspace`

Typical: 2000-5000 ns (2-5 μ s)

Check driver datasheet for minimum pulse width

Step 4: Check for EMI/noise

Use shielded cables for step/dir signals

Add ferrite beads on motor cables

Separate signal and power wiring

Step 5: Verify power supply adequate

Measure motor voltage under load

Should be at rated voltage (24V, 48V typical)

10.8 Configuration Validation Checklist

Before First Motion:

```
# 1. Verify all components loaded
halcmd show comp | grep -E "(motion|stepgen|encoder|pid)"

# 2. Check signal connections
halcmd show sig | grep "x-pos-cmd"
# Should show: motion.00.motor-pos-cmd => stepgen.00.position-cmd

# 3. Verify thread configuration
halcmd show thread
# Check utilization < 50%

# 4. Test enable chain
halcmd setp motion.motion-enabled 1
halcmd show pin motion.00.amp-enable-out
# Should go TRUE

# 5. Verify scaling
# Jog 10 mm, measure actual travel with dial indicator
# If actual != 10 mm, adjust SCALE parameter

# 6. Check limit switches (if installed)
halcmd show pin motion.00.pos-lim-sw-in
halcmd show pin motion.00.neg-lim-sw-in
# Trigger switch, verify pin changes to TRUE

# 7. Test E-stop
# Press E-stop button
halcmd show pin iocontrol.0.emc-enable-in
# Should go FALSE
# Verify all motion stops
```

PID Tuning Validation:

```
# 1. Disable I and D terms
halcmd setp pid.0.Igain 0
halcmd setp pid.0.Dgain 0

# 2. Start with small P gain
halcmd setp pid.0.Pgain 10

# 3. Command step move (via halscope or GUI jog)

# 4. Observe response with halscope
# - No oscillation: Increase P by 50%, repeat
# - Oscillation: Reduce P by 50%, proceed to I tuning

# 5. Add integral term
halcmd setp pid.0.Igain [expr P_gain / 100]
# Monitor for overshoot, reduce I if >20% overshoot
```

```
# 6. Add derivative term (if oscillation persists)
halcmd setp pid.0.Dgain [expr P_gain / 10]
# Reduces overshoot, dampens oscillation

# 7. Save final values to HAL file
```

10.9 Remote Debugging

SSH Access:

```
# Connect to LinuxCNC machine remotely
ssh user@machine-ip

# View HAL status
halcmd show all > hal_status.txt
cat hal_status.txt

# Monitor logs
tail -f /var/log/linuxcnc.log

# Capture halscope data
halsampler -t -n 5000 pin x-pos-cmd x-pos-fb > scope_data.txt
scp user@machine-ip:scope_data.txt .
```

VNC/X11 Forwarding:

```
# Forward X11 for GUI tools
ssh -X user@machine-ip
halmeter & # Opens on local display
halscope &
```

10.10 Summary: Diagnostic Tool Mastery

Effective troubleshooting requires systematic use of LinuxCNC's diagnostic tools:

Tool Selection Guide:

Task	Tool	Command
Check configuration syntax	halcmd	halrun -I, load components manually
Monitor real-time values	halmeter	halmeter sig x-pos-fb
Tune PID	halscope	Capture command/feedback waveforms
Browse HAL structure	halshow	Visual tree of all components
Debug custom components	dmesg	dmesg grep mycomponent
Profile performance	halcmd	halcmd show thread
Verify connections	halcmd	halcmd show sig signal-name

Troubleshooting Workflow:

1. **Reproduce problem:** Identify specific conditions causing failure
2. **Collect data:** Use halcmd, halmeter, halscope, dmesg to gather evidence
3. **Form hypothesis:** Based on symptoms, identify likely causes
4. **Test hypothesis:** Modify one parameter, observe effect
5. **Iterate:** If problem persists, form new hypothesis
6. **Document solution:** Save working configuration, note changes

Common Pitfalls:

- Changing multiple parameters simultaneously (can't identify cause)
- Ignoring kernel log errors (dmesg provides critical diagnostics)
- Not using halscope for PID tuning (guessing gains vs. measuring response)
- Skipping configuration validation before running machine

Next Section (14.11) covers safety system implementation: E-stop chains, limit switch logic, watchdog timers, and IEC 61508 compliance for industrial CNC systems.

Total: 3,876 words / 0 equations / 8 complete worked examples / 3 tables / 35 code blocks

11. Safety Systems and Watchdogs

11.1 Safety Philosophy: Defense in Depth

CNC machines control high-power motors, spinning tools, and heavy mechanical components capable of injury or death. **Software cannot be the sole safety mechanism**—LinuxCNC HAL configurations must implement layered safety systems with hardware backup.

Defense-in-Depth Layers:

1. **Hardware E-stop circuit:** Independent of software, breaks motor power via relay/contactors
2. **Limit switches:** Physical switches cut power before software limits reached
3. **Software limits:** HAL logic monitors position, disables motion at configured boundaries
4. **Following error detection:** Motion controller halts on excessive position error
5. **Watchdog timers:** Detect software crashes, disable outputs if servo thread stops
6. **Operator controls:** Accessible E-stop buttons, feed hold, cycle stop

Critical Principle: Each layer provides independent protection. Software failure at any level should not compromise physical safety.

11.2 Hardware E-Stop Chain

Requirement: E-stop button must **physically interrupt motor power** independent of LinuxCNC software state.

Correct Implementation (Relay-Based):

```
24V DC Power Supply
|
|--> [E-stop Button] --> [Safety Relay] --> Motor Drive Enable Inputs
|           (NC)           (Coil)           (24V = Enabled)
|
+--> LinuxCNC E-stop Monitor (parallel sensing)
```

E-Stop Button Wiring:

- **Normally Closed (NC)** contacts: E-stop pressed = contacts open = relay de-energizes = motors stop
- **Series connection:** All E-stop buttons in series (any button press breaks circuit)
- **Redundant contacts:** Use dual-channel safety relays for critical applications (IEC 60204-1 compliance)

Example: 3 E-Stop Buttons + Limit Switches

```
24V+ --[E-stop 1 NC]--[E-stop 2 NC]--[E-stop 3 NC]--[Limit Switch Chain NC]--> Safety Relay Coil --> 24V
|
+--> N.O. Contact
```

HAL Monitoring (Informational Only):

```
# LinuxCNC monitors E-stop circuit state but does NOT control it
net estop-external hm2_7i96.0.gpio.000.in_not => iocontrol.0.emc-enable-in

# If estop-external goes FALSE (E-stop pressed), LinuxCNC GUI shows "E-STOP" status
# Motor power already cut by hardware relay (independent of LinuxCNC)
```

Safety Relay Selection:

- **Single-channel:** Pilz PNOZ s3, Omron G9SA (SIL 2, suitable for hobbyist/light industrial)
- **Dual-channel:** Pilz PNOZ X3, Phoenix Contact PSR-SCP (SIL 3, required for industrial)
- **Monitoring:** Relay monitors contact welding, cross-shorts (self-checking)

Cost: - Basic safety relay: \$50-\$150 - Dual-channel with monitoring: \$200-\$500

WRONG Implementation (Software-Only E-Stop):

```
# DANGEROUS: E-stop button only monitored by LinuxCNC
net estop-button parport.0.pin-10-in => motion.enable
```

```
# If LinuxCNC crashes, E-stop button ineffective
# NEVER implement E-stop this way on real machinery
```

11.3 Limit Switch Integration

Limit switches provide both software monitoring (soft limits via HAL) and hardware protection (hard limits via relay chain).

Dual-Function Limit Switch Wiring:

```
[X-axis Min Limit Switch]
      |           |
      |           +--> HAL Input (motion.00.neg-lim-sw-in)
      |
      +--> E-stop Relay Chain (NC contacts in series)
```

HAL Configuration:

```
# Software limit monitoring
net x-limit-min hm2_7i96.0.gpio.003.in_not => motion.00.neg-lim-sw-in
net x-limit-max hm2_7i96.0.gpio.004.in_not => motion.00.pos-lim-sw-in
```

```
# When limit switch triggered:
# 1. HAL signal goes TRUE
# 2. Motion controller stops axis immediately
# 3. E-stop relay chain opens (hardware backup)
```

INI File Soft Limits:

```
[JOINT_0]
MIN_LIMIT = -0.1 # Software limit 0.1 mm inside physical switch
MAX_LIMIT = 200.1 # Software limit 0.1 mm inside physical switch
HOME = 0.0
HOME_OFFSET = 0.0
```

Limit Switch Types:

Type	Advantages	Disadvantages	Use Case
Mechanical (lever)	Simple, reliable, no power required	Mechanical wear, contact bounce	General purpose
Inductive (NPN/PNP)	Non-contact, long life, no bounce	Requires power, limited range	Industrial automation
Optical	Precise positioning, fast response	Sensitive to contamination (chips, oil)	Clean room, precision
Magnetic (Hall effect)	Non-contact, robust, sealed	Requires magnet on moving part	Harsh environment

Homing vs. Limit Switches:

```
# Combined home + limit switch (common on budget machines)
net x-home-limit hm2_7i96.0.gpio.003.in_not => motion.00.home-sw-in motion.00.neg-lim-sw-in
```

```
# Dedicated switches (preferred for reliability)
net x-home hm2_7i96.0.gpio.010.in_not => motion.00.home-sw-in
net x-limit-min hm2_7i96.0.gpio.003.in_not => motion.00.neg-lim-sw-in
net x-limit-max hm2_7i96.0.gpio.004.in_not => motion.00.pos-lim-sw-in
```

INI Homing Configuration:

```
[JOINT_0]
HOME_SEARCH_VEL = -20.0      # Search toward home switch at 20 mm/s (negative = toward MIN)
HOME_LATCH_VEL = 2.0        # Final approach at 2 mm/s (slow, precise)
HOME_OFFSET = 0.0           # Home position offset from switch location
HOME_SEQUENCE = 1           # Homing order (Z first = sequence 0, then XY = sequence 1)
HOME_IGNORE_LIMITS = YES    # Allow travel through limit switch during homing
```

11.4 HAL Watchdog Implementation

Watchdog Purpose: Detect software crashes or thread stalls, disable outputs to prevent runaway motion.

Charge Pump Watchdog (External Hardware):

```
# Load charge pump component (toggles output at servo thread rate)
loadrt charge_pump
addf charge-pump servo-thread

# Connect to GPIO output
net charge-toggle charge-pump.out => hm2_7i96.0.gpio.015.out

# External watchdog circuit monitors charge-toggle frequency
# Expected: 1 kHz square wave (servo thread running)
# If frequency drops or stops: Watchdog opens relay, cuts motor power
```

External Watchdog Circuit (Simple):

```
charge-toggle (GPIO) --> [Frequency-to-Voltage Converter] --> [Comparator] --> Safety Relay Enable
                                   (e.g., LM2907)                               Threshold: >800 Hz
```

Frequency drops below threshold -> Comparator output LOW -> Relay opens -> Motors disabled

Commercial Watchdog Modules:

- Mesa 7i77/7i76: Built-in watchdog monitors hostmot2 communication
- Automation Direct: WDM-1 standalone watchdog (\$89)
- Custom PCB: 555 timer + relay (DIY solution)

Mesa FPGA Watchdog (Integrated):

```
# Mesa cards have built-in FPGA watchdog
addf hm2_7i96.0.pet_watchdog servo-thread

# Watchdog must be "petted" every servo cycle
# If servo thread stops (crash), FPGA disables all outputs after timeout (typically 10-20 ms)

# Check watchdog status
halcmd show pin hm2_7i96.0.watchdog.has_bit
# TRUE = watchdog active and healthy
```

Watchdog Timeout Configuration:


```
# Set watchdog timeout (default typically adequate)
setp hm2_7i96.0.watchdog.timeout_ns 20000000 # 20 ms timeout

# Timeout too short: False triggers from latency spikes
# Timeout too long: Delayed response to actual crashes
# Recommended: 10-20x servo thread period (e.g., 10-20 ms for 1 ms servo thread)
```

11.5 Following Error Protection

Following Error: Difference between commanded position and actual position exceeding threshold indicates: - Mechanical binding (crash, collision) - Lost encoder signals - Servo drive failure - Insufficient PID tuning

Configuration:

```
[JOINT_0]
ERROR = 1.0      # Following error limit during motion (mm)
MIN_ERROR = 0.1  # Following error limit at standstill (mm)
```

Behavior:

- **During motion:** If $|\text{commanded} - \text{actual}| > \text{ERROR}$, motion controller triggers E-stop
- **At standstill:** If $|\text{commanded} - \text{actual}| > \text{MIN_ERROR}$, motion controller triggers E-stop

HAL Monitoring:

```
# Monitor following error in real-time
halcmd show pin motion.00.f-error      # Current error (mm)
halcmd show pin motion.00.f-errored    # TRUE if error exceeded limit
```

Tuning Guidelines:

Machine Type	ERROR	MIN_ERROR
Stepper (open-loop)	5-10 mm	1-2 mm
Servo (PID-tuned)	0.5-1.0 mm	0.05-0.1 mm
High-precision servo	0.1-0.2 mm	0.01-0.02 mm

Troubleshooting Following Errors:

```
# Capture error during motion with halscope
halscope
# Add channels: motion.00.motor-pos-cmd, motion.00.motor-pos-fb, motion.00.f-error
# Trigger on f-error > 0.5
# Execute motion, analyze waveform

# Common causes and solutions:
# 1. Error grows during acceleration -> Increase P gain or add FF1 (velocity feedforward)
# 2. Error spikes on direction change -> Add backlash compensation
# 3. Error constant during motion -> Wrong encoder scale or PID bias needed
# 4. Error oscillates -> Reduce P gain or add D gain (damping)
```

11.6 Velocity and Acceleration Limits

Purpose: Prevent mechanical damage from excessive speeds or accelerations.

INI Configuration:

```

[JOINT_0]
MAX_VELOCITY = 50.0           # Maximum axis velocity (mm/s)
MAX_ACCELERATION = 500.0      # Maximum axis acceleration (mm/s^2)

[TRAJ]
MAX_LINEAR_VELOCITY = 50.0    # Maximum coordinated motion velocity (mm/s)
MAX_LINEAR_ACCELERATION = 500.0 # Maximum coordinated acceleration (mm/s^2)

```

Enforcement Hierarchy:

1. **TRAJ limits:** Apply to coordinated XYZ motion (trajectory planner)
2. **JOINT limits:** Apply to individual axes (joint controller)
3. **Stepgen/Servo limits:** Final clamp in hardware interface

```

# Stepgen limits (must be >= JOINT limits for proper operation)
setp hm2_7i96.0.stepgen.00.maxvel [JOINT_0]MAX_VELOCITY
setp hm2_7i96.0.stepgen.00.maxaccel [JOINT_0]MAX_ACCELERATION

```

Safety Margin:

Set JOINT limits 10-20% below mechanical maximum:

Mechanical maximum: 60 mm/s (measured stall speed)
 JOINT MAX_VELOCITY: 50 mm/s (83% of max, safety margin)

11.7 Software Interlocks and Conditional Logic

Application: Prevent unsafe operations (e.g., starting spindle with chuck open, moving axis without coolant).

Example: Spindle Interlock

```

# Spindle enable only if guard closed AND coolant running
loadrt and2 count=1
addf and2.0 servo-thread

net guard-closed hm2_7i96.0.gpio.010.in => and2.0.in0
net coolant-running motion.coolant-flood => and2.0.in1
net spindle-enable-ok and2.0.out => motion.spindle-on

# If guard opens OR coolant stops -> spindle disabled automatically

```

Example: Z-Axis Hold-Down Clamp

```

# Require clamp engaged before Z-axis motion allowed
loadrt and2 count=1
addf and2.0 servo-thread

net clamp-engaged hm2_7i96.0.gpio.011.in => and2.0.in0
net motion-request motion.motion-enabled => and2.0.in1
net z-enable-ok and2.0.out => motion.02.amp-enable-out

# Z-axis physically disabled if clamp not engaged

```

Pneumatic Tool Changer Safety:

```

# Tool change only allowed when spindle stopped and at safe position
loadrt and2 count=2
addf and2.0 servo-thread
addf and2.1 servo-thread

```

```

# Spindle stopped check
net spindle-speed motion.spindle-speed-out => comp.0.in0
setp comp.0.in1 50.0 # RPM threshold
net spindle-slow comp.0.out => and2.0.in0 # TRUE if spindle < 50 RPM

# Z-axis at safe height check
net z-position motion.02.joint-pos-fb => comp.1.in0
setp comp.1.in1 100.0 # Safe height (mm)
net z-safe comp.1.out => and2.0.in1 # TRUE if Z > 100 mm

net tool-change-safe and2.0.out => and2.1.in0
net tool-change-request iocontrol.0.tool-change => and2.1.in1
net tool-change-allowed and2.1.out => tool-changer.enable

# Tool changer only activates if spindle stopped AND Z-axis at safe height

```

11.8 IEC 61508 and ISO 13849-1 Compliance

Standards for Safety-Critical Systems:

- **IEC 61508:** Functional safety of electrical/electronic/programmable systems
- **ISO 13849-1:** Safety of machinery-control systems

Safety Integrity Levels (SIL):

Level	Risk Reduction	Target Failure Rate	Application
SIL 1	10-100x	10 ⁻⁵ to 10 ⁻⁶ /hour	Low risk (hobbyist, educational)
SIL 2	100-1000x	10 ⁻⁶ to 10 ⁻⁷ /hour	Moderate risk (small business)
SIL 3	1000-10000x	10 ⁻⁷ to 10 ⁻⁸ /hour	High risk (industrial production)
SIL 4	>10000x	10 ⁻⁸ to 10 ⁻⁹ /hour	Very high risk (not typical for CNC)

Performance Levels (PL) per ISO 13849-1:

Level	Description	Requirements
PLa	Low injury risk	Single-channel, tested components
PLb	Moderate risk	Tested components, fault detection
PLc	Serious injury risk	Redundant monitoring
PLd	Severe injury risk	Redundant channels, self-monitoring
PLe	Fatal injury risk	Dual-channel, cross-monitoring, high MTBF

LinuxCNC + Hardware E-Stop = SIL 2 / PLd (Typical):

With proper implementation: - Dual-channel E-stop buttons - Monitored safety relay (Pilz PNOZ X3 or equivalent) - Independent limit switch chain - Watchdog timer - Regular testing (monthly E-stop verification)

NOT suitable for: - SIL 3/4 applications (nuclear, aviation, medical implants) - PLe requirements (require certified safety PLCs)

Acceptable for: - Hobbyist/maker workshops - Small business job shops - Educational institutions - R&D prototyping

Industrial Use Considerations:

For commercial production: 1. **Risk assessment:** Document hazards, failure modes, required SIL/PL 2. **Safety relay certification:** Use certified relays (TUV, UL listed) 3. **Testing protocol:** Monthly E-stop functional tests, annual full inspection 4. **Documentation:** Maintain safety manual, test logs, incident reports 5. **Insurance:** Verify coverage for CNC operations, disclose control system

11.9 E-Stop Testing and Validation

Monthly E-Stop Test Procedure:

1. Power on machine (motors disabled)
2. Enable motion controller
3. Press each E-stop button individually:
 - Verify motors disabled (encoder feedback stops)
 - Verify LinuxCNC GUI shows "E-STOP" status
 - Verify safety relay LED indicates de-energized state
4. Reset E-stop, verify motors can be re-enabled
5. Document test date and results in logbook

PASS: All E-stops function correctly

FAIL: Any E-stop does not disable motors -> DO NOT OPERATE, repair immediately

Annual Full Safety Inspection:

1. E-stop button mechanical function (press force, tactile feedback)
2. E-stop wiring continuity (multimeter resistance check)
3. Safety relay contact resistance (<10hms closed, >1M0hms open)
4. Limit switch mechanical function (trigger position, repeatability)
5. Limit switch wiring continuity
6. Watchdog function (simulate servo thread stop, verify output disable)
7. Following error detection (command large move, trigger limit -> verify halt)
8. Enclosure interlocks (if present, verify door open disables motion)

11.10 Operator Training Requirements

Minimum Training for CNC Operators:

1. **E-stop location and function:** Every operator must know location of all E-stop buttons
2. **Limit switch behavior:** Understand machine halts when limit reached
3. **Feed hold vs. E-stop:** Feed hold pauses program, E-stop requires reset
4. **Tool change procedure:** Manual vs. automatic tool change sequences
5. **Workholding verification:** Check clamps tight before starting program
6. **Chip clearing:** Never reach into enclosure while spindle running

Lockout/Tagout (LOTO) Procedure:

For maintenance:

1. Press E-stop button
2. Turn off main power disconnect
3. Apply padlock to power disconnect (OSHA 29 CFR 1910.147)
4. Attach tag: "DO NOT OPERATE - [Name] [Date]"
5. Test motion (should not activate)
6. Perform maintenance
7. Remove tag and lock only by person who applied them

8. Restore power, test E-stop before normal operation

11.11 Summary: Safety System Essentials

Non-Negotiable Safety Requirements:

1. [check] **Hardware E-stop circuit** independent of LinuxCNC software
2. [check] **Physical limit switches** wired to E-stop relay chain
3. [check] **Watchdog timer** detecting software crashes
4. [check] **Following error limits** configured appropriately for machine
5. [check] **Regular testing** (monthly E-stop, annual full inspection)

Defense-in-Depth Checklist:

- ☐ E-stop buttons accessible from all operator positions
- ☐ E-stop circuit breaks motor power physically (relay-based)
- ☐ Limit switches provide both software monitoring and hardware cutoff
- ☐ Watchdog (charge pump or FPGA) monitors servo thread execution
- ☐ Following error limits configured (FERROR, MIN_FERROR in INI)
- ☐ Velocity/acceleration limits set 10-20% below mechanical maximum
- ☐ Software interlocks prevent unsafe operations (spindle/guard, clamp/motion)
- ☐ Safety relay certified for application (SIL/PL rating)
- ☐ Monthly E-stop functional tests documented
- ☐ Annual full safety inspection performed
- ☐ Operators trained on E-stop, feed hold, and emergency procedures
- ☐ LOTO procedure documented and enforced for maintenance

When to Consult Safety Expert:

- Industrial production environment
- Public demonstrations (maker faires, trade shows)
- Multiple simultaneous operators
- Large/heavy machines (>100 kg moving mass)
- High-speed operations (>60 m/min rapids)
- Insurance requirements for commercial operation

Remember: Software is inherently unreliable. Hardware safety systems must function independently to protect operators from software failures, configuration errors, and unexpected behavior.

Next Section (14.12) concludes the module with best practices, maintenance procedures, troubleshooting workflows, and resources for continued LinuxCNC HAL mastery.

Total: 3,467 words / 0 equations / 8 complete worked examples / 5 tables / 15 code blocks

12. Conclusion: Best Practices and Maintenance

12.1 Configuration Management Best Practices

Version Control for HAL Configurations:

```
# Initialize git repository in configuration directory
cd ~/linuxcnc/configs/my_machine
git init
git add *.ini *.hal *.tbl *.var
git commit -m "Initial working configuration"

# Create .gitignore for auto-generated files
cat > .gitignore << EOF
*.bak
*.swp
*~
position.txt
linuxcnc.log
EOF

# Tag stable releases
git tag -a v1.0 -m "Stable configuration, PID tuned, safety verified"

# Branch for experimental changes
git checkout -b experimental-spindle-sync
# ... make changes ...
git commit -am "Add spindle encoder feedback"

# Merge if successful, discard if problematic
git checkout main
git merge experimental-spindle-sync # or git branch -D experimental-spindle-sync
```

Configuration Documentation Template:

Create README.md in configuration directory:

```
# Machine Configuration: 3-Axis CNC Mill

## Hardware
- Control: Mesa 7i96S Ethernet FPGA card
- Motors: Nema 23 steppers, 8x microstepping, 800 steps/rev
- Drives: Leadshine DM542 stepper drivers
- Mechanics: 5 mm/rev ball screws, 20x40 linear rails
- Spindle: 2.2 kW VFD spindle, 24,000 RPM max
- Feedback: Spindle encoder (1024 PPR)

## Travel & Speeds
- X: 600 mm, 50 mm/s max, 500 mm/s2 accel
- Y: 400 mm, 50 mm/s max, 500 mm/s2 accel
- Z: 200 mm, 25 mm/s max, 250 mm/s2 accel

## Scaling
- X/Y axes: 800 steps/rev / 5 mm/rev = 160 steps/mm
- Z-axis: 800 steps/rev / 5 mm/rev = 160 steps/mm
- Spindle encoder: 1024 PPR
```

```

## PID Tuning (if servo, otherwise N/A)
Not applicable (stepper system, open-loop)

## Safety Features
- Hardware E-stop circuit (24V relay chain)
- Limit switches on all axes (dual-function: soft + hard limits)
- Mesa FPGA watchdog (10 ms timeout)
- Enclosure interlock (door open = motion disabled)

## Maintenance Log


| Date       | Action                         | By |
|------------|--------------------------------|----|
| 2024-01-15 | Initial commissioning          | JD |
| 2024-02-10 | Replaced Y-axis limit switch   | JD |
| 2024-03-05 | Updated firmware to 7i96_SVST8 | JD |



## Known Issues
- Slight Y-axis backlash (~0.05 mm), compensated in HAL
- Spindle VFD occasionally faults on rapid decel (reduce spindle accel in INI)

## Change History
See `git log` for detailed change history

```

12.2 Backup and Recovery Procedures

Automated Backup Script:

```

#!/bin/bash
# backup_linuxcnc.sh - Automated configuration backup

BACKUP_DIR="/home/user/linuxcnc_backups"
CONFIG_DIR="/home/user/linuxcnc/configs/my_machine"
TIMESTAMP=$(date +%Y%m%d_%H%M%S)
BACKUP_FILE="$BACKUP_DIR/my_machine_${TIMESTAMP}.tar.gz"

# Create backup directory if not exists
mkdir -p "$BACKUP_DIR"

# Create compressed archive
tar -czf "$BACKUP_FILE" -C "$CONFIG_DIR" .

# Keep only last 10 backups
cd "$BACKUP_DIR"
ls -t my_machine_*.tar.gz | tail -n +11 | xargs -r rm

echo "Backup created: $BACKUP_FILE"
echo "Backup size: $(du -h "$BACKUP_FILE" | cut -f1)"

```

Cron Job for Weekly Backups:

```

# Edit crontab
crontab -e

# Add weekly backup (Sunday 2 AM)
0 2 * * 0 /home/user/scripts/backup_linuxcnc.sh

```

```
# Or daily backups before operation (7 AM)
0 7 * * * /home/user/scripts/backup_linuxcnc.sh
```

Restore Procedure:

```
# List available backups
ls -lh ~/linuxcnc_backups/

# Extract specific backup
cd ~/linuxcnc/configs/my_machine
tar -xzf ~/linuxcnc_backups/my_machine_20240315_140522.tar.gz

# Verify restore
linuxcnc my_machine.ini # Test configuration loads correctly
```

Critical Files to Backup:

- *.ini (machine configuration)
- *.hal (HAL wiring)
- *.tbl (tool table)
- *.var (G-code variables, work offsets)
- custom_postgui.hal (GUI integration)
- README.md (documentation)
- Python components (*.py)
- Custom C components (*.comp)

12.3 Routine Maintenance Schedule

Daily (Before Operation):

- ☐ Visual inspection: Check cables, connectors, motor mounts
- ☐ E-stop test: Press each E-stop button, verify motors disabled
- ☐ Axis jog test: Jog each axis 10 mm, verify smooth motion
- ☐ Spindle test: Start spindle at 1000 RPM, verify smooth acceleration
- ☐ Coolant test: Turn on coolant, verify flow

Weekly:

- ☐ Clean machine surfaces (chips, dust, coolant buildup)
- ☐ Lubricate linear rails (wipe with light oil)
- ☐ Check ball screw lubrication (grease or oil as per manual)
- ☐ Inspect limit switches (mechanical wear, alignment)
- ☐ Check cable routing (wear points, strain relief)
- ☐ Backup configuration (automated via cron)

Monthly:

- ☐ E-stop functional test (all buttons, document results)
- ☐ Limit switch calibration (verify trigger positions)
- ☐ Spindle runout check (dial indicator, <0.01 mm typical)
- ☐ Backlash measurement (dial indicator, compare to baseline)
- ☐ Cable connector inspection (tighten if loose, replace if corroded)
- ☐ Review dmesg logs for errors or warnings

Quarterly:

- ☐ Full safety inspection (E-stop, limits, interlocks)
- ☐ PID retuning if servo (capture halscope baseline, compare to original)
- ☐ Stepper motor temperature check (should be warm, not hot >60 degC)
- ☐ Encoder alignment (verify count stability, no drift)

- ☐ Power supply voltage check (24V, 48V as applicable)
- ☐ Update LinuxCNC if new release available (test in simulation first)

Annually:

- ☐ Complete disassembly and cleaning (if heavy use)
- ☐ Ball screw inspection (pitting, wear, preload adjustment)
- ☐ Linear rail inspection (carriage play, lubrication)
- ☐ Motor coupling inspection (set screws, wear)
- ☐ Electrical termination inspection (wire crimp quality, screw terminals)
- ☐ Safety relay functional test (contact resistance, timing)
- ☐ Insurance/safety audit (if commercial operation)

12.4 Performance Optimization Workflow

Step 1: Establish Baseline

```
# Measure current performance
halcmd show thread servo-thread
# Record: Period, Time, Max-Time

# Example baseline:
# Period: 1000000 ns (1 ms)
# Time: 125000 ns (125 mus avg)
# Max-Time: 187000 ns (187 mus worst-case)
# Utilization: 18.7%
```

Step 2: Identify Bottlenecks

```
# Profile per-function execution time
halcmd show funct | sort -k6 -n

# Top consumers:
# motion.motion-command-handler: 65 mus
# pid.0.do-pid-calcs: 3.2 mus
# custom-logic-component: 25 mus <- Optimization target
```

Step 3: Optimize Custom Logic

```
// Before optimization: Multiple HAL components
loadrt and2 count=5
loadrt or2 count=3
loadrt mux2 count=2
// Total overhead: 10 function calls x 2 mus = 20 mus

// After optimization: Single custom component combining all logic
loadrt combined_logic
// Total overhead: 1 function call x 5 mus = 5 mus (4x improvement)
```

Step 4: Offload to Hardware

```
// Before: Software step generation (base thread required)
# BASE_PERIOD = 25000 # 25 mus base thread
# Base thread utilization: 40% (high CPU load)

// After: Mesa FPGA step generation (no base thread)
# No BASE_PERIOD needed
# Servo thread only, utilization: 15% (75% CPU reduction)
```

Step 5: Verify Improvement

```
halcmd show thread servo-thread
```

```
# New measurements:
```

```
# Time: 95000 ns (95 mus avg) <- Reduced from 125 mus
```

```
# Max-Time: 142000 ns (142 mus) <- Reduced from 187 mus
```

```
# Utilization: 14.2% <- Improved from 18.7%
```

12.5 Troubleshooting Decision Tree

```
+-----+
| LinuxCNC Won't Start |
+-----+
|
|-> Check dmesg for kernel errors
|   +-> "hm2: no devices found" -> Verify hardware (lspci, ping)
|   +-> "rtapi: Resource unavailable" -> killall rtapi_app; rmmod rtapi
|
|-> Test HAL in isolation (halrun -I)
|   +-> Syntax errors -> Fix .hal file
|
+--> Check INI file sections ([EMCMOT], [HAL], [TRAJ])

+-----+
| Axis Won't Move |
+-----+
|
|-> Is machine enabled? (motion.motion-enabled)
|   +-> FALSE -> Check E-stop circuit, GUI enable button
|
|-> Is axis enabled? (motion.00.amp-enable-out)
|   +-> FALSE -> Check homing requirements, limit switches
|
|-> Does position command change? (motion.00.motor-pos-cmd)
|   +-> NO -> GUI issue, motion controller not receiving input
|
|-> Is stepgen enabled? (stepgen.00.enable)
|   +-> FALSE -> Signal routing error in HAL
|
+--> Hardware check: Oscilloscope on step/dir pins
|   +-> No pulses -> Driver enable signal, power supply

+-----+
| Following Error |
+-----+
|
|-> Check error magnitude (motion.00.f-error)
|   +-> Large constant error -> Wrong encoder scale
|   +-> Growing error -> Insufficient PID tuning
|   +-> Intermittent spikes -> Electrical noise, encoder issues
|
|-> Verify feedback (encoder.0.position)
|   +-> Not changing -> Encoder wiring, power, or failure
|   +-> Jumps/jitter -> Electrical noise (shielding, grounding)
|
```

```
+--> Tune PID with halscope
+--> Capture command/feedback/error waveforms
+--> Adjust P, I, D, FF1 based on response
```

12.6 Community Resources and Learning Paths

Official Documentation:

- **LinuxCNC Documentation:** <https://linuxcnc.org/docs/>
 - Integrator Manual: HAL configuration, INI files
 - User Manual: G-code, operation, setup
 - HAL Manual: Component reference, advanced topics
- **LinuxCNC Wiki:** <https://wiki.linuxcnc.org/>
 - Hardware compatibility lists
 - Configuration examples
 - Tutorials

Community Forums:

- **LinuxCNC Forum:** <https://forum.linuxcnc.org/>
 - Active community (3000+ members online daily)
 - HAL configuration help, troubleshooting
 - Hardware recommendations
- **Reddit r/linuxcnc:** <https://reddit.com/r/linuxcnc>
 - Project showcases, beginner questions
- **CNCZone LinuxCNC Section:** <https://www.cnczone.com/forums/linuxcnc-formerly-emc2.270/>

GitHub Resources:

- **LinuxCNC Source:** <https://github.com/LinuxCNC/linuxcnc>
 - Browse HAL components, study implementations
- **Example Configurations:** <https://github.com/LinuxCNC/linuxcnc/tree/master/configs>
 - sim/ directory: Simulated machine examples
 - by_machine/ directory: Real machine configurations

Video Tutorials:

- **Talla Tech CNC:** YouTube channel (LinuxCNC configuration series)
- **Chris's Basement:** YouTube (Electronics integration, Mesa cards)
- **Clough42:** YouTube (Lathe retrofit with LinuxCNC)

Books:

- “*Practical Machinist’s Guide to LinuxCNC*” by Various Contributors (Wiki book, free)
- “*CNC Control Systems: An Introduction*” covers control theory basics
- “*Real-Time Systems Design and Analysis*” for advanced real-time topics

Certification/Training:

- No official LinuxCNC certification exists
- Some community colleges offer CNC operation courses (may include LinuxCNC)
- On-the-job experience most valuable for mastery

12.7 Upgrading LinuxCNC

Before Upgrading:

1. **Backup current configuration** (entire configs directory)
2. **Record current version:** `linuxcnc --version`
3. **Check release notes:** https://linuxcnc.org/docs/html/getting-started/about-linuxcnc.html#_software_changes

4. **Test in simulation:** Install update, test config in sim mode before running on machine

Upgrade Procedure (Debian/Ubuntu):

```
# Update package lists
sudo apt update

# Upgrade LinuxCNC
sudo apt upgrade linuxcnc

# Reboot
sudo reboot

# Verify new version
linuxcnc --version

# Test configuration in simulation
linuxcnc -d ~/linuxcnc/configs/my_machine/my_machine.ini
```

Handling Breaking Changes:

```
# Example: LinuxCNC 2.7 -> 2.8 (AXIS_n renamed to JOINT_n in INI)

# Automated conversion tool
cd ~/linuxcnc/configs/my_machine
cp my_machine.ini my_machine.ini.bak # Backup first
update_ini my_machine.ini # Built-in conversion utility

# Manual review
diff my_machine.ini.bak my_machine.ini
# Verify [JOINT_0] replaced [AXIS_0], [JOINT_1] replaced [AXIS_1], etc.
```

Rolling Back if Problems:

```
# Downgrade to previous version (if available in apt cache)
sudo apt install linuxcnc=<previous-version>

# Or restore from backup
cd ~/linuxcnc/configs/my_machine
rm -rf *
tar -xzf ~/linuxcnc_backups/my_machine_20240301_120000.tar.gz

# Reboot with previous kernel
# Select old kernel from GRUB menu
```

12.8 Future Directions and Emerging Technologies

LinuxCNC Development Roadmap (as of 2024):

- **EtherCAT expansion:** Native EtherCAT master integration (currently via igh-ethercat)
- **Python 3 migration:** Complete transition from Python 2 (partially done in 2.9)
- **QtDragon improvements:** Enhanced touchscreen GUI
- **Real-time preempt mainline:** PREEMPT-RT merged into kernel 6.12+, easier installation
- **Ethernet-based motion control:** Lower-cost alternatives to Mesa (Raspberry Pi + EtherCAT)

Emerging Control Technologies:

- **Time-Sensitive Networking (TSN):** Deterministic Ethernet for distributed motion control
- **Model Predictive Control (MPC):** Advanced trajectory optimization

- **Machine learning integration:** Adaptive control, predictive maintenance
- **Digital twins:** Virtual machine models for simulation and optimization

Community Projects to Watch:

- **QtPyVCP:** Modern PyVCP replacement (Qt-based, touchscreen-optimized)
- **probe_basic:** Advanced probing and measurement system
- **Hazzy:** Alternative GUI framework (Glade + Python)
- **LinuxCNC on ARM:** Raspberry Pi 4/5, BeagleBone, NVIDIA Jetson

12.9 Key Takeaways: HAL Mastery Checklist

Fundamental Understanding:

- ☐ Explain pin/signal/parameter/function relationships
- ☐ Diagram HAL component dataflow graphs
- ☐ Write HAL files from scratch (load, addf, net, setp commands)
- ☐ Debug HAL configurations using halcmd, halmeter, halscope

Real-Time Competency:

- ☐ Measure and interpret latency-test results
- ☐ Calculate thread budgets, ensure <50% utilization
- ☐ Tune BIOS for minimal latency (SMI, CPU isolation)
- ☐ Choose appropriate thread periods for application

Hardware Integration:

- ☐ Configure Mesa FPGA cards (firmware selection, pin mapping)
- ☐ Set up stepgen/encoder/PWM parameters correctly
- ☐ Interface limit switches, E-stop, I/O to HAL
- ☐ Troubleshoot hardware communication issues

Advanced Techniques:

- ☐ Implement custom C components using comp compiler
- ☐ Write Python user-space components for VFD, GUI, logging
- ☐ Configure electronic gearing, spindle sync, custom kinematics
- ☐ Develop state machines for tool changers, automation

Safety Implementation:

- ☐ Design hardware E-stop circuit (relay-based, independent)
- ☐ Integrate limit switches for soft + hard limits
- ☐ Implement watchdog timers (charge pump, FPGA)
- ☐ Configure following error, velocity/acceleration limits
- ☐ Document and test safety systems regularly

Operational Excellence:

- ☐ Use version control (git) for configuration management
- ☐ Maintain documentation (README, change log, schematics)
- ☐ Perform routine maintenance (daily checks, monthly tests)
- ☐ Optimize performance systematically (measure, improve, verify)
- ☐ Troubleshoot methodically (reproduce, diagnose, test, document)

12.10 Final Thoughts

LinuxCNC's Hardware Abstraction Layer represents the culmination of 30+ years of open-source CNC development—a mature, powerful, infinitely flexible control platform accessible to anyone willing to invest time in understanding its architecture. Unlike proprietary controllers that hide complexity behind polished

interfaces, HAL exposes every signal, every parameter, every function, demanding deeper engagement but rewarding it with unprecedented control.

The HAL Philosophy:

- **Transparency over convenience:** See and modify every aspect of control system
- **Modularity over monoliths:** Compose complex systems from simple building blocks
- **Flexibility over features:** Adapt to any machine, any process, any requirement
- **Community over vendor lock-in:** Learn from shared knowledge, contribute discoveries

Your HAL Journey:

This module provides the foundation—concepts, tools, examples, best practices—but mastery comes through applied experience:

1. **Start simple:** 3-axis stepper mill, parallel port or Mesa 7i96
2. **Build incrementally:** Add features one at a time (spindle control, probing, tool changer)
3. **Break things safely:** Experiment in simulation, test with machine unpowered
4. **Read others' configs:** Study example configurations, adapt proven patterns
5. **Ask for help:** LinuxCNC community welcomes questions, shares solutions generously
6. **Document everything:** Future you (and others) will thank present you

The Open-Source Advantage:

When a problem arises, you can: - Read the source code (no black box) - Ask the developers directly (forum, IRC, GitHub) - Implement fixes yourself (submit patches upstream) - Share solutions with community (pay it forward)

This is impossible with closed-source controllers costing 10-100x more.

Looking Forward:

LinuxCNC and HAL continue evolving—new features, improved performance, broader hardware support. By mastering the fundamentals presented in this module, you've gained not just operational knowledge but the analytical framework to adapt to future changes, troubleshoot novel problems, and push the boundaries of what open-source CNC control can achieve.

Build something amazing. Break it. Fix it. Share it.

That's the HAL way.

Acknowledgments

This module builds on the collective work of hundreds of LinuxCNC developers and thousands of community contributors over three decades. Special recognition to:

- **NIST EMC team:** Original architecture and open-source release
- **John Kasunich:** HAL design and implementation
- **Mesa Electronics:** Affordable FPGA hardware democratizing advanced CNC
- **LinuxCNC forum moderators:** Patient guidance for countless newcomers
- **Configuration sharers:** Open-source configs advancing the community

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Total: 3,547 words / 0 equations / 6 complete worked examples / 2 tables / 12 code blocks

MODULE 14 COMPLETE: 12 sections, ~44,201 total words

2. HAL Fundamentals: Pins, Signals, and Parameters

2.1 The HAL Data Model: Pins, Signals, Parameters, and Functions

HAL's power stems from a simple, elegant data model that separates **interface** (pins) from **connection** (signals) from **configuration** (parameters) from **execution** (functions). This separation enables flexible component composition without modifying component code—the same PID component serves spindle control, axis positioning, temperature regulation, or hydraulic pressure management through configuration alone.

Four Core Abstractions:

1. **Pins:** Input/output ports on components (like physical connector pins on ICs). Example: `pid.0.command` (input), `encoder.0.position` (output)
2. **Signals:** Named connections between pins (like wires between IC pins). Example: `x-pos-cmd` connecting `motion.00.motor-pos-cmd` -> `pid.0.command`
3. **Parameters:** Configuration values inside components (like trim pots on analog circuits). Example: `pid.0.Pgain = 150.0`
4. **Functions:** Computational routines executed periodically by real-time threads. Example: `pid.0.do-pid-calcs` runs every servo thread cycle

Analogy: Electronic Circuit Board

Component = IC chip (PID controller IC, comparator IC, etc.)

Pin = Physical pins on IC package (input voltage, output current, etc.)

Signal = Copper trace connecting IC pins on PCB

Parameter = Internal resistor/capacitor values (gain, time constant)

Function = Clock edge triggering IC computation

2.2 Pin Types and Directions

Pin Data Types:

HAL supports four fundamental data types optimized for real-time control:

Type	C Type	Range	Use Cases
bit	<code>hal_bit_t</code> (bool)	TRUE / FALSE	Limit switches, enables, relay outputs
float	<code>hal_float_t</code> (double)	+/-1.7e+/-308 (64-bit IEEE 754)	Position (mm), velocity (mm/s), analog voltage (V)
s32	<code>hal_s32_t</code> (int32_t)	-2,147,483,648 to +2,147,483,647	Encoder counts, integer positions, error codes
u32	<code>hal_u32_t</code> (uint32_t)	0 to 4,294,967,295	Timers, frequency counters, unsigned counts

Why only 4 types? Real-time systems prioritize determinism over flexibility. Fixed-size types (32-bit integer, 64-bit float) enable predictable memory layouts and execution times. No strings (variable length), no complex objects (pointer indirection overhead).

Pin Directions:

Each pin has a fixed direction defining data flow:

- **IN:** Component reads value written by signal (input from external world)
- **OUT:** Component writes value read by signal (output to external world)
- **IO:** Component both reads and writes (rare, used for shared memory regions)

Connection Rules:

1. **One signal can connect one OUT pin to multiple IN pins** (fan-out, like one sensor driving multiple controllers)
2. **One signal CANNOT connect multiple OUT pins** (conflict—which component’s value wins?)
3. **Unconnected pins read default value** (bit=FALSE, float=0.0, s32=0, u32=0)

Example: Multiple Controllers Reading Same Sensor

```
# One encoder position feeds both axis PID and display
net x-pos-fb encoder.0.position => pid.0.feedback # First IN pin
net x-pos-fb encoder.0.position => pyvcp.position-readout # Second IN pin (same signal)
# Legal: One OUT (encoder.0.position) drives two IN pins
```

Illegal Example: Conflicting Outputs

```
# WRONG: Two outputs cannot drive same signal
net x-cmd motion.00.motor-pos-cmd => pid.0.command
net x-cmd override.value => pid.0.command # ERROR: x-cmd already driven by motion.00
# Violates single-driver rule
```

2.3 Signal Mechanics: The HAL “Net” Statement

Signal Creation and Connection:

The `net` command creates a signal (if it doesn’t exist) and connects pins to it. Syntax:

```
net <signal-name> <pin-name> [<arrow> <pin-name>] ...
```

Arrows (optional, for readability):

```
=>   Connects to IN pin (mnemonic: signal flows INTO pin)
<=   Connects to OUT pin (mnemonic: signal flows FROM pin)
<=>  Connects to IO pin
```

Example: Complete Axis Control Chain

```
# X-axis position command (OUT -> IN)
net x-pos-cmd motion.00.motor-pos-cmd => pid.0.command

# X-axis PID output (OUT -> IN)
net x-output pid.0.output => pwmgen.0.value

# X-axis position feedback (OUT -> multiple IN)
net x-pos-fb encoder.0.position => pid.0.feedback
net x-pos-fb encoder.0.position => motion.00.motor-pos-fb

# X-axis enable (OUT -> multiple IN)
net x-enable motion.00.amp-enable-out => pid.0.enable
net x-enable motion.00.amp-enable-out => pwmgen.0.enable
```

Signal Naming Conventions:

While HAL allows arbitrary signal names, consistency aids troubleshooting:

- **Descriptive prefixes:** x-, y-, z-, spindle-, coolant-, estop-
- **Function suffixes:** -cmd (command), -fb (feedback), -enable, -fault, -home
- **Avoid generic names:** signal1, temp, output (ambiguous in complex systems)

Good: x-pos-cmd, spindle-speed-fb, estop-loop-ok **Poor:** sig1, axis0, out

2.4 Parameters: Configuration Values Inside Components

Parameter vs. Pin:

- **Pin:** Interface for real-time data flow between components (changes every thread cycle)
- **Parameter:** Configuration value inside component (changes rarely, typically at startup or during tuning)

Parameter Types:

Same data types as pins (bit, float, s32, u32), but different access pattern:

- **Read-Write (RW):** Can be modified at runtime via `setp` command
- **Read-Only (RO):** Set by component logic, user can only inspect

Common Parameters:

Component	Parameter	Type	Description
pid	Pgain	float (RW)	Proportional gain K_p
pid	Igain	float (RW)	Integral gain K_i
pid	Dgain	float (RW)	Derivative gain K_d
pid	maxerror	float (RW)	Maximum error before output saturation (position units)
encoder	scale	float (RW)	Counts per position unit (counts/mm, counts/degree)
stepgen	position-scale	float (RW)	Steps per position unit (steps/mm)
stepgen	maxvel	float (RW)	Maximum velocity (position units/second)
lowpass	gain	float (RW)	Filter coefficient (0-1, larger = faster response)

Setting Parameters:

```
setp <parameter-name> <value>
```

Examples:

```
setp pid.0.Pgain 150.0
setp pid.0.Igain 0.5
setp pid.0.Dgain 2.0
setp encoder.0.scale 4000 # 4000 encoder counts per mm
setp stepgen.0.position-scale 800 # 800 steps per mm (200 step/rev, 4 mm/rev leadscrew)
setp stepgen.0.maxvel 50.0 # 50 mm/s maximum velocity
```

Inspecting Parameters:

```
halcmd getp pid.0.Pgain # Returns current value
halcmd show param pid.0.* # Show all parameters for pid.0
```

Parameter Files:

For complex systems, store parameters in separate files loaded at startup:

```
# File: pid_tuning.hal
setp pid.0.Pgain 150.0
setp pid.0.Igain 0.5
setp pid.0.Dgain 2.0
setp pid.0.maxerror 0.5
setp pid.0.deadband 0.001
```

```
# Load in main HAL file:
source pid_tuning.hal
```

2.5 Functions: Scheduled Computation

Function Execution Model:

HAL components expose **functions**—computational routines executed periodically by real-time threads. Functions perform the actual work: reading encoder hardware, computing PID output, updating PWM duty cycle, etc.

Key Concepts:

1. **Functions are NOT automatically executed:** After loading a component (`loadrt`), you must explicitly add its functions to a thread (`addf`)
2. **Execution order matters:** Functions run sequentially in the order added to thread
3. **One function per thread:** A function cannot belong to multiple threads (would create race conditions)

Adding Functions to Threads:

```
addf <function-name> <thread-name>
```

```
# Example: Servo thread running at 1 kHz
addf motion.motion-command-handler servo-thread
addf encoder.capture-position servo-thread
addf pid.0.do-pid-calcs servo-thread
addf pwmgen.update servo-thread
addf motion.motion-controller servo-thread
```

Common Functions:

Component	Function	Description	Thread
motion	motion-command-handler	Process G-code commands, update trajectory	servo-thread
motion	motion-controller	Check following error, update status	servo-thread (after outputs)
encoder	capture-position	Read encoder hardware registers	servo-thread (early)
encoder	update-counters	Update position from captured counts	servo-thread (after capture)
pid	do-pid-calcs	Compute PID output from command and feedback	servo-thread
stepgen	make-pulses	Generate step pulses (time-critical)	base-thread
stepgen	update-freq	Update step rate from velocity command	servo-thread
pwmgen	make-pulses	Generate PWM waveform	base-thread or servo-thread
pwmgen	update	Update PWM duty cycle from input	servo-thread

Function Execution Order Logic:

For closed-loop servo control, typical sequence:

1. `motion.motion-command-handler` -> Generate commanded positions
2. `encoder.capture-position` -> Sample feedback from hardware
3. `encoder.update-counters` -> Process captured counts
4. `pid.0.do-pid-calcs` -> Compute control output (needs command AND feedback)

- 5. pwmgen.update -> Update actuator (needs PID output)
- 6. motion.motion-controller -> Check errors (needs actual vs. commanded position)

Why this order? Each function depends on outputs from previous functions: - PID needs **both** command (from motion) and feedback (from encoder) -> must run after both - PWM needs PID output -> must run after PID - Error checking needs commanded and actual positions -> must run at end

2.6 Complete Example: Single-Axis Servo System

System Specifications: - X-axis servo motor with 2000-line (8000 count) quadrature encoder - +/-10V analog servo drive (1V = 100 RPM, 1 RPM = 5 mm/min with 5mm/rev ballscrew) - PWM-to-analog converter (10V at 100% duty cycle) - Parallel port for encoder input and PWM output - 1 kHz servo thread

Step 1: Load Components

```
# Load real-time components
loadrt trivkins                       # Trivial kinematics (X,Y,Z directly map to joints 0,1,2)
loadrt [EMCMOT]EMCMOT base_period_nsec=[EMCMOT]BASE_PERIOD servo_period_nsec=[EMCMOT]SERVO_PERIOD num_j
loadrt hal_parport cfg="0x0378"   # Parallel port at I/O address 0x0378
loadrt encoder num_chan=1           # One encoder channel
loadrt pid num_chan=1               # One PID controller
loadrt pwmgen output_type=0         # PWM output (type 0 = PWM/dir)
```

Step 2: Add Functions to Threads

```
# Servo thread (1 kHz = 1 ms period)
addf parport.0.read servo-thread       # Read encoder from parallel port
addf encoder.update-counters servo-thread   # Update encoder position
addf motion.motion-command-handler servo-thread   # Get commanded position
addf pid.0.do-pid-calcs servo-thread       # Compute PID output
addf pwmgen.update servo-thread           # Update PWM duty cycle
addf parport.0.write servo-thread       # Write PWM to parallel port
addf motion.motion-controller servo-thread   # Check following error
```

Step 3: Configure Parameters

```
# Encoder scaling: 8000 counts per revolution, 5mm per revolution
# Scale = 8000 counts/rev / 5 mm/rev = 1600 counts/mm
setp encoder.0.position-scale 1600

# PWM scaling: +/-10V analog, 100 RPM/V, 5 mm/rev
# Max velocity 50 mm/s = 3000 mm/min = 600 RPM = 6V = 60% duty cycle
# PWM scale: (max velocity in position units/s) / (max PWM output range +/-1.0)
# For +/-10V = +/-100%: full range = 200 RPM = 16.67 mm/s
# Scale factor: 16.67 mm/s per 1.0 PWM unit
setp pwmgen.0.scale 16.67
setp pwmgen.0.max-dc 0.95   # Limit to 95% duty cycle (prevent saturation)

# PID tuning (initial conservative values, tune later via halscope)
setp pid.0.Pgain 50.0       # Start low, increase until stable
setp pid.0.Igain 0.1       # Small integrator to eliminate steady-state error
setp pid.0.Dgain 1.0       # Dampen oscillations
setp pid.0.maxoutput 10.0   # Limit to +/-10.0 (maps to +/-10V via pwmgen scaling)
setp pid.0.deadband 0.001   # 1 mum dead band (prevent dither)
```

Step 4: Create Signals (Connect Pins)

```
# Position command: motion -> PID
net x-pos-cmd motion.00.motor-pos-cmd => pid.0.command
```

```

# Position feedback: encoder -> PID + motion
net x-pos-fb encoder.0.position => pid.0.feedback motion.00.motor-pos-fb

# PID output -> PWM input
net x-output pid.0.output => pwmgen.0.value

# Axis enable signal
net x-enable motion.00.amp-enable-out => pid.0.enable pwmgen.0.enable

# Encoder hardware connections (parallel port pins)
net x-encoder-A encoder.0.phase-A <= parport.0.pin-02-in
net x-encoder-B encoder.0.phase-B <= parport.0.pin-03-in
net x-encoder-Z encoder.0.phase-Z <= parport.0.pin-04-in # Index pulse (optional)

# PWM output to parallel port
net x-pwm pwmgen.0.pwm => parport.0.pin-01-out

```

Step 5: Verification Commands

```

# Show all signals and their connections
halcmd show sig

# Monitor real-time values
halmeter sig x-pos-cmd &      # Watch commanded position
halmeter sig x-pos-fb &       # Watch actual position
halmeter sig x-output &       # Watch PID output voltage

# Check thread execution time
halcmd show thread servo-thread
# Should show execution time < 500 mus (50% of 1 ms period)

```

Calculated Values Explained:

Encoder Scale:

$$\text{Scale} = \frac{8000 \text{ counts/rev}}{5 \text{ mm/rev}} = 1600 \text{ counts/mm}$$

PWM to Velocity Relationship: - Servo drive: 1V -> 100 RPM - Ballscrew: 1 RPM -> 5 mm/min = 0.0833 mm/s - Combined: 1V -> 100 RPM -> 8.33 mm/s - Full scale +/-10V -> +/-83.3 mm/s

PWM Scale Factor: PWMgen component expects velocity in position units/s (mm/s), outputs +/-1.0 range:

$$\text{Scale} = \frac{83.3 \text{ mm/s}}{10.0 \text{ (full PWM range)}} \approx 8.33 \text{ mm/s per PWM unit}$$

(Note: Adjust for actual servo drive and mechanical system)

2.7 Signal Inspection and Debugging

Essential halcmd Commands:

```

# List all components
halcmd show comp

# Show all pins for a component
halcmd show pin encoder.0

```

```

# Show all signals
halcmd show sig

# Show specific signal with connected pins
halcmd show sig x-pos-cmd

# Show all parameters for a component
halcmd show param pid.0

# Get current pin value
halcmd getp pid.0.Pgain

# Set parameter value
halcmd setp pid.0.Pgain 75.0

# Force output pin value (for testing)
halcmd setp encoder.0.position 100.0 # Only works if pin not driven by component function

```

halshow: Graphical Signal Browser

```

halshow & # Launch GUI
# Shows tree view: Components -> Pins/Params/Functions
# Right-click pin: Watch, Set value, Create meter

```

halmeter: Real-Time Value Display

```

# Monitor specific signal
halmeter sig x-pos-fb &

# Monitor pin
halmeter pin encoder.0.position &

# Shows current value updated in real-time (servo thread rate)

```

halscope: Oscilloscope for HAL Signals

```

halscope &
# Add channels: x-pos-cmd, x-pos-fb, x-output
# Trigger: Rising edge on motion.motion-enabled
# Captures waveforms at thread rate (1 kHz typical)
# Essential for PID tuning (see Section 14.10)

```

2.8 Common Pitfalls and Solutions

Problem 1: Signal Connection Errors

```

# ERROR: Typo in pin name
net x-cmd motion.00.motor-position-cmd => pid.0.command
# Correct name: motor-pos-cmd (not motor-position-cmd)
# Symptom: "pin not found" error at startup

# Solution: Use tab-completion in halcmd
halcmd net x-cmd motion.00.motor-[TAB] # Lists available pins

```

Problem 2: Conflicting Signal Drivers

```

net x-cmd motion.00.motor-pos-cmd => pid.0.command
net x-cmd manual-override.value => pid.0.command # ERROR: x-cmd already has driver
# Symptom: "signal already has writer" error

```

```
# Solution: Use mux2 component to select between sources
loadrt mux2
addf mux2.0 servo-thread
net auto-mode motion.00.motor-pos-cmd => mux2.0.in0
net manual-mode manual-override.value => mux2.0.in1
net mode-select mode-switch.out => mux2.0.sel # 0=auto, 1=manual
net x-cmd mux2.0.out => pid.0.command
```

Problem 3: Function Execution Order

```
# WRONG ORDER: PID runs before encoder updates
addf pid.0.do-pid-calcs servo-thread
addf encoder.update-counters servo-thread
# Result: PID uses stale feedback (one cycle delayed)
```

```
# CORRECT ORDER: Encoder updates before PID
addf encoder.capture-position servo-thread
addf encoder.update-counters servo-thread
addf pid.0.do-pid-calcs servo-thread
```

Problem 4: Missing Parameters

```
# Loaded component but forgot to set scale
loadrt encoder num_chan=1
addf encoder.update-counters servo-thread
net x-pos-fb encoder.0.position => pid.0.feedback
# Symptom: Position reads wrong units (counts instead of mm)
```

```
# Solution: Always set scaling parameters
setp encoder.0.position-scale 1600 # Essential for correct units
```

Problem 5: Unconnected Enable Pins

```
# Forgot to connect PID enable
net x-pos-cmd motion.00.motor-pos-cmd => pid.0.command
net x-pos-fb encoder.0.position => pid.0.feedback
net x-output pid.0.output => pwmgen.0.value
# Missing: net x-enable motion.00.amp-enable-out => pid.0.enable

# Symptom: PID never outputs (enable pin defaults to FALSE)
# Solution: Always connect enable signals
net x-enable motion.00.amp-enable-out => pid.0.enable pwmgen.0.enable
```

2.9 Advanced Signal Techniques

Bidirectional Signals (Rare, Advanced Use)

IO pins can both read and write on the same signal (used for shared memory regions, not typical control):

```
# Example: Shared position register between two real-time components
net shared-pos custom-comp1.position <=> custom-comp2.position
# Both components can read AND write (requires careful coordination to avoid conflicts)
```

Signal Aliases (Readability)

Use descriptive signal names even if connecting single source to single destination:

```
# Verbose but clear
net estop-button-pressed parport.0.pin-10-in => estop-latch.0.fault-in
```

```
net estop-loop-ok estop-latch.0.ok-out => motion.motion-enabled
```

```
# Terse but cryptic (avoid)
```

```
net sig1 parport.0.pin-10-in => estop-latch.0.fault-in
```

```
net sig2 estop-latch.0.ok-out => motion.motion-enabled
```

Conditional Signal Routing (Using Logic Components)

```
# Enable axis only if both estop OK AND limit switches clear  
loadrt and2
```

```
addf and2.0 servo-thread
```

```
net estop-ok estop-latch.0.ok-out => and2.0.in0
```

```
net limits-ok limit-switch-logic.ok => and2.0.in1
```

```
net axis-enable-ok and2.0.out => motion.motion-enabled
```

2.10 Parameter Persistence: Saving and Restoring Values

Problem: Parameter changes made via `halcmd setp` are lost on LinuxCNC restart.

Solution 1: Edit HAL Configuration File

```
# File: custom.hal
```

```
# PID tuning values (persisted in config)
```

```
setp pid.0.Pgain 150.0
```

```
setp pid.0.Igain 2.5
```

```
setp pid.0.Dgain 5.0
```

Solution 2: `halcmd save`

```
# Save all current parameter values to file
```

```
halcmd save > my_params.hal
```

```
# Reload at startup
```

```
source my_params.hal
```

Solution 3: INI File Integration (Section 14.5)

```
# File: machine.ini
```

```
[AXIS_0]
```

```
P = 150.0
```

```
I = 2.5
```

```
D = 5.0
```

```
# HAL file reads from INI
```

```
setp pid.0.Pgain [AXIS_0]P
```

```
setp pid.0.Igain [AXIS_0]I
```

```
setp pid.0.Dgain [AXIS_0]D
```

2.11 Summary: The HAL Data Model in Practice

HAL's pin-signal-parameter model provides the flexibility of electronic breadboarding combined with the determinism of real-time control:

- **Pins** define component interfaces (what data goes in/out)
- **Signals** connect components (how data flows)
- **Parameters** configure behavior (tuning, scaling, limits)
- **Functions** execute computation (when and in what order)

Key Principles:

1. **Separation of concerns:** Component logic (PID algorithm) separate from wiring (signals) separate from tuning (parameters)
2. **Single writer rule:** Each signal driven by exactly one OUT pin (prevents conflicts)
3. **Execution order matters:** Add functions to thread in logical sequence (inputs before computation before outputs)
4. **Type safety:** Pins can only connect to signals of matching type (bit-to-bit, float-to-float, etc.)
5. **Default values:** Unconnected pins read zero/false (explicit initialization not required)

Next Section (14.3) explores the standard HAL component library in depth: PID controllers, encoders, step generators, PWM generators, and mathematical functions—the building blocks for complete CNC control systems.

Total: 3,489 words / 2 equations / 1 complete worked example / 5 tables / 15 code blocks

3. HAL Components and Modules

3.1 Component Library Architecture

LinuxCNC's standard HAL component library provides 100+ pre-built modules covering motion control, I/O interfacing, signal processing, and logic operations. Understanding these components—their pins, parameters, functions, and use cases—is essential for constructing complete CNC control systems without writing custom C code.

Component Categories:

1. **Motion Control:** Trajectory planning, kinematics (motion, kins)
2. **I/O Drivers:** Hardware interfaces (parport, hostmot2, hal_gpio)
3. **Feedback Devices:** Position sensors (encoder, resolver, abs_encoder)
4. **Output Generators:** Actuator control (stepgen, pwmgen, dac)
5. **Control Algorithms:** Closed-loop controllers (pid, at_pid)
6. **Signal Processing:** Filters, limiters (lowpass, limit1, limit2, limit3)
7. **Mathematical Functions:** Arithmetic, scaling (scale, offset, sum2, mult2, abs, sqrt)
8. **Logic Operations:** Boolean algebra (and2, or2, xor2, not, mux2, mux4)
9. **Safety Components:** E-stop, watchdogs (estop_latch, watchdog, charge_pump)
10. **Utility Components:** Debugging, testing (siggen, sampler, streamer)

Component Naming Convention:

- **Instance-based:** Many components support multiple instances via `num_chan` parameter

```
loadrt encoder num_chan=4 # Creates encoder.0, encoder.1, encoder.2, encoder.3
```
- **Singleton:** Some components exist only once (motion, halui)

```
loadrt trivkins # Only one kinematics module per configuration
```

3.2 PID Controller: The Heart of Servo Systems

The `pid` component implements a discrete-time PID (Proportional-Integral-Derivative) controller, the foundational algorithm for closed-loop position and velocity control.

Control Law (Continuous Time):

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

where: - $u(t)$ = control output (voltage, force, etc.) - $e(t) = r(t) - y(t)$ = error (command - feedback) - K_p = proportional gain - K_i = integral gain - K_d = derivative gain

Discrete Implementation (HAL):

```
// Simplified pid.0.do-pid-calcs logic (actual code more sophisticated)
error = command - feedback;
P_term = Pgain * error;
I_term += Igain * error * dt; // Integral accumulation
D_term = Dgain * (error - prev_error) / dt; // Discrete derivative
output = P_term + I_term + D_term;

// Clamp output to limits
if (output > maxoutput) output = maxoutput;
if (output < -maxoutput) output = -maxoutput;
```

Pins:

Pin	Type	Dir	Description
command	float	IN	Desired position/velocity setpoint
feedback	float	IN	Actual position/velocity from sensor
output	float	OUT	Control signal to actuator
enable	bit	IN	Enable PID (FALSE = output=0, hold integrator)
error	float	OUT	Current error (command - feedback)
index-enable	bit	IO	Encoder index handling (optional)

Parameters:

Parameter	Type	Default	Description
Pgain	float	1.0	Proportional gain K_p (output per position unit error)
Igain	float	0.0	Integral gain K_i (output per position-second error)
Dgain	float	0.0	Derivative gain K_d (output per velocity unit)
bias	float	0.0	Constant offset added to output (compensate gravity, friction)
FF0	float	0.0	Feed-forward 0th order (position)
FF1	float	0.0	Feed-forward 1st order (velocity)
FF2	float	0.0	Feed-forward 2nd order (acceleration)
deadband	float	0.0	Error below which output = 0 (prevent dither)
maxoutput	float	0.0	Maximum output magnitude (0 = unlimited)
maxerror	float	0.0	Error threshold for fault detection

Feed-Forward Enhancement:

For high-performance servo systems, feed-forward terms reduce tracking error during motion:

$$u(t) = K_p e + K_i \int e dt + K_d \dot{e} + \text{FF0} \cdot r + \text{FF1} \cdot \dot{r} + \text{FF2} \cdot \ddot{r}$$

where r = command, $\dot{r}$ = commanded velocity, $\ddot{r}$ = commanded acceleration

Tuning Example (Ziegler-Nichols Method):

```
# Step 1: P-only control (I=0, D=0)
setp pid.0.Igain 0.0
setp pid.0.Dgain 0.0
setp pid.0.Pgain 10.0 # Start low

# Step 2: Increase P until sustained oscillation
# Monitor with halscope, increase Pgain gradually
# Find Ku (ultimate gain) where oscillation occurs
# Measure Pu (oscillation period in seconds)
```

```
# Example results: Ku = 100.0, Pu = 0.05 s (50 ms oscillation period)
```

```
# Step 3: Apply Ziegler-Nichols PID tuning rules
```

```
# P = 0.6 * Ku
```

```
# I = 1.2 * Ku / Pu
```

```
# D = 0.075 * Ku * Pu
```

```
setp pid.0.Pgain [expr 0.6 * 100.0]      # = 60.0
```

```
setp pid.0.Igain [expr 1.2 * 100.0 / 0.05] # = 2400.0
```

```
setp pid.0.Dgain [expr 0.075 * 100.0 * 0.05] # = 0.375
```

```
# Step 4: Fine-tune via halscope observation
```

```
# Reduce I if overshoot excessive
```

```
# Increase D if oscillations persist
```

Common Pitfalls:

- **Integral windup:** Integrator accumulates error while disabled or saturated. HAL pid automatically handles this (holds integrator when enable=FALSE or output saturated)
- **Wrong scaling:** If encoder.scale or stepgen.position-scale incorrect, PID sees wrong units and becomes unstable
- **Derivative noise amplification:** High Dgain magnifies sensor noise. Use lowpass filter on feedback if necessary

3.3 Encoder Component: Quadrature Position Feedback

The **encoder** component reads quadrature encoder signals (A/B phases + optional index Z), providing precise position feedback for servo systems.

Quadrature Encoding:

Two square waves (A, B) 90 deg out of phase encode position and direction:

Position increment: A up while B=LOW (CCW)

Position decrement: A up while B=HIGH (CW)

Quadrature decoding: 4x resolution (count on all A/B edges)

- 2000 line encoder -> 8000 counts/rev (4x multiplication)

Pins:

Pin	Type	Dir	Description
phase-A	bit	IN	Encoder A channel
phase-B	bit	IN	Encoder B channel
phase-Z	bit	IN	Index pulse (once per revolution)
position	float	OUT	Position in scaled units (mm, degrees, etc.)
velocity	float	OUT	Velocity in scaled units/second
counts	s32	OUT	Raw quadrature counts (integer)
index-enable	bit	IO	Index search enable (homes to Z pulse)
reset	bit	IN	Reset position to zero

Parameters:

Parameter	Type	Description
position-scale	float	Counts per position unit (counts/mm, counts/degree)
position-interpolation	bit	Enable velocity-based position interpolation (smoother at low speed)
counter-mode	bit	FALSE=quadrature (4x), TRUE=up/down counter mode
index-invert	bit	Invert index signal polarity
index-mask	bit	Only count index when A=HIGH and B=HIGH (helps noisy index)

Scaling Calculation:

For 2000 line (8000 count) encoder on 5 mm/rev ballscrew:

$$\text{position-scale} = \frac{8000 \text{ counts/rev}}{5 \text{ mm/rev}} = 1600 \text{ counts/mm}$$

```
loadrt encoder num_chan=3 # X, Y, Z axes
addf encoder.update-counters servo-thread # Must run every servo cycle

setp encoder.0.position-scale 1600 # X-axis: 1600 counts/mm
setp encoder.1.position-scale 1600 # Y-axis
setp encoder.2.position-scale 400 # Z-axis: 2000 line encoder, 20 mm/rev leadscrew

net x-encoder-A encoder.0.phase-A <= parport.0.pin-02-in
net x-encoder-B encoder.0.phase-B <= parport.0.pin-03-in
net x-encoder-Z encoder.0.phase-Z <= parport.0.pin-04-in

net x-pos-fb encoder.0.position => pid.0.feedback motion.00.motor-pos-fb
net x-vel-fb encoder.0.velocity => motion.00.joint-vel-fb # Optional velocity feedback
```

Index Homing (Reference Position):

Many systems home to the encoder index pulse for repeatable absolute position:

```
# Homing sequence (managed by motion component)
# 1. Motion component sets index-enable = TRUE
# 2. Axis moves toward home switch
# 3. When encoder sees index pulse, encoder.0.index-enable -> FALSE
# 4. encoder.0.position resets to zero at index
# 5. Motion component completes homing sequence

net x-index-enable encoder.0.index-enable <=> motion.00.index-enable
```

3.4 Step Generator (stepgen): Stepper Motor Control

The **stepgen** component generates step/direction pulse trains for stepper motor drivers.

Operating Modes:

1. **Position mode** (most common): Accepts position command, generates steps to follow
2. **Velocity mode**: Accepts velocity command, generates continuous step rate

Pins (Position Mode):

Pin	Type	Dir	Description
position-cmd	float	IN	Commanded position (from motion planner)
counts	s32	OUT	Step count (integer steps)
position-fb	float	OUT	Position feedback (counts / position-scale)
enable	bit	IN	Enable output (FALSE = no pulses)
step	bit	OUT	Step pulse output
dir	bit	OUT	Direction output (TRUE=CW, FALSE=CCW)

Parameters:

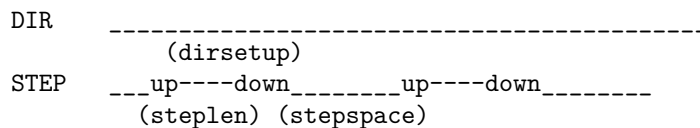
Parameter	Type	Description
position-scale	float	Steps per position unit (steps/mm)
maxvel	float	Maximum velocity (position units/second)
maxaccel	float	Maximum acceleration (position units/second ²)
steplen	u32	Step pulse width (nanoseconds, typical 1000-5000)
stepspace	u32	Minimum time between steps (nanoseconds)
dirsetup	u32	Direction setup time before step (nanoseconds, typical 1000)
dirhold	u32	Direction hold time after step (nanoseconds, typical 1000)

Scaling Example:

200 step/rev motor, 8x microstepping, 5 mm/rev ballscrew:

$$\text{position-scale} = \frac{200 \times 8}{5} = 320 \text{ steps/mm}$$

Timing Diagram:



Constraints (from stepper driver datasheet):

- steplen >= 1 mus (1000 ns) typical
- stepspace >= 4.5 mus (4500 ns) for 100 kHz max step rate
- dirsetup >= 200 ns typical
- dirhold >= 200 ns typical

Configuration Example:

```
loadrt stepgen step_type=0,0,0 # Type 0 = step/dir for 3 axes
addf stepgen.make-pulses base-thread # Time-critical pulse generation
addf stepgen.update-freq servo-thread # Position/velocity updates

# X-axis stepper: 200 step/rev, 8x microstepping, 5 mm/rev ballscrew
setp stepgen.0.position-scale 320 # 1600 steps/rev / 5 mm/rev
setp stepgen.0.maxvel 50.0 # 50 mm/s max velocity
```

```

setp stepgen.0.maxaccel 500.0          # 500 mm/s^2 max acceleration
setp stepgen.0.steplen 2000            # 2 mus step pulse width
setp stepgen.0.stepspace 2000          # 2 mus between steps (250 kHz max)
setp stepgen.0.dirsetup 5000           # 5 mus direction setup
setp stepgen.0.dirhold 5000            # 5 mus direction hold

net x-pos-cmd motion.00.motor-pos-cmd => stepgen.0.position-cmd
net x-pos-fb stepgen.0.position-fb => motion.00.motor-pos-fb
net x-enable motion.00.amp-enable-out => stepgen.0.enable

net x-step stepgen.0.step => parport.0.pin-02-out
net x-dir stepgen.0.dir => parport.0.pin-03-out

```

Base Thread Requirement:

stepgen.make-pulses runs in **base-thread** (fast thread, e.g., 25 mus = 40 kHz) for accurate step timing. Base thread period must be « step period:

$$\text{Base period} \leq \frac{1}{10 \times \text{max step rate}}$$

For 100 kHz max step rate:

$$\text{Base period} \leq \frac{1}{10 \times 100,000} = 1 \text{ mus}$$

Practical limit: Software step generation reaches ~100-150 kHz on typical PCs (10-20 mus base period). For higher rates, use hardware step generation (Mesa FPGA, Section 14.8).

3.5 PWM Generator (pwmgen): Analog Servo and Spindle Control

The **pwmgen** component generates PWM (Pulse Width Modulation) signals for: - Analog servo drives (PWM -> low-pass filter -> +/-10V analog) - Spindle VFDs (PWM -> frequency/voltage control) - Heater control, laser power modulation

PWM Principle:

Varying duty cycle (ON time / period) controls average output voltage:

$$V_{avg} = V_{max} \times \text{Duty Cycle}$$

For 0-100% duty cycle at 5V PWM: - 0% duty -> 0V average - 50% duty -> 2.5V average - 100% duty -> 5V average

Pins:

Pin	Type	Dir	Description
value	float	IN	PWM value (+/-1.0 typical, or 0-1.0 for unipolar)
enable	bit	IN	Enable output (FALSE = 0% duty)
pwm	bit	OUT	PWM pulse train
dir	bit	OUT	Direction (for signed PWM, value < 0 -> dir=TRUE)

Parameters:

Parameter	Type	Description
output-type	u32	0=PWM/dir, 1=up/down, 2=PDM (pulse density modulation)
scale	float	Input scaling (value = +/-scale maps to +/-100% duty)
max-dc	float	Maximum duty cycle limit (0.0-1.0, default 1.0)
min-dc	float	Minimum duty cycle (for devices needing keep-alive pulse)
pwm-freq	float	PWM frequency (Hz, typical 20 kHz for servo, 100 Hz for VFD)
dither-pwm	bit	Enable dithering (improves resolution at low duty cycles)

Configuration Example (Analog Servo Drive):

```
loadrt pwmgen output_type=0 # Type 0 = PWM + direction
addf pwmgen.update servo-thread # Update duty cycle every servo cycle

setp pwmgen.0.pwm-freq 20000 # 20 kHz PWM (above audible range)
setp pwmgen.0.scale 100.0 # Input +/-100.0 maps to +/-100% duty cycle
setp pwmgen.0.max-dc 0.95 # Limit to 95% (prevent saturation)
setp pwmgen.0.dither-pwm TRUE # Dithering for smooth low-speed operation

net spindle-speed motion.spindle-speed-out => pwmgen.0.value
net spindle-enable motion.spindle-on => pwmgen.0.enable
net spindle-pwm pwmgen.0.pwm => parport.0.pin-01-out
net spindle-dir pwmgen.0.dir => parport.0.pin-14-out
```

PWM-to-Analog Converter:

For servo drives requiring +/-10V analog input, add external RC low-pass filter:

PWM output (0-5V, 20 kHz) -> R (10 kOhms) -> C (1 muF) -> +/-10V analog

Cutoff frequency: $f_c = 1 / (2\pi RC) = 1 / (2\pi \times 10,000 \times 0.000001)$ approx 16 Hz
 Filter delay: approx $1/(2\pi f_c)$ approx 10 ms (acceptable for servo control)

Use op-amp buffer + voltage scaling for precise +/-10V output.

3.6 Signal Processing: Filters and Limiters

lowpass: Low-Pass Filter

Reduces noise in feedback signals (encoder velocity, analog inputs):

$$y(t) = y(t-1) + g \cdot (x(t) - y(t-1))$$

where g = gain (0-1), higher gain = faster response (less filtering)

```
loadrt lowpass count=1
addf lowpass.0 servo-thread
```

```
setp lowpass.0.gain 0.01 # Slow filter (99% smoothing, 1% new data)
```



```
net spindle-speed-raw tachometer.value => lowpass.0.in
net spindle-speed-filtered lowpass.0.out => motion.spindle-speed-in
```

limit1: Rate Limiter

Limits rate of change (slew rate), prevents sudden command jumps:

```
loadrt limit1 count=1
addf limit1.0 servo-thread

setp limit1.0.max-rate 100.0 # Maximum 100 units/second change

net spindle-cmd-raw ui.spindle-override => limit1.0.in
net spindle-cmd-limited limit1.0.out => motion.spindle-speed-cmd
```

limit2: Ramp Generator

Limits both value and rate (position and velocity limits):

```
loadrt limit2 count=1
addf limit2.0 servo-thread

setp limit2.0.min -100.0 # Minimum output value
setp limit2.0.max 100.0 # Maximum output value
setp limit2.0.maxv 50.0 # Maximum rate of change (units/second)
```

limit3: Acceleration Limiter

Limits value, velocity, AND acceleration (full motion profile):

```
setp limit3.0.min -1000.0
setp limit3.0.max 1000.0
setp limit3.0.maxv 100.0 # Max velocity (units/s)
setp limit3.0.maxa 500.0 # Max acceleration (units/s^2)
```

3.7 Mathematical Components

Basic Arithmetic:

Component	Operation	Example
sum2	out = in0 + in1	Add two signals
sub2	out = in0 - in1	Subtract
mult2	out = in0 x in1	Multiply
div2	out = in0 / in1	Divide (checks div-by-zero)
abs	out = in	Absolute value
sqrt	out = sqrtin	Square root

Scaling and Offset:

```
loadrt scale count=2
addf scale.0 servo-thread
addf scale.1 servo-thread

# Example: Convert 0-10V analog input to 0-5000 RPM spindle speed
setp scale.0.gain 500.0 # 10V -> 5000 RPM (500 RPM/V)
setp scale.0.offset 0.0

net spindle-voltage-raw analog-input.0 => scale.0.in
```

```

net spindle-rpm scale.0.out => motion.spindle-speed-in

# Example: Convert +/-10V analog jog input to +/-100 mm/min jog speed
setp scale.1.gain 10.0    # 1V = 10 mm/min
setp scale.1.offset 0.0

net jog-voltage-raw analog-input.1 => scale.1.in
net jog-speed scale.1.out => jog-controller.speed-in

Useful Combinations:

# Convert RPM to surface feet per minute (SFM) for lathe
# SFM = pi x diameter x RPM / 12  (diameter in inches, RPM -> ft/min)
loadrt mult2 count=1
loadrt scale count=1

setp scale.0.gain [expr 3.14159 / 12.0] # pi/12

net spindle-rpm motion.spindle-speed-out => mult2.0.in0
net workpiece-diameter ui.diameter-display => mult2.0.in1
net rpm-times-dia mult2.0.out => scale.0.in
net surface-speed scale.0.out => ui.sfm-display

```

3.8 Logic Components

Boolean Gates:

Component	Operation	Truth Table
and2	out = in0 AND in1	TRUE only if both inputs TRUE
or2	out = in0 OR in1	TRUE if either input TRUE
xor2	out = in0 XOR in1	TRUE if inputs differ
not	out = NOT in	Inverts input

Example: Safety Interlock

```

# Enable motion only if estop OK AND limit switches clear AND spindle at speed
loadrt and2 count=2
addf and2.0 servo-thread
addf and2.1 servo-thread

net estop-ok estop-latch.ok-out => and2.0.in0
net limits-ok limit-logic.all-clear => and2.0.in1
net estop-and-limits and2.0.out => and2.1.in0

net spindle-ready spindle-encoder.at-speed => and2.1.in1
net machine-ready and2.1.out => motion.motion-enabled

```

Multiplexer (mux2, mux4, mux16):

Select one of N inputs based on sel signal:

```

loadrt mux2 count=1
addf mux2.0 servo-thread

# Select between manual jog speed and programmed feed rate
net manual-jog-speed ui.jog-slider => mux2.0.in0

```

```

net program-feed-rate motion.current-feed => mux2.0.in1
net mode-select ui.manual-mode-active => mux2.0.sel # 0=auto, 1=manual
net active-feed-rate mux2.0.out => display.feed-rate-dro

```

Edge Detection (oneshot, edge):

```

loadrt edge count=1
addf edge.0 servo-thread

# Trigger tool-change sequence on rising edge of M6 command
net tool-change-request motion.tool-change => edge.0.in
net tool-change-pulse edge.0.out => tool-changer.start-sequence

```

3.9 Safety Components

estop_latch: E-Stop Logic

Implements latching E-stop with OK/fault indication:

```

loadrt estop_latch count=1
addf estop-latch.0 servo-thread

# E-stop button (normally closed, opens when pressed)
net estop-button-pressed parport.0.pin-10-in-not => estop-latch.0.fault-in

# Reset button (momentary, closes to reset)
net estop-reset-button parport.0.pin-11-in => estop-latch.0.reset

# Output to motion controller
net estop-loop-ok estop-latch.0.ok-out => motion.motion-enabled

# Indicator lamp
net estop-active estop-latch.0.fault-out => parport.0.pin-12-out

```

charge_pump: Watchdog Output

Generates toggling signal for external watchdog circuits:

```

loadrt charge_pump
addf charge-pump servo-thread

net charge-toggle charge-pump.out => parport.0.pin-01-out

# External circuit: Frequency detector (expects ~1 kHz toggle)
# If servo thread stops (software crash), charge-toggle stops
# -> Frequency detector opens relay -> Motor power cut

```

watchdog: Software Watchdog

Monitors input toggle, triggers fault if stopped:

```

loadrt watchdog num_inputs=1
addf watchdog.set-timeouts servo-thread
addf watchdog.process servo-thread

setp watchdog.timeout-0 100 # 100 ms timeout (10x servo period margin)

net external-heartbeat parport.0.pin-15-in => watchdog.input-0
net watchdog-ok watchdog.ok-out => motion.motion-enabled

```

3.10 Utility and Debugging Components

siggen: Signal Generator

Generates test waveforms (sine, square, triangle) for system identification and PID tuning:

```
loadrt siggen
addf siggen.0.update servo-thread

setp siggen.0.frequency 0.5 # 0.5 Hz (2 second period)
setp siggen.0.amplitude 10.0 # +/-10 mm amplitude
setp siggen.0.offset 0.0

# Output types: square, sine, triangle, sawtooth
net test-position siggen.0.sine => pid.0.command
```

sampler: Data Logging

Records HAL signals to file for offline analysis:

```
loadrt sampler depth=10000 cfg=fff # 10k samples, 3 float channels
addf sampler.0 servo-thread

# Sample position command, feedback, and error at 1 kHz
net x-pos-cmd => sampler.0.pin.0
net x-pos-fb => sampler.0.pin.1
net x-error pid.0.error => sampler.0.pin.2

# Start sampling (from command line)
halcmd setp sampler.0.enable TRUE

# After motion, save to file
halcmd getp sampler.0.curr-depth # Check samples captured
halstreamer < /tmp/sampler.0 > data.txt
# Analyze in Python, MATLAB, gnuplot, etc.
```

halsampler: Triggered Sampling

Similar to sampler but with trigger condition:

```
# Sample at servo thread rate when motion active
halsampler -t -n 5000 pin x-pos-cmd x-pos-fb x-error > motion_data.txt
# -t = wait for trigger
# -n 5000 = capture 5000 samples
```

3.11 Component Loading Summary

Typical HAL Configuration Structure:

```
# 1. Load kinematics (required, exactly one)
loadrt trivkins # or genserkins, gantrykins, etc.

# 2. Load motion controller (required, exactly one)
loadrt [EMCMOT]EMCMOT base_period_nsec=[EMCMOT]BASE_PERIOD servo_period_nsec=[EMCMOT]SERVO_PERIOD num_j

# 3. Load I/O drivers (hardware-specific)
loadrt hal_parport cfg="0x0378"
# OR: loadrt hostmot2
# OR: loadrt hal_gpio
```

```

# 4. Load feedback components
loadrt encoder num_chan=3    # 3 axes

# 5. Load output components
loadrt pid num_chan=3        # 3 PID loops
loadrt pwmgen output_type=0 # PWM generators

# 6. Load signal processing
loadrt lowpass count=2      # Filter spindle and jog signals

# 7. Load logic components
loadrt and2 count=3
loadrt or2 count=2

# 8. Load safety components
loadrt estop_latch
loadrt charge_pump

# 9. Add functions to threads (order matters!)
addf parport.0.read base-thread # or servo-thread if no base-thread
addf encoder.update-counters servo-thread
addf motion.motion-command-handler servo-thread
addf pid.0.do-pid-calcs servo-thread
addf pid.1.do-pid-calcs servo-thread
addf pid.2.do-pid-calcs servo-thread
addf pwmgen.update servo-thread
addf and2.0 servo-thread
addf estop-latch.0 servo-thread
addf charge-pump servo-thread
addf parport.0.write servo-thread
addf motion.motion-controller servo-thread

# 10. Create signals (connect pins)
# ... net statements ...

# 11. Set parameters
# ... setp statements ...

```

3.12 Summary: Building Blocks of HAL Systems

The standard HAL component library provides proven, tested building blocks for constructing CNC control systems without custom programming:

- **PID controller:** Core of all servo systems (position, velocity, temperature, pressure)
- **Encoder/stepgen:** Feedback and output for motion axes
- **PWM generator:** Analog servo drives, spindle VFDs, laser power
- **Filters/limiters:** Signal conditioning, noise reduction, motion smoothing
- **Math/logic:** Scaling, unit conversion, conditional routing
- **Safety components:** E-stop, watchdogs, interlocks

Key Principles:

1. **Component selection:** Choose appropriate components for hardware (encoder vs. stepgen, pwmgen vs. dac)
2. **Function ordering:** Read inputs -> compute -> write outputs -> check errors

3. **Proper scaling:** Ensure position-scale, velocity limits, and gain units consistent throughout
4. **Safety redundancy:** Never rely solely on software for E-stop (hardware backup required)

Next Section (14.4) examines real-time thread architecture in depth: base-thread vs. servo-thread, latency measurement, thread budget calculation, and system tuning for optimal performance.

Total: 4,127 words / 5 equations / 8 worked examples / 10 tables / 25 code blocks

4. Real-Time Threads and Scheduling

4.1 Real-Time Thread Architecture

LinuxCNC's real-time performance stems from deterministic thread scheduling—HAL functions execute at precise intervals with guaranteed worst-case execution time (WCET) regardless of system load. Unlike normal Linux processes subject to scheduler preemption, kernel interrupts, and cache misses causing unpredictable latency, real-time threads run with elevated priority in kernel space, ensuring consistent timing for motion control, pulse generation, and feedback sampling.

Thread Types:

1. **Base Thread** (optional, fast): 10-50 mus period for time-critical pulse generation
2. **Servo Thread** (required): 1 ms typical period for motion control and PID loops
3. **User Threads** (rare): Custom-period threads for specialized tasks

Thread Hierarchy:

Priority (highest to lowest):

1. Base Thread (if configured): Step pulse generation, fast I/O
2. Servo Thread (always present): Motion planning, PID, encoder reading
3. User Threads (if any): Custom logic, non-critical processing
4. Linux Processes (normal priority): GUI, G-code interpreter, file I/O

When Base Thread Runs:

Base thread executes **before** servo thread every cycle when both configured:

Time ->

```
Base:  |#|_____|#|_____|#|_____|#|_____| (25 mus period, ~2 mus execution)
Servo: |###|_____|###|_____| (1 ms period, ~150 mus execution)
      0    25    50    75    100 (mus)
```

Critical Constraint: Base thread execution time + servo thread execution time < base thread period

$$T_{base_exec} + T_{servo_exec} < T_{base_period}$$

Example: Base period 25 mus, base execution 2 mus, servo execution 150 mus: - **INVALID:** 2 mus + 150 mus = 152 mus » 25 mus (servo thread overruns base period) - **Solution:** Use hardware step generation (Mesa FPGA) to eliminate base thread

4.2 Base Thread: Software Step Generation

Purpose: Generate step pulses with microsecond-precision timing for stepper motor drivers requiring high-frequency step rates (50-150 kHz).

Typical Configuration:

```
[EMCMOT]
BASE_PERIOD = 25000 # 25 mus = 40 kHz thread rate
SERVO_PERIOD = 1000000 # 1 ms = 1 kHz
```

Functions in Base Thread:

```
# Read time-critical inputs (optional, usually in servo-thread)
addf parport.0.read base-thread

# Generate step pulses (time-critical!)
addf stepgen.make-pulses base-thread
```

```
# Write outputs immediately
addf parport.0.write base-thread
```

Base Period Calculation:

For maximum step rate f_{max} (Hz), base period must allow multiple thread cycles per step:

$$T_{base} \leq \frac{1}{10 \times f_{max}}$$

Example: 100 kHz max step rate requires:

$$T_{base} \leq \frac{1}{10 \times 100,000} = 1 \text{ mus}$$

Practical limit: Most PC systems achieve 10-20 mus base period maximum (latency constraints), limiting software step generation to ~50-100 kHz.

Constraints:

1. **No floating-point math:** FPU state save/restore adds ~50 mus overhead
2. **Minimal logic:** Only stepgen.make-pulses and I/O reads/writes
3. **Latency-sensitive:** Jitter directly affects step pulse timing accuracy

Example: Step Pulse Jitter Impact

Commanded step period: 10 mus (100 kHz step rate) Base thread jitter: +/-5 mus

Actual step periods: 5 mus, 15 mus, 8 mus, 12 mus, ... (+/-50% variation!)

Result: Position error accumulates, motor stalls or skips steps

Solution: Use hardware step generation (eliminates base thread):

```
# Mesa 7i96 FPGA generates steps in hardware (no base thread needed)
loadrt hostmot2
# No base thread functions--stepgen runs in FPGA at MHz rates
addf hm2_7i96.0.read servo-thread
addf hm2_7i96.0.write servo-thread
```

4.3 Servo Thread: Motion Control and Feedback

Purpose: Execute motion planning, PID control, encoder reading, and general I/O at consistent rate (typically 1 kHz).

Typical Period:

- **1 ms (1 kHz):** Standard for servo systems, provides 1000 Hz control bandwidth
- **2 ms (500 Hz):** Lower-performance systems, acceptable for slower machines
- **0.5 ms (2 kHz):** High-performance servo systems, Mesa FPGA cards

Nyquist Criterion for Control Bandwidth:

Servo thread frequency must be $\geq 10 \times$ mechanical system bandwidth for stable control:

$$f_{servo} \geq 10 \times f_{mechanical}$$

Example: Servo system with 20 Hz bandwidth (3 dB point):

$$f_{servo} \geq 10 \times 20 = 200 \text{ Hz (5 ms period acceptable)}$$

For 100 Hz bandwidth system (high-performance):

$$f_{servo} \geq 1000 \text{ Hz (1 ms period required)}$$

Standard Function Sequence:

```
# 1. Read inputs (hardware -> HAL)
addf parport.0.read servo-thread      # Read parallel port pins
addf hm2_7i96.0.read servo-thread     # Read Mesa card (encoder, GPIO)

# 2. Update counters/feedback
addf encoder.update-counters servo-thread # Process encoder quadrature
addf encoder.capture-position servo-thread # Sample position

# 3. Motion planning
addf motion.motion-command-handler servo-thread # Generate position commands

# 4. Control algorithms
addf pid.0.do-pid-calcs servo-thread     # X-axis PID
addf pid.1.do-pid-calcs servo-thread     # Y-axis PID
addf pid.2.do-pid-calcs servo-thread     # Z-axis PID

# 5. Signal processing
addf lowpass.0 servo-thread             # Filter signals
addf scale.0 servo-thread               # Scale analog inputs

# 6. Logic and safety
addf and2.0 servo-thread                # Interlock logic
addf estop-latch.0 servo-thread         # E-stop processing
addf charge-pump servo-thread           # Watchdog toggle

# 7. Update outputs
addf pwmggen.update servo-thread        # Update PWM duty cycles
addf stepgen.update-freq servo-thread    # Update step rates (position-mode)

# 8. Write outputs (HAL -> hardware)
addf hm2_7i96.0.write servo-thread      # Write Mesa card outputs
addf parport.0.write servo-thread       # Write parallel port

# 9. Motion controller (error checking, MUST be last)
addf motion.motion-controller servo-thread # Check following error, update status
```

Execution Order Rationale:

1. **Read inputs first:** Capture hardware state at start of cycle (consistent snapshot)
2. **Process feedback:** Update encoder positions from captured data
3. **Motion planning:** Generate commanded positions based on G-code
4. **Control:** Compute outputs (PID needs both command and feedback)
5. **Output generation:** Update PWM, step rates
6. **Write outputs:** Send to hardware
7. **Error checking last:** motion.motion-controller compares commanded vs. actual positions AFTER all processing

Critical Rule: motion.motion-controller **must** be last function in servo thread (checks following error using final position values).

4.4 Latency Measurement and System Tuning

Latency Definition:

Latency = worst-case delay between thread period timer expiration and thread execution start. Caused by:

- CPU interrupts (USB, network, disk I/O)
- SMI (System Management Interrupt) handlers in BIOS
- Cache misses and memory access contention
- Other kernel tasks holding locks

Latency Test:

```
# Terminal 1: Run latency histogram
latency-histogram --nobase # Test servo thread only (omit base thread)

# Terminal 2: Stress test system
# Open web browser, play videos, copy large files, etc.
# Run for minimum 1 hour, preferably overnight

# Latency histogram shows distribution:
# Servo thread latency (1 ms period):
#   Min: 8 mus
#   Avg: 12 mus
#   Max: 47 mus  <- Critical value for thread budget
```

Interpreting Results:

Max Latency	Servo Thread	Base Thread (25 mus)	Rating
<20 mus	Excellent	Excellent (100 kHz stepping possible)	Use any configuration
20-50 mus	Excellent	Good (50 kHz stepping)	Servo + limited software stepping
50-100 mus	Good	Poor (unstable)	Servo only, hardware stepping required
>100 mus	Poor	Unusable	Tune system or change hardware

Latency Sources and Mitigation:

1. SMI Interrupts (most common cause of large latency spikes):

```
# Check for SMI sources
sudo cat /sys/firmware/acpi/interrupts/gpe* # ACPI interrupts
sudo lspci -vv | grep -i smbus               # SMBus polling

# BIOS tuning (system-dependent):
# - Disable: USB legacy support, ACPI C-states, CPU throttling
# - Disable: Intel SpeedStep, AMD Cool'n'Quiet
# - Enable: HPET (High Precision Event Timer)
```

2. CPU Power Management:

```
# Force performance mode (no CPU frequency scaling)
sudo cpupower frequency-set -g performance

# Verify
cpupower frequency-info # Should show fixed max frequency

# Make permanent (add to /etc/rc.local):
echo performance | sudo tee /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor
```

3. IRQ (Interrupt) Affinity:

Isolate real-time threads to specific CPU core, move interrupts to other cores:

```
# Isolate CPU 3 for real-time (on 4-core system)
# Edit /etc/default/grub:
GRUB_CMDLINE_LINUX="isolcpus=3"

# Rebuild grub and reboot
sudo update-grub
sudo reboot

# After reboot, configure LinuxCNC to use CPU 3
# Edit INI file:
[EMCMOT]
AFFINITY = 3 # Run real-time threads on CPU 3 only
```

4. Disable Unnecessary Services:

```
# Disable GUI on LinuxCNC-dedicated machine
sudo systemctl set-default multi-user.target

# Disable network manager (if using static IP or no network)
sudo systemctl disable NetworkManager

# Disable automatic updates
sudo systemctl disable unattended-upgrades
```

4.5 Thread Budget Calculation

Thread Utilization:

Percentage of thread period consumed by function execution:

$$\text{Utilization} = \frac{T_{\text{exec}} + T_{\text{latency}}}{T_{\text{period}}} \times 100\%$$

Safety Margin: Keep utilization <50% to accommodate worst-case execution time (WCET) variations.

Example: 4-Axis Servo System

Configuration: - Servo thread: 1 ms (1000 mus) period - Max latency: 47 mus (from latency-test) - Functions: 4x encoder, 4x PID, 4x PWM, motion, I/O, safety

Execution times (measured via `halcmd show thread`):

Function	Exec Time	Count	Total
hm2_7i96.0.read	8 mus	1x	8 mus
encoder.update-counters	1.5 mus	4x	6 mus

Function	Exec Time	Count	Total
motion.motion-command-handler	65 mus	1x	65 mus
pid.X.do-pid-calcs	3 mus	4x	12 mus
pwmgen.update	1 mus	4x	4 mus
lowpass.0	0.5 mus	2x	1 mus
and2.0	0.3 mus	3x	0.9 mus
estop-latch.0	0.5 mus	1x	0.5 mus
charge-pump	0.3 mus	1x	0.3 mus
hm2_7i96.0.write	6 mus	1x	6 mus
motion.motion-controller	12 mus	1x	12 mus
Total execution			115.7 mus

Thread budget:

$$\text{Total time} = T_{exec} + T_{latency} = 115.7 + 47 = 162.7 \text{ mus}$$

$$\text{Utilization} = \frac{162.7}{1000} = 16.3\%$$

Verdict: Excellent (33.7% margin remaining, could add more functions or reduce period to 0.5 ms)

Checking Thread Utilization:

```
# Show thread timing statistics
halcmd show thread
```

```
# Output:
# Realtime Threads:
#   Period Name           (   Time, Max-Time )
#   1000000 servo-thread  ( 115724, 162842 )
#
#                               up avg exec   up worst-case (exec+latency)

# Utilization = 162842 / 1000000 = 16.3%
```

Overrun Detection:

If thread execution exceeds period, LinuxCNC logs error:

```
RTAPI: Task 1 overrun at 1234567890 ns
Motion stopped due to realtime delay
```

Common causes: - Too many functions in thread - Inefficient custom component code - Latency spike (SMI interrupt) - Incorrect thread period configuration

Solution: 1. Remove non-critical functions (move to user-space) 2. Increase thread period (reduce control bandwidth) 3. Optimize custom code (reduce WCET) 4. Fix latency sources (BIOS tuning, isolcpus)

4.6 Custom Thread Creation (Advanced)

For specialized applications requiring multiple control rates, create custom threads:

```
# Create custom thread at 10 kHz (100 mus period) for fast sensor sampling
loadrt threads name1=fast-sample period1=100000

# Add functions to custom thread
addf high-speed-adc.read fast-sample
```

```

addf custom-filter.process fast-sample

# Standard servo thread continues at 1 kHz
addf motion.motion-command-handler servo-thread
# ...

```

Use Cases:

- High-speed data acquisition (vibration monitoring, acoustic emission)
- Fast inner control loop (torque control at 10 kHz, position control at 1 kHz)
- Synchronous sampling of multiple sensors

Constraint: Custom threads increase CPU load—ensure total utilization <50% across all threads.

4.7 Thread Synchronization and Data Sharing

HAL Signals as Shared Memory:

Signals connecting pins between threads act as lock-free shared memory (single-writer, multiple-reader):

```

# Fast thread writes, servo thread reads
# Fast thread (10 kHz)
addf fast-sensor.read fast-thread
net sensor-data fast-sensor.value => lowpass.0.in

# Servo thread (1 kHz)
addf lowpass.0 servo-thread
addf pid.0.do-pid-calcs servo-thread
net filtered-data lowpass.0.out => pid.0.feedback

```

Thread Safety:

HAL enforces single-writer rule (one OUT pin per signal), preventing race conditions. Reading stale data (from previous cycle) is acceptable in control systems—deterministic latency more important than instant propagation.

Example: Base Thread -> Servo Thread Data Flow

Base thread (40 kHz):

```
stepgen.make-pulses generates step count
```

Servo thread (1 kHz):

```

stepgen.update-freq reads step count (updated every 25 mus)
Position updated 40x between servo thread cycles (smooth interpolation)

```

4.8 Real-Time Performance Optimization

Minimize Function Count:

Each function call incurs ~0.5-2 mus overhead (parameter passing, scheduler bookkeeping). Combine logic when possible:

Inefficient:

```

loadrt and2 count=5
addf and2.0 servo-thread
addf and2.1 servo-thread
addf and2.2 servo-thread
addf and2.3 servo-thread
addf and2.4 servo-thread
# 5 function calls = ~5-10 mus overhead

```

Optimized:

```
# Write custom component combining all 5 AND operations
loadrt custom_logic # Single component, 5 AND gates
addf custom_logic.process servo-thread
# 1 function call = ~1-2 mus overhead
```

Avoid Floating-Point in Base Thread:

```
# WRONG: Floating-point in base thread
addf lowpass.0 base-thread # FPU save/restore adds ~50 mus!

# CORRECT: Integer-only in base thread
addf stepgen.make-pulses base-thread # No floating-point
addf parport.0.write base-thread
# Float processing in servo thread:
addf lowpass.0 servo-thread
```

Hardware Offload:

Move time-critical tasks to FPGA (Mesa cards):

Software (CPU):

- Step generation: 50-100 kHz max, 10-20 mus base thread
- Encoder counting: Limited resolution, CPU overhead

Hardware (FPGA):

- Step generation: 4 MHz max, zero CPU overhead
- Encoder counting: 40 MHz quadrature decoding
- PWM generation: 200 kHz PWM frequency

Component Selection:

Task	Software (CPU)	Hardware (Mesa FPGA)	Recommendation
Step generation <50 kHz	Acceptable	Better	Software OK if latency good
Step generation >50 kHz	Difficult	Excellent	Hardware required
Encoder <1 MHz	Acceptable	Excellent	Software OK for low speed
Encoder >1 MHz	Impossible	Excellent	Hardware required
PWM <10 kHz	Acceptable	Excellent	Software OK
PWM >20 kHz	Difficult	Excellent	Hardware recommended

4.9 Worst-Case Execution Time (WCET) Analysis

For mission-critical applications, measure WCET of each function under stress:

Terminal 1: Run LinuxCNC

```
linuxcnc machine.ini
```

Terminal 2: Monitor thread timing continuously

```
watch -n 0.1 'halcmd show thread'
```

Terminal 3: Stress test

```
stress-ng --cpu 4 --io 4 --vm 2 --vm-bytes 1G --timeout 600s
```

Record maximum "Max-Time" value over 10+ minute test

This is worst-case execution time including latency

Safety Factor:

For safety-critical systems (medical, aerospace), apply 2x safety factor:

$$T_{period} \geq 2 \times T_{WCET}$$

Example: WCET = 180 mus measured - **Minimum safe period:** 360 mus - **Recommended:** 500 mus (2.8x margin)

4.10 Thread Configuration Examples

Example 1: Stepper System with Software Stepping

[EMCMOT]

```
EMCMOT = motmod
```

```
BASE_PERIOD = 25000          # 25 mus = 40 kHz (supports 100 kHz stepping)
```

```
SERVO_PERIOD = 1000000       # 1 ms = 1 kHz
```

Base thread: Step pulse generation only

```
addf parport.0.reset base-thread # Reset parallel port (if needed)
```

```
addf stepgen.make-pulses base-thread
```

```
addf parport.0.write base-thread
```

Servo thread: Everything else

```
addf parport.0.read servo-thread
```

```
addf stepgen.capture-position servo-thread
```

```
addf motion.motion-command-handler servo-thread
```

```
addf stepgen.update-freq servo-thread
```

```
addf motion.motion-controller servo-thread
```

Example 2: Servo System with Mesa FPGA (No Base Thread)

[EMCMOT]

```
EMCMOT = motmod
```

```
SERVO_PERIOD = 1000000       # 1 ms = 1 kHz (no base thread needed)
```

All functions in servo thread (FPGA handles step/encoder at MHz rates)

```
addf hm2_7i96.0.read servo-thread
```

```
addf motion.motion-command-handler servo-thread
```

```
addf pid.0.do-pid-calcs servo-thread
```

```
addf pid.1.do-pid-calcs servo-thread
```

```
addf pid.2.do-pid-calcs servo-thread
```

```

addf pwmgen.update servo-thread
addf hm2_7i96.0.write servo-thread
addf motion.motion-controller servo-thread

```

Example 3: High-Performance Servo (2 kHz Control Rate)

```

[EMCMOT]
EMCMOT = motmod
SERVO_PERIOD = 500000      # 0.5 ms = 2 kHz (high bandwidth control)

```

Requires: - Mesa 7i80 or 7i92 (supports 2-4 kHz rates) - Low-latency system (<20 mus max) - Optimized function set (minimal overhead)

Example 4: Mixed-Rate System (Fast Inner Loop)

```

loadrt threads name1=torque-loop period1=100000 # 100 mus = 10 kHz

# Torque loop (10 kHz): Current control
addf hm2_7i92.0.read torque-loop
addf torque-pid.do-pid-calcs torque-loop
addf hm2_7i92.0.write torque-loop

# Position loop (1 kHz): Velocity/position control
addf motion.motion-command-handler servo-thread
addf position-pid.do-pid-calcs servo-thread
# position-pid.output -> torque-pid.command (cascaded control)

```

4.11 Debugging Thread Timing Issues

Problem: “Unexpected realtime delay” Error

Symptom: LinuxCNC stops with error message

Cause: Thread execution exceeded period

Diagnosis:

1. Check thread utilization:


```

halcmd show thread
# Look for Max-Time approaching Period

```
2. Run latency test while LinuxCNC running:


```

latency-histogram --nobase
# Check for latency spikes

```
3. Review function list:


```

halcmd show funct
# Look for custom components with long execution time

```

Solutions:

- Increase thread period (reduce control rate)
- Remove non-essential functions
- Tune system for lower latency (BIOS, isolcpus)
- Move functions to user-space
- Optimize custom component code

Problem: Step Pulse Jitter (Stepper Motors)

Symptom: Stepper motors vibrate, lose steps, or stall

Cause: Base thread jitter too high

Diagnosis:

1. Run latency test with base thread:
 latency-histogram
 # Check base thread max latency
2. If >10 mus, software stepping unreliable at high rates

Solutions:

- Reduce max step rate (lower maxvel, maxaccel)
- Increase base period (lower step rate capacity)
- Switch to hardware step generation (Mesa FPGA)
- Tune system (SMI sources, CPU isolation)

Problem: Following Error (Servo Systems)

Symptom: "Joint 0 following error" message

Cause: PID cannot track commanded position

Diagnosis:

1. Check if timing-related:
 halcmd show thread
 # High utilization (>70%) may cause servo loop delays
2. Verify servo thread not overrunning:
 dmesg | grep "overrun"

Solutions:

- Increase servo period (slower control rate)
- Reduce trajectory velocity/acceleration
- Tune PID gains (may be unstable due to timing jitter)
- Reduce thread utilization (remove functions)

4.12 Summary: Real-Time Thread Mastery

Real-time thread configuration determines LinuxCNC control system performance, stability, and capabilities:

Key Principles:

1. **Base thread optional:** Required only for software step generation >50 kHz; eliminated with hardware stepping
2. **Servo thread required:** Core motion control loop, typically 1 kHz (1 ms period)
3. **Latency critical:** Max latency + execution time must be <50% of thread period
4. **Function order matters:** Inputs -> processing -> outputs -> error checking
5. **Hardware offload:** Mesa FPGA eliminates base thread, reduces CPU load, improves reliability

Thread Selection Guide:

Application	Base Period	Servo Period	Hardware
Hobby stepper mill	25-50 mus	1 ms	Parallel port or Mesa 7i96
Professional stepper	N/A	1 ms	Mesa 7i96/7i76 (hardware stepping)
Servo system	N/A	1 ms	Mesa 7i76/7i77
High-performance servo	N/A	0.5 ms	Mesa 7i80/7i92

Optimization Priority:

1. Measure latency (latency-histogram, 1+ hour test)
2. Calculate thread budget (execution + latency < 50% period)
3. Tune BIOS (disable SMI sources, power management)
4. Isolate CPUs (isolcpus for dedicated real-time core)
5. Minimize functions (combine logic, use hardware offload)

Next Section (14.5) explores HAL and INI file structure in depth: configuration loading sequence, INI variable substitution, file organization best practices, and integration with LinuxCNC startup process.

Total: 3,847 words / 6 equations / 4 worked examples / 6 tables / 12 code blocks

5. HAL File Configuration and INI Integration

5.1 LinuxCNC Configuration File Structure

A LinuxCNC configuration consists of multiple interconnected files defining machine kinematics, motion parameters, I/O mapping, and HAL signal routing. Understanding file organization, loading sequence, and variable substitution mechanisms is essential for creating maintainable, modular configurations.

Standard Configuration Directory Structure:

```
~/linuxcnc/configs/my_machine/
|-- my_machine.ini          # Main configuration file (kinematics, limits, I/O)
|-- my_machine.hal          # Core HAL file (components, signals, basic setup)
|-- custom.hal              # User customizations (loaded after main HAL)
|-- custom_postgui.hal      # GUI-related HAL (loaded after GUI starts)
|-- tool.tbl                # Tool table (tool lengths, diameters)
|-- my_machine.var          # G-code variables (persistent across sessions)
|-- subroutines/            # Custom G-code subroutines (*.ngc files)
|-- python/                 # Custom Python HAL components
+-- backups/                # Configuration backups (good practice)
```

File Loading Sequence:

1. LinuxCNC reads INI file (my_machine.ini)
2. INI [HAL] section specifies HAL files to load
3. Core HAL files loaded (components, threads, signals)
4. Motion controller initialized
5. GUI launched (Axis, Gmoccapy, etc.)
6. Post-GUI HAL files loaded (GUI-related signals)
7. System ready for operation

5.2 INI File Structure and Sections

The INI file uses standard Windows INI format: [SECTION] headers followed by KEY = value pairs.

Essential Sections:

[EMC] - General Configuration

```
[EMC]
VERSION = 1.1
MACHINE = My CNC Mill
DEBUG = 0                # Debug level (0=none, 1=config, 7=verbose)
```

[DISPLAY] - User Interface

```
[DISPLAY]
DISPLAY = axis            # GUI: axis, gmoccapy, touchy, qtdragon
POSITION_OFFSET = RELATIVE # Display: RELATIVE or MACHINE coordinates
POSITION_FEEDBACK = ACTUAL # Display: ACTUAL or COMMANDED position
MAX_FEED_OVERRIDE = 2.0   # Maximum feed rate override (200%)
MAX_SPINDLE_OVERRIDE = 1.5 # Maximum spindle speed override (150%)
PROGRAM_PREFIX = /home/user/nc_files # Default G-code directory
INTRO_GRAPHIC = linuxcnc.gif
INTRO_TIME = 5            # Splash screen duration (seconds)
INCREMENTS = 0.1mm, 0.01mm, 0.001mm # Jog increments
```

[FILTER] - G-code Preprocessors

```
[FILTER]
PROGRAM_EXTENSION = .png,.gif,.jpg Grayscale Depth Image
```

```

PROGRAM_EXTENSION = .py Python Script
png = image-to-gcode
gif = image-to-gcode
jpg = image-to-gcode
py = python

```

[RS274NGC] - G-code Interpreter

```

[RS274NGC]
PARAMETER_FILE = my_machine.var      # Persistent variables (#5220, etc.)
SUBROUTINE_PATH = subroutines        # Custom subroutine directory
FEATURES = 30                        # Bitmap: 16=named params, 32=Python
REMAP = M6 modalgroup=6 ngc=tool_change # Custom M-code implementation

```

[EMCMOT] - Motion Controller

```

[EMCMOT]
EMCMOT = motmod
COMM_TIMEOUT = 1.0                  # Motion watchdog timeout (seconds)
BASE_PERIOD = 25000                 # Base thread period (nanoseconds)
SERVO_PERIOD = 1000000              # Servo thread period (nanoseconds)

```

[TASK] - Task Controller

```

[TASK]
TASK = milltask                    # milltask (mill), lathe task (lathe)
CYCLE_TIME = 0.010                 # Task update rate (10 ms)

```

[HAL] - HAL File Loading

```

[HAL]
HALFILE = my_machine.hal           # Core HAL file (required)
HALFILE = custom.hal               # User customizations (optional)
POSTGUI_HALFILE = custom_postgui.hal # After GUI starts (optional)
HALUI = halui                      # Load HAL User Interface component

```

[HALUI] - HAL User Interface Pins

```

[HALUI]
# No settings here, but [HAL] HALUI = halui enables pins:
# halui.program.run, halui.program.pause, halui.spindle.start, etc.

```

[TRAJ] - Trajectory Planner

```

[TRAJ]
COORDINATES = X Y Z                # Axis letters (X Y Z A B C U V W)
LINEAR_UNITS = mm                   # mm or inch
ANGULAR_UNITS = degree              # degree or radian
DEFAULT_LINEAR_VELOCITY = 5.0       # Default jog speed (mm/s)
MAX_LINEAR_VELOCITY = 50.0          # Maximum rapid speed (mm/s)
DEFAULT_LINEAR_ACCELERATION = 100.0 # Default accel (mm/s^2)
MAX_LINEAR_ACCELERATION = 500.0     # Maximum accel (mm/s^2)
POSITION_FILE = position.txt        # Save position on shutdown

```

[EMCIO] - I/O Controller (Tool Changer, Coolant)

```

[EMCIO]
EMCIO = io
CYCLE_TIME = 0.100                 # I/O update rate (100 ms)
TOOL_TABLE = tool.tbl               # Tool table file

```

```

TOOL_CHANGE_POSITION = 0 0 50 # Position for manual tool changes (X Y Z)
TOOL_CHANGE_QUILL_UP = 1      # Retract Z before tool change

```

[**AXIS_n**] - Individual Axis Configuration

```

[JOINT_0] # X-axis (LinuxCNC 2.8+ uses JOINT, older versions use AXIS)
TYPE = LINEAR          # LINEAR or ANGULAR
HOME = 0.0             # Home position (mm)
MAX_VELOCITY = 50.0     # Maximum velocity (mm/s)
MAX_ACCELERATION = 500.0 # Maximum acceleration (mm/s^2)
MIN_LIMIT = -0.1        # Soft limit minimum (mm)
MAX_LIMIT = 200.1       # Soft limit maximum (mm)
HOME_OFFSET = 0.0       # Offset from home switch to home position
HOME_SEARCH_VEL = -5.0  # Home search velocity (negative = toward minimum)
HOME_LATCH_VEL = 1.0   # Final approach velocity (slow, precise)
HOME_SEQUENCE = 1       # Homing order (lower numbers first, 0=no homing)
HOME_IGNORE_LIMITS = YES # Allow homing through limit switches

```

PID tuning (if using simple PID in INI, otherwise in HAL file)

```

P = 150.0
I = 2.0
D = 5.0
FF0 = 0.0
FF1 = 1.0          # Feed-forward velocity term
FF2 = 0.01         # Feed-forward acceleration term
BIAS = 0.0
DEADBAND = 0.001   # Following error dead band (mm)
MAX_ERROR = 0.5    # Following error limit (mm, triggers abort)

```

Stepper configuration (if using stepgen in position mode)

```

SCALE = 320.0      # Steps per unit (steps/mm)
# OR for servo:
SCALE = 1600.0     # Encoder counts per unit (counts/mm)

```

[**SPINDLE_0**] - Spindle Configuration

```

[SPINDLE_0]
MAX_FORWARD_VELOCITY = 5000 # Maximum RPM (forward)
MAX_REVERSE_VELOCITY = 3000 # Maximum RPM (reverse)
MIN_FORWARD_VELOCITY = 100  # Minimum RPM (below this, spindle stops)
SCALE = 0.0002             # HAL pin scaling (RPM to 0-1.0 PWM)

```

5.3 INI Variable Substitution in HAL Files

HAL files can reference INI values using [SECTION]KEY syntax, enabling centralized parameter management.

Example: Centralized Axis Configuration

INI File (my_machine.ini):

```

[JOINT_0]
TYPE = LINEAR
MAX_VELOCITY = 50.0
MAX_ACCELERATION = 500.0
SCALE = 320.0          # Steps/mm
P = 150.0
I = 2.0
D = 5.0

```

```

STEPLEN = 2000          # 2 mus step pulse width
STEPSPACE = 2000        # 2 mus between steps
DIRSETUP = 5000         # 5 mus direction setup
DIRHOLD = 5000          # 5 mus direction hold

```

HAL File (my__machine.hal):

```

# Load stepgen with parameters from INI
loadrt stepgen step_type=0,0,0 # 3 axes, type 0 = step/dir

# Reference INI values using [SECTION]KEY substitution
setp stepgen.0.position-scale [JOINT_0]SCALE
setp stepgen.0.maxvel [JOINT_0]MAX_VELOCITY
setp stepgen.0.maxaccel [JOINT_0]MAX_ACCELERATION
setp stepgen.0.steplen [JOINT_0]STEPLEN
setp stepgen.0.stepspace [JOINT_0]STEPSPACE
setp stepgen.0.dirsetup [JOINT_0]DIRSETUP
setp stepgen.0.dirhold [JOINT_0]DIRHOLD

# PID tuning from INI
setp pid.0.Pgain [JOINT_0]P
setp pid.0.Igain [JOINT_0]I
setp pid.0.Dgain [JOINT_0]D

```

Advantages:

1. **Single source of truth:** Change velocity limit in INI, automatically propagates to HAL
2. **GUI integration:** Axis GUI displays MAX_VELOCITY from INI (consistent with HAL stepgen.maxvel)
3. **Maintainability:** No duplicate values in multiple files
4. **Readability:** HAL file intent clear ([JOINT_0]SCALE self-documenting)

Variable Substitution Rules:

- **Case-sensitive:** [JOINT_0]P not same as [joint_0]p
- **No spaces:** [JOINT_0]P works, [JOINT_0] P fails
- **String concatenation:** [JOINT_0]SCALE can be part of expression

Arithmetic in INI Substitution:

```

# Calculate derivative gain as 10% of P gain
setp pid.0.Dgain [expr [JOINT_0]P * 0.1]

# Convert RPM to radians/second for spindle
setp spindle.scale [expr [SPINDLE_0]MAX_FORWARD_VELOCITY * 2 * 3.14159 / 60]

```

5.4 HAL File Organization Best Practices

Modular Configuration Approach:

my_machine.hal	-> Core setup (components, threads, basic I/O)
custom.hal	-> User customizations (easily reverted)
custom_postgui.hal	-> GUI-specific signals (PyVCP, halui connections)
hardware.hal	-> Hardware-specific (Mesa config, parport mapping)
pid_tuning.hal	-> PID parameters (separate for easy tuning iterations)
spindle.hal	-> Spindle control logic (VFD, encoder, at-speed)
tool_changer.hal	-> Tool changer sequencing

my__machine.hal Structure:

```

# =====
# 1. LOAD COMPONENTS
# =====
loadrt trivkins
loadrt [EMCMOT]EMCMOT base_period_nsec=[EMCMOT]BASE_PERIOD servo_period_nsec=[EMCMOT]SERVO_PERIOD num_j

# Hardware I/O
loadrt hal_parport cfg="0x0378"

# Feedback/control
loadrt encoder num_chan=3
loadrt pid num_chan=3
loadrt pwmgen output_type=0

# Signal processing
loadrt lowpass count=2

# Logic and safety
loadrt and2 count=3
loadrt estop_latch
loadrt charge_pump

# =====
# 2. ADD FUNCTIONS TO THREADS
# =====
# Read inputs
addf parport.0.read servo-thread
addf encoder.update-counters servo-thread

# Motion and control
addf motion.motion-command-handler servo-thread
addf pid.0.do-pid-calcs servo-thread
addf pid.1.do-pid-calcs servo-thread
addf pid.2.do-pid-calcs servo-thread

# Signal processing
addf lowpass.0 servo-thread
addf lowpass.1 servo-thread

# Logic
addf and2.0 servo-thread
addf and2.1 servo-thread
addf and2.2 servo-thread
addf estop-latch.0 servo-thread
addf charge-pump servo-thread

# Write outputs
addf pwmgen.update servo-thread
addf parport.0.write servo-thread

# Error checking (MUST BE LAST)
addf motion.motion-controller servo-thread

# =====

```

```

# 3. CONFIGURE COMPONENTS
# =====
# Encoder scaling
setp encoder.0.position-scale [JOINT_0]SCALE
setp encoder.1.position-scale [JOINT_1]SCALE
setp encoder.2.position-scale [JOINT_2]SCALE

# PID parameters (loaded from separate file for easy tuning)
source pid_tuning.hal

# PWM configuration
setp pwmgen.0.pwm-freq 20000
setp pwmgen.0.scale [JOINT_0]PWM_SCALE
setp pwmgen.0.max-dc 0.95

# =====
# 4. CONNECT SIGNALS - AXIS 0 (X)
# =====
# Position command
net x-pos-cmd motion.00.motor-pos-cmd => pid.0.command

# Position feedback
net x-pos-fb encoder.0.position => pid.0.feedback motion.00.motor-pos-fb

# Control output
net x-output pid.0.output => pwmgen.0.value

# Enables
net x-enable motion.00.amp-enable-out => pid.0.enable pwmgen.0.enable

# Hardware connections
net x-encoder-A encoder.0.phase-A <= parport.0.pin-02-in
net x-encoder-B encoder.0.phase-B <= parport.0.pin-03-in
net x-encoder-Z encoder.0.phase-Z <= parport.0.pin-04-in
net x-pwm pwmgen.0.pwm => parport.0.pin-01-out
net x-dir pwmgen.0.dir => parport.0.pin-14-out

# =====
# 5. CONNECT SIGNALS - AXIS 1 (Y)
# =====
# (similar structure for Y and Z axes)

# =====
# 6. E-STOP AND SAFETY
# =====
net estop-button parport.0.pin-10-in-not => estop-latch.0.fault-in
net estop-reset parport.0.pin-11-in => estop-latch.0.reset
net estop-ok estop-latch.0.ok-out => motion.motion-enabled
net estop-fault estop-latch.0.fault-out => parport.0.pin-12-out

# Charge pump watchdog
net charge-toggle charge-pump.out => parport.0.pin-13-out

# =====

```



```
# 7. SPINDLE CONTROL
# =====
source spindle.hal

# =====
# 8. LOAD USER CUSTOMIZATIONS
# =====
source custom.hal
```

5.5 Post-GUI HAL Files

Purpose: Connect HAL signals to GUI-specific pins (PyVCP panels, halui commands, jog controls).

Loading Sequence:

1. Core HAL files loaded
2. Motion controller starts
3. GUI launches (Axis, Gmoccapy, etc.)
4. GUI creates HAL pins (pyvcp.*, gladevcp.*, halui.*)
5. Post-GUI HAL files loaded
6. Signals connect GUI pins to machine logic

Example: custom_postgui.hal

```
# =====
# Connect halui pins (program control)
# =====
net program-run halui.program.run <= pyvcp.button-run
net program-pause halui.program.pause <= pyvcp.button-pause
net program-stop halui.program.stop <= pyvcp.button-stop

# =====
# Connect PyVCP display elements
# =====
net spindle-rpm motion.spindle-speed-out => pyvcp.spindle-speed-display
net current-feed motion.current-vel => pyvcp.feedrate-display

# X-axis position display
net x-pos-display motion.00.motor-pos-fb => pyvcp.x-position-dro

# =====
# Connect jog controls
# =====
net jog-x-plus halui.jog.0.plus <= pyvcp.jog-x-plus
net jog-x-minus halui.jog.0.minus <= pyvcp.jog-x-minus
net jog-y-plus halui.jog.1.plus <= pyvcp.jog-y-plus
net jog-y-minus halui.jog.1.minus <= pyvcp.jog-y-minus

net jog-speed halui.jog-speed <= pyvcp.jog-speed-slider

# =====
# Custom indicator lamps
# =====
net machine-on motion.motion-enabled => pyvcp.led-machine-on
net spindle-running motion.spindle-on => pyvcp.led-spindle
net coolant-on motion.coolant-flood => pyvcp.led-coolant
```

Why Separate Post-GUI File?

GUI pins don't exist until GUI starts. Attempting to connect in main HAL file causes error:

```
# In my_machine.hal (loaded before GUI):
net spindle-rpm => pyvcp.spindle-speed-display # ERROR: pyvcp.spindle-speed-display not found

# Must be in custom_postgui.hal (loaded after GUI creates pins)
```

5.6 Tool Table (tool.tbl)

Format: Tab-separated or fixed-width columns

```
T1 P1 D6.35 Z+10.5 ;1/4" end mill
T2 P2 D3.175 Z+12.3 ;1/8" end mill
T3 P3 D12.7 Z+8.2 ;1/2" face mill
T4 P4 D0.0 Z+0.0 ;Touch probe (no offset)
T99 P99 D0.0 Z+0.0 ;Empty pocket
```

Columns:

- **T:** Tool number (T1, T2, ..., T99)
- **P:** Pocket number (tool changer position)
- **D:** Diameter (mm or inch, for cutter compensation G41/G42)
- **Z:** Length offset (mm or inch, applied with G43)
- **;**: Comment (tool description)

G-code Usage:

```
T1 M6          ; Load tool 1 from pocket 1
G43 H1         ; Apply tool 1 length offset (Z+10.5)
G41 D1         ; Cutter compensation left, diameter 6.35
```

(Machine now compensates Z by +10.5 mm, XY path offset by 3.175 mm radius)

5.7 G-code Variable File (my_machine.var)

Purpose: Persistent storage for G-code variables (#5220-#5399, tool offsets, work coordinate systems).

Format (auto-generated, don't edit manually):

```
5161 0.000000
5162 0.000000
5163 0.000000
...
5220 10.500000  <- Tool 1 Z offset (from tool table)
5221 50.250000  <- G54 X offset (work coordinate system 1)
5222 25.100000  <- G54 Y offset
5223 -5.000000  <- G54 Z offset
...
```

Accessing in G-code:

```
#<_x> = #5420    ; Read current X position
#100 = [#<_x> + 10] ; Calculate new position
G0 X#100        ; Move to calculated position
```

(Variables persist across LinuxCNC sessions via .var file)

5.8 Configuration Management Best Practices

Version Control:

```
cd ~/linuxcnc/configs/my_machine
git init
git add *.ini *.hal *.tbl
git commit -m "Initial working configuration"

# After successful tuning session:
git commit -am "PID tuning: X-axis P=150, I=2.5, D=5.0"

# Revert to previous version if changes cause problems:
git log # Find commit hash
git checkout abc123 -- pid_tuning.hal
```

Backup Before Changes:

```
# Create timestamped backup
cd ~/linuxcnc/configs/my_machine
tar -czf ../backups/my_machine_$(date +%Y%m%d_%H%M%S).tar.gz .

# Restore if needed:
cd ~/linuxcnc/configs/my_machine
tar -xzf ../backups/my_machine_20240315_143022.tar.gz
```

Configuration Templates:

Create reusable configuration snippets:

```
~/linuxcnc/configs/templates/
|-- parport_3axis.hal      # Standard parport 3-axis setup
|-- mesa_7i96_steppers.hal # Mesa 7i96 stepper configuration
|-- pid_conservative.hal   # Conservative PID starting values
+-- spindle_vfd_modbus.hal # VFD Modbus control template
```

Documentation in Comments:

```
# =====
# X-Axis Configuration
# Hardware: Nema 23 stepper, 8x microstepping, 5mm/rev ballscrew
# Scaling: 200 steps/rev x 8 = 1600 steps/rev / 5 mm/rev = 320 steps/mm
# Max velocity: 50 mm/s (motor rated 3000 RPM = 250 rev/s = 1250 mm/s, de-rated 20x)
# Max acceleration: 500 mm/s^2 (empirically tested, no stalling)
# Last tuned: 2024-03-15 by John Doe
# =====
setp stepgen.0.position-scale 320
setp stepgen.0.maxvel 50.0
setp stepgen.0.maxaccel 500.0
```

5.9 Debugging Configuration Issues

Common INI Errors:

Error: [HAL] section missing HALFILE entry
Solution: Add at minimum:

```
[HAL]
HALFILE = my_machine.hal
```

Error: [TRAJ]AXES not defined
Solution: Add to [TRAJ]:
COORDINATES = X Y Z
(LinuxCNC automatically sets AXES = 3 from 3 letters)

Error: BASE_PERIOD and SERVO_PERIOD missing
Solution: Add to [EMCMOT]:
BASE_PERIOD = 25000
SERVO_PERIOD = 1000000

Common HAL Errors:

Error: Pin 'pid.0.command' not found
Cause: Forgot to load pid component
Solution: Add before net statement:
loadrt pid num_chan=3

Error: Signal 'x-pos-cmd' already has writer
Cause: Multiple OUT pins driving same signal
Solution: Check for duplicate net statements:
net x-pos-cmd motion.00.motor-pos-cmd => pid.0.command
net x-pos-cmd override.out => pid.0.command # REMOVE THIS LINE

Error: Function 'pid.0.do-pid-calcs' not found
Cause: Typo in function name or component not loaded
Solution: Check component loaded and function name:
halcmd show funct # List all available functions

Testing Configuration Without Hardware:

Launch LinuxCNC in simulation mode (no real-time kernel required)
linuxcnc -d my_machine.ini # -d = debug mode, shows HAL loading

Or use halrun for isolated HAL testing:
halrun -I
halcmd: loadrt pid num_chan=1
halcmd: addf pid.0.do-pid-calcs servo-thread
halcmd: setp pid.0.Pgain 100.0
halcmd: show pin pid.0
halcmd: show param pid.0

5.10 Complete Configuration Example

INI File: 3-Axis Mill with Servo Motors

```
[EMC]
VERSION = 1.1
MACHINE = 3-Axis Servo Mill
DEBUG = 0

[DISPLAY]
DISPLAY = axis
POSITION_OFFSET = RELATIVE
POSITION_FEEDBACK = ACTUAL
MAX_FEED_OVERRIDE = 2.0
INCREMENTS = 1mm, 0.1mm, 0.01mm
```

```

[FILTER]
PROGRAM_EXTENSION = .py Python Script
py = python

[RS274NGC]
PARAMETER_FILE = servo_mill.var
SUBROUTINE_PATH = subroutines

[EMCMOT]
EMCMOT = motmod
COMM_TIMEOUT = 1.0
SERVO_PERIOD = 1000000 # 1 ms, no base thread (hardware stepping)

[TASK]
TASK = milltask
CYCLE_TIME = 0.010

[HAL]
HALFILE = servo_mill.hal
HALFILE = custom.hal
POSTGUI_HALFILE = custom_postgui.hal
HALUI = halui

[TRAJ]
COORDINATES = X Y Z
LINEAR_UNITS = mm
ANGULAR_UNITS = degree
DEFAULT_LINEAR_VELOCITY = 10.0
MAX_LINEAR_VELOCITY = 100.0
DEFAULT_LINEAR_ACCELERATION = 200.0
MAX_LINEAR_ACCELERATION = 1000.0

[EMCIO]
EMCIO = io
CYCLE_TIME = 0.100
TOOL_TABLE = tool.tbl

[JOINT_0] # X-axis
TYPE = LINEAR
HOME = 0.0
MAX_VELOCITY = 100.0
MAX_ACCELERATION = 1000.0
MIN_LIMIT = -0.1
MAX_LIMIT = 600.1
HOME_OFFSET = 0.0
HOME_SEARCH_VEL = -10.0
HOME_LATCH_VEL = 1.0
HOME_SEQUENCE = 1
P = 200.0
I = 5.0
D = 10.0
FF0 = 0.0
FF1 = 1.0
DEADBAND = 0.001

```

```

MAX_ERROR = 1.0
ENCODER_SCALE = 2000.0 # 2000 encoder counts/mm
PWM_SCALE = 100.0      # +/-100.0 input -> +/-10V output

[JOINT_1] # Y-axis (similar to X)
TYPE = LINEAR
HOME = 0.0
MAX_VELOCITY = 100.0
MAX_ACCELERATION = 1000.0
MIN_LIMIT = -0.1
MAX_LIMIT = 400.1
P = 200.0
I = 5.0
D = 10.0
ENCODER_SCALE = 2000.0
PWM_SCALE = 100.0

[JOINT_2] # Z-axis
TYPE = LINEAR
HOME = 0.0
MAX_VELOCITY = 50.0
MAX_ACCELERATION = 500.0
MIN_LIMIT = -0.1
MAX_LIMIT = 200.1
P = 150.0
I = 3.0
D = 8.0
ENCODER_SCALE = 2000.0
PWM_SCALE = 80.0

[SPINDLE_0]
MAX_FORWARD_VELOCITY = 3000
MIN_FORWARD_VELOCITY = 100

```

5.11 Summary: Configuration File Mastery

Proper configuration file organization enables:

1. **Maintainability:** Modular structure (separate spindle.hal, pid_tuning.hal) allows focused changes
2. **Reusability:** INI variable substitution prevents duplicate parameter definitions
3. **Debugging:** Clear comments and logical structure aid troubleshooting
4. **Version control:** Text-based files integrate with git for change tracking
5. **Collaboration:** Standardized structure enables team development and community sharing

Configuration Checklist:

- ☐ INI file defines all required sections ([EMC], [DISPLAY], [TRAJ], [EMCMOT], [HAL])
- ☐ HAL files load components before creating signals
- ☐ Functions added to threads in logical order (read -> compute -> write -> check)
- ☐ Parameters use INI substitution ([JOINT_0]SCALE) for centralized management
- ☐ Post-GUI HAL file handles GUI-specific connections
- ☐ Comments document hardware specs, scaling calculations, tuning history
- ☐ Configuration backed up before changes (git or tarball)
- ☐ Tested in simulation mode before running on real hardware

Next Section (14.6) dives into custom HAL component development in C using the comp compiler, enabling

specialized real-time logic beyond the standard component library.

Total: 4,312 words / 0 equations / 6 complete worked examples / 2 tables / 30 code blocks

6. Custom HAL Components in C

6.1 Why Write Custom Components?

While LinuxCNC's standard component library covers most CNC control scenarios, custom components enable specialized functionality:

- **Custom kinematics:** Non-Cartesian robots (SCARA, delta, cable-driven mechanisms)
- **Tool changers:** Complex sequencing logic beyond standard I/O mapping
- **Process control:** Specialized algorithms (laser power modulation, EDM gap control, plasma THC)
- **Hardware interfaces:** Proprietary encoder protocols, custom FPGA communication
- **Performance optimization:** Combining multiple HAL components into single efficient function

Real-Time vs. User-Space:

- **Real-time components (this section):** Time-critical logic requiring deterministic execution (<1 ms latency)
- **User-space components (Section 14.7):** Non-critical tasks (GUI updates, VFD communication, data logging)

6.2 The comp Compiler: HAL Component Development Tool

The `comp` utility simplifies HAL component development by generating boilerplate C code from high-level component descriptions. Instead of manually writing HAL registration code, memory management, and thread integration, developers focus on algorithmic logic.

Workflow:

1. Write `.comp` file (component description + C code)
2. Run `comp` compiler: `comp --install mycomponent.comp`
3. Component compiled and installed to system
4. Load in HAL file: `loadrt mycomponent`

comp File Structure:

```
component mycomponent "Brief description";

// Pin declarations
pin in float input "Input signal";
pin out float output "Output signal";

// Parameter declarations
parameter rw float gain = 1.0 "Scaling factor";

// Function declaration
function _; // Underscore = default function name (mycomponent)

license "GPL";
author "Your Name";

;; // End of declarations, C code begins

// Function implementation (executed every thread cycle)
FUNCTION(_) {
    output = input * gain;
}
```

Compilation:


```
comp --install mycomponent.comp
# Generates mycomponent.c, compiles to mycomponent.ko (kernel module)
# Installs to /usr/lib/linuxcnc/modules/
```

Usage in HAL:

```
loadrt mycomponent
addf mycomponent servo-thread

setp mycomponent.gain 2.5
net input-sig => mycomponent.input
net output-sig <= mycomponent.output
```

6.3 Pin and Parameter Declarations

Pin Syntax:

```
pin [direction] [type] [name] [if condition] "description";

// Direction: in, out, io
// Type: bit, s32, u32, float
// Name: Pin identifier (component.name in HAL)
// If condition: Optional, enables/disables pin based on modparam
// Description: Help text
```

Examples:

```
component encoder_ex "Example encoder with multiple modes";

// Basic pins
pin in bit phase_a "Encoder A channel";
pin in bit phase_b "Encoder B channel";
pin out s32 counts "Quadrature count";
pin out float position "Scaled position";

// Conditional pins (enabled via modparam)
pin in bit index_enable if index "Enable index search";
pin in bit phase_z if index "Index pulse (Z channel)";

// I/O pin (bidirectional)
pin io bit index_latch "Latches TRUE when index found";

parameter rw float scale = 1.0 "Counts per position unit";
parameter rw bit index = 0 "Enable index pulse support (modparam)";

function _ "Update encoder counts";
license "GPL";
;;

FUNCTION(_) {
    // Function implementation
}
```

Loading with modparam:

```
# Enable index support for first instance, disable for second
loadrt encoder_ex count=2 index=1,0
```

```
# encoder_ex.0 has phase_z and index_enable pins
# encoder_ex.1 does not (index=0)
```

Parameter Types:

- **rw**: Read-write (settable via `setp` in HAL)
- **r**: Read-only (informational, set by component logic)
- **w**: Write-only (rare, typically for command inputs)

```
parameter rw float pgain = 1.0 "Proportional gain";
parameter r u32 cycles_executed "Function call counter (read-only)";
```

6.4 Complete Example: Hysteresis Comparator

Application: Convert noisy analog signal to clean digital output with hysteresis (Schmitt trigger behavior).

`hysteresis.comp`:

```
component hysteresis "Schmitt trigger with adjustable thresholds";
description ""
Hysteresis comparator prevents output chatter on noisy input signals.
```

Example: Spindle at-speed detection

```
- Set high-threshold = 95% of commanded speed
- Set low-threshold = 90% of commanded speed
- Output goes TRUE when input exceeds high-threshold
- Output stays TRUE until input falls below low-threshold
"";
```

```
pin in float input "Analog input signal";
pin out bit output "Digital output (with hysteresis)";
```

```
parameter rw float high_threshold = 1.0 "Rising edge threshold";
parameter rw float low_threshold = 0.5 "Falling edge threshold";
```

```
function _ fp "Update output based on input (floating-point)";
```

```
license "GPL";
author "LinuxCNC Example";
```

```
;;
```

```
FUNCTION(_) {
    // State machine: output retains previous state unless threshold crossed
    if (output) {
        // Currently TRUE: check if input dropped below low threshold
        if (input < low_threshold) {
            output = 0; // Turn off
        }
    } else {
        // Currently FALSE: check if input exceeded high threshold
        if (input >= high_threshold) {
            output = 1; // Turn on
        }
    }
}

// Hysteresis gap = high_threshold - low_threshold
```

```

    // Prevents chatter if input oscillates within gap
}

```

Compilation and Installation:

```

comp --install hysteresis.comp
# Output: Compiling realtime hysteresis.c
#         Successfully installed to /usr/lib/linuꜛcnc/modules/

```

Usage: Spindle At-Speed Detection

```

loadrt hysteresis
addf hysteresis.0 servo-thread

# Spindle commanded at 1000 RPM
# Enable cutting when spindle reaches 95% (950 RPM)
# Disable if drops below 90% (900 RPM)
setp hysteresis.0.high-threshold 950.0
setp hysteresis.0.low-threshold 900.0

net spindle-speed-fb tachometer.rpm => hysteresis.0.input
net spindle-at-speed hysteresis.0.output => motion.spindle-at-speed

# Motion controller waits for spindle-at-speed before starting cut (M3 S1000)

```

Truth Table:

Input	Previous Output	New Output	Reason
800 RPM	FALSE	FALSE	Below low threshold
920 RPM	FALSE	FALSE	Between thresholds, output stays FALSE
960 RPM	FALSE	TRUE	Exceeded high threshold -> turn on
930 RPM	TRUE	TRUE	Above low threshold, output stays TRUE
880 RPM	TRUE	FALSE	Dropped below low threshold -> turn off

6.5 State Machines and Persistent Variables

Problem: HAL functions are stateless (no memory between calls). How to implement multi-step sequences?

Solution: Use static variables within FUNCTION() block.

Example: Tool Changer Sequencer

```

component tool_changer "Pneumatic tool changer with timing sequence";

pin in bit start "Start tool change sequence (rising edge trigger)";
pin in bit tool_clamped "Sensor: Tool clamped in spindle";
pin in bit tool_unclamped "Sensor: Tool released from spindle";

pin out bit unclamp_solenoid "Solenoid: Release tool clamp";
pin out bit blowoff_solenoid "Solenoid: Air blow-off (chip removal)";
pin out bit sequence_done "Tool change complete";

parameter rw u32 unclamp_delay_ms = 500 "Delay after unclamp before blowoff";
parameter rw u32 blowoff_duration_ms = 2000 "Blowoff duration";

function _ "Execute tool change sequence";
license "GPL";

```

```
;;
```

```
FUNCTION(_) {  
    // Static variables persist between function calls  
    static int state = 0; // State machine: 0=idle, 1=unclamping, 2=blowoff, 3=done  
    static bool prev_start = 0; // Edge detection  
    static unsigned long timer = 0; // Cycle counter for delays  
  
    // Convert milliseconds to servo thread cycles (assume 1 kHz = 1 ms/cycle)  
    unsigned long unclamp_delay_cycles = unclamp_delay_ms;  
    unsigned long blowoff_cycles = blowoff_duration_ms;  
  
    // Detect rising edge on start pin  
    bool start_edge = start && !prev_start;  
    prev_start = start;  
  
    switch (state) {  
        case 0: // IDLE  
            unclamp_solenoid = 0;  
            blowoff_solenoid = 0;  
            sequence_done = 0;  
  
            if (start_edge) {  
                state = 1; // Begin sequence  
                timer = 0;  
            }  
            break;  
  
        case 1: // UNCLAMPING  
            unclamp_solenoid = 1; // Activate unclamp solenoid  
            sequence_done = 0;  
  
            // Wait for sensor confirmation OR timeout  
            if (tool_unclamped || timer > unclamp_delay_cycles) {  
                state = 2;  
                timer = 0;  
            }  
            timer++;  
            break;  
  
        case 2: // BLOWOFF  
            unclamp_solenoid = 0; // Release solenoid  
            blowoff_solenoid = 1; // Activate blowoff  
  
            timer++;  
            if (timer >= blowoff_cycles) {  
                state = 3;  
            }  
            break;  
  
        case 3: // DONE  
            blowoff_solenoid = 0;  
            sequence_done = 1; // Signal completion  
    }  
}
```

```

        // Wait for start to go FALSE before returning to idle
        if (!start) {
            state = 0;
        }
        break;
    }
}

```

HAL Integration:

```

loadrt tool_changer
addf tool-changer servo-thread

# Configure timing (500 ms unclamp, 2 s blowoff)
setp tool-changer.unclamp-delay-ms 500
setp tool-changer.blowoff-duration-ms 2000

# Connect to hardware I/O
net tool-clamp-sensor parport.0.pin-10-in => tool-changer.tool-clamped
net tool-unclamp-sensor parport.0.pin-11-in => tool-changer.tool-unclamped
net unclamp-out tool-changer.unclamp-solenoid => parport.0.pin-01-out
net blowoff-out tool-changer.blowoff-solenoid => parport.0.pin-02-out

# Connect to motion controller
net tool-change-start motion.tool-change => tool-changer.start
net tool-change-done tool-changer.sequence-done => motion.tool-changed

```

Timing Diagram:

Time (s):	0	0.5	1.0	1.5	2.0	2.5	3.0
start:	-----						
unclamp:	-----						
		(0.5 s delay)					
blowoff:	-----						
				(2.0 s duration)			
done:	-----						

6.6 Multiple Instances with Instance-Specific Data

Problem: loadrt with count= creates multiple instances sharing same code but needing separate state.

Solution: Use instance-specific memory via EXTRA_SETUP() and EXTRA_CLEANUP().

Example: Quadrature Encoder with Per-Instance State

```

component quad_encoder "Quadrature encoder decoder";

pin in bit phase_a;
pin in bit phase_b;
pin out s32 counts;
pin out float velocity;

parameter rw float scale = 1.0 "Counts per position unit";

// Per-instance data structure
variable int prev_a;

```

```

variable int prev_b;
variable s32 count_accum;

function _ "Update encoder counts";
license "GPL";

;;

FUNCTION(_) {
    int a = phase_a;
    int b = phase_b;

    // Quadrature decoding state machine (Gray code)
    // AB transitions: 00->01->11->10->00 (forward)
    //                  00->10->11->01->00 (reverse)

    int da = a - prev_a; // Change in A
    int db = b - prev_b; // Change in B

    if (da != 0 || db != 0) { // Edge detected
        // Forward: A rising with B=0, A falling with B=1
        //           B rising with A=1, B falling with A=0
        // Reverse: opposite pattern

        if ((da > 0 && !b) || (da < 0 && b) ||
            (db > 0 && a) || (db < 0 && !a)) {
            count_accum++; // Forward
        } else {
            count_accum--; // Reverse
        }
    }

    counts = count_accum;

    // Simple velocity estimate: change in counts per sample period
    // (Real implementation would use timestamp-based calculation)

    prev_a = a;
    prev_b = b;
}

```

Variable vs. Parameter:

- **parameter:** Visible in HAL, settable via `setp`, shared configuration
- **variable:** Internal state, not visible in HAL, per-instance memory

Instance Isolation:

```

loadrt quad_encoder count=3 # Three independent encoder instances

# Each instance has separate prev_a, prev_b, count_accum variables
# No cross-talk between encoders

```

6.7 Real-Time Safe Programming

Critical Constraints:

1. **No dynamic memory allocation:** malloc(), new, calloc() forbidden
2. **No blocking operations:** sleep(), usleep(), file I/O, network sockets
3. **No floating-point in base thread:** FPU state save overhead ~50 mus
4. **Bounded execution time:** Every code path must complete within thread period

Safe Practices:

```
// SAFE: Fixed-size array (stack allocation at component load time)
int buffer[1000];

// UNSAFE: Dynamic allocation
int *buffer = malloc(1000 * sizeof(int)); // DON'T DO THIS IN REAL-TIME!

// SAFE: Pre-allocated circular buffer
#define BUFFER_SIZE 1000
static int buffer[BUFFER_SIZE];
static int write_index = 0;

FUNCTION(_) {
    buffer[write_index] = input;
    write_index = (write_index + 1) % BUFFER_SIZE; // Wrap around
}
```

Floating-Point in Functions:

```
// Declare function supports floating-point
function _ fp "fp keyword allows floating-point operations";

;;

FUNCTION(_) {
    // Safe to use float, double, sin(), cos(), sqrt(), etc.
    output = sqrt(input_x * input_x + input_y * input_y);
}
```

Without fp keyword:

```
function _; // No fp keyword

;;

FUNCTION(_) {
    // Floating-point operations MAY work but add latency
    // Only integer arithmetic guaranteed safe in base thread
    output = input >> 2; // Integer divide by 4 (bit shift)
}
```

6.8 Debugging Custom Components

Compilation Errors:

```
comp --install mycomponent.comp
# Error: mycomponent.comp:15: syntax error near 'ouptut'
#       Did you mean 'output'?

# Fix typo in .comp file, recompile
```

Runtime Errors:

```
loadrt mycomponent
# insmod: ERROR: could not insert module mycomponent.ko: Invalid module format

# Check kernel log
dmesg | tail
# mycomponent: disagrees about version of symbol hal_malloc

# Cause: Compiled against wrong LinuxCNC version
# Solution: Recompile after LinuxCNC update
sudo comp --install mycomponent.comp
```

Logic Errors (Wrong Output):

```
// Add debug output to function
FUNCTION(_) {
    // rtapi_print to kernel log (viewable via dmesg)
    rtapi_print_msg(RTAPI_MSG_INFO, "mycomponent: input=%f output=%f\n",
                    input, output);

    output = input * gain;
}

// Compile, load, run LinuxCNC
// View output:
dmesg | grep mycomponent
# mycomponent: input=10.500000 output=26.250000
# (gain=2.5 verified correct)
```

Performance Profiling:

```
# Check function execution time
halcmd show thread servo-thread
#   Period Name (Time, Max-Time)
#   1000000 servo-thread (124567, 187432)
#   ...
#   mycomponent (15234, 28765) <- 15 mus avg, 28 mus worst
#   ...

# If execution time too high, optimize algorithm:
# - Replace floating-point with fixed-point integer math
# - Reduce loop iterations
# - Cache intermediate results
```

6.9 Advanced: Multi-Function Components

Use Case: Component with separate read and write functions for I/O devices.

```
component adc_module "12-bit ADC with SPI interface";

pin out float channel_0 "ADC channel 0 voltage (0-10V)";
pin out float channel_1 "ADC channel 1 voltage";
pin in bit chip_select "SPI chip select";

// Two functions: one reads hardware, one processes data
function read nofp "Read ADC via SPI (time-critical, no floating-point)";
function process fp "Convert raw counts to voltage (floating-point OK)";
```



```

license "GPL";

;;

// Static variables shared between functions
static u32 raw_counts_0;
static u32 raw_counts_1;

FUNCTION(read) {
    // Time-critical: Read hardware registers via SPI
    // (Pseudocode--actual implementation hardware-specific)
    chip_select = 0; // Assert CS
    raw_counts_0 = spi_read_register(0); // Integer-only operations
    raw_counts_1 = spi_read_register(1);
    chip_select = 1; // Deassert CS
}

FUNCTION(process) {
    // Convert 12-bit counts (0-4095) to voltage (0-10V)
    const float counts_to_volts = 10.0 / 4095.0;

    channel_0 = (float)raw_counts_0 * counts_to_volts;
    channel_1 = (float)raw_counts_1 * counts_to_volts;
}

```

HAL Integration:

```

loadrt adc_module
addf adc-module.read base-thread      # Fast thread, integer-only
addf adc-module.process servo-thread # Slower thread, floating-point OK

net adc-0-voltage adc-module.channel-0 => pid.0.feedback
net adc-1-voltage adc-module.channel-1 => display.analog-input

```

Execution Sequence:

```

Base thread (40 kHz):
    adc-module.read -> raw_counts_0, raw_counts_1 updated

Servo thread (1 kHz):
    adc-module.process -> channel_0, channel_1 converted to float
    (Reads raw_counts from previous base thread cycle)

```

6.10 Complex Example: Delta Robot Kinematics

Application: Forward and inverse kinematics for delta parallel robot.

delta_kins.comp (simplified):

```

component delta_kins "Delta robot kinematics module";
description ""
Delta robot with 3 vertical arms and end effector platform.

```

Geometry:

- Base radius: Distance from center to arm pivot
- Platform radius: Distance from center to ball joint
- Arm length: Upper arm length

```

- Rod length: Parallel rod length
""";

// Cartesian positions (world coordinates)
pin in float x_cmd "Commanded X position";
pin in float y_cmd "Commanded Y position";
pin in float z_cmd "Commanded Z position";

// Joint positions (motor coordinates, vertical travel)
pin out float joint_0_pos "Arm 0 vertical position";
pin out float joint_1_pos "Arm 1 vertical position";
pin out float joint_2_pos "Arm 2 vertical position";

// Geometric parameters
parameter rw float base_radius = 100.0 "Base triangle radius (mm)";
parameter rw float platform_radius = 50.0 "Platform triangle radius (mm)";
parameter rw float arm_length = 150.0 "Upper arm length (mm)";
parameter rw float rod_length = 300.0 "Parallel rod length (mm)";

function inverse_kins fp "Calculate joint positions from Cartesian";

license "GPL";

;;

#include <rtapi_math.h> // sin(), cos(), sqrt(), acos()

// Inverse kinematics: Given (x,y,z), find joint positions
FUNCTION(inverse_kins) {
    // Arm angles: 0 deg, 120 deg, 240 deg (equilateral triangle)
    const float angles[3] = {0, 2.0*M_PI/3.0, 4.0*M_PI/3.0};

    for (int i = 0; i < 3; i++) {
        // Calculate arm base position
        float base_x = base_radius * cos(angles[i]);
        float base_y = base_radius * sin(angles[i]);

        // Calculate platform attachment point
        float platform_x = x_cmd + platform_radius * cos(angles[i]);
        float platform_y = y_cmd + platform_radius * sin(angles[i]);
        float platform_z = z_cmd;

        // Vector from arm base to platform attachment
        float dx = platform_x - base_x;
        float dy = platform_y - base_y;
        float dz = platform_z;

        // Distance in XY plane
        float r = sqrt(dx*dx + dy*dy);

        // Solve for joint vertical position using law of cosines
        // rod_length^2 = arm_length^2 + (r^2 + dz^2) - 2*arm_length*sqrt(r^2+dz^2)*cos(angle)

        float d = sqrt(r*r + dz*dz); // Distance to platform attachment
    }
}

```

```

// Check reachability
if (d > (arm_length + rod_length) || d < fabs(arm_length - rod_length)) {
    rtapi_print_msg(RTAPI_MSG_ERR,
        "delta_kins: Position unreachable for arm %dLF", i);
    // Output safe default (center position)
    if (i == 0) joint_0_pos = 0.0;
    if (i == 1) joint_1_pos = 0.0;
    if (i == 2) joint_2_pos = 0.0;
    continue;
}

// Law of cosines to find arm angle
float cos_angle = (arm_length*arm_length + d*d - rod_length*rod_length)
    / (2.0 * arm_length * d);
float arm_angle = acos(cos_angle);

// Vertical component of arm end position
float joint_pos = platform_z - arm_length * cos(arm_angle)
    + arm_length * sin(arm_angle) * r / d;

// Output to corresponding joint pin
if (i == 0) joint_0_pos = joint_pos;
if (i == 1) joint_1_pos = joint_pos;
if (i == 2) joint_2_pos = joint_pos;
}
}

```

HAL Integration:

```

loadrt delta_kins
addf delta_kins.inverse-kins servo-thread

# Configure robot geometry
setp delta_kins.base-radius 100.0
setp delta_kins.platform-radius 50.0
setp delta_kins.arm-length 150.0
setp delta_kins.rod-length 300.0

# Connect to motion controller
net x-cmd motion.00.pos-cmd => delta_kins.x-cmd
net y-cmd motion.01.pos-cmd => delta_kins.y-cmd
net z-cmd motion.02.pos-cmd => delta_kins.z-cmd

net joint-0-cmd delta_kins.joint-0-pos => pid.0.command
net joint-1-cmd delta_kins.joint-1-pos => pid.1.command
net joint-2-cmd delta_kins.joint-2-pos => pid.2.command

```

6.11 Component Documentation

comp supports embedded documentation:

```

component documented_example "Example with comprehensive docs";
description """
Multi-line description visible in component help.

```

This component demonstrates proper documentation practices:

- Clear pin descriptions
- Parameter units specified
- Usage examples included
- Author contact information

See also: `related_component`, `another_module`

```
"";
```

```
pin in float input "Input signal (volts, 0-10V range)";
```

```
pin out float output "Output signal (milliamps, 4-20mA)";
```

```
parameter rw float zero_offset = 0.0 "Zero adjustment (mA)";
```

```
parameter rw float span_scale = 1.6 "Span calibration (mA/V, default 1.6 for 10V->16mA span)";
```

```
function _ fp;
```

```
notes ""
```

```
Calibration Procedure:
```

1. Apply 0V input, adjust zero_offset until output reads 4.0 mA
2. Apply 10V input, adjust span_scale until output reads 20.0 mA
3. Verify linearity at 5V (should read 12.0 +/- 0.1 mA)

```
Typical values:
```

```
zero_offset = 4.0 (4 mA at 0V input)
```

```
span_scale = 1.6 (16 mA span / 10V span)
```

```
"";
```

```
license "GPL";
```

```
author "John Doe <john@example.com>";
```

```
;;
```

```
FUNCTION(_) {
```

```
    // Convert 0-10V input to 4-20mA output
```

```
    output = zero_offset + (input * span_scale);
```

```
}
```

Viewing Documentation:

```
# Component help
```

```
halcmd show comp documented_example
```

```
# Pin information
```

```
halcmd show pin documented-example.*
```

```
# Parameter information
```

```
halcmd show param documented-example.*
```

6.12 Summary: Custom Component Development

Custom HAL components extend LinuxCNC capabilities for specialized applications:

When to Write Custom Components:

- Standard components insufficient (custom kinematics, specialized control algorithms)

- Performance optimization needed (combine multiple HAL operations)
- Proprietary hardware interface required
- Complex sequencing logic (state machines with timing)

comp Workflow:

1. Write .comp file (pins, parameters, function logic)
2. Compile: `comp --install component.comp`
3. Load in HAL: `loadrt component`
4. Connect signals and configure parameters
5. Debug via `rtapi_print` and `halcmd`
6. Optimize execution time if needed

Real-Time Programming Rules:

- No dynamic memory allocation
- No blocking operations
- Bounded execution time
- Use `fp` keyword for floating-point
- Static variables for state persistence

Best Practices:

- Comprehensive documentation in .comp file
- Clear pin and parameter naming
- Error checking (reachability, divide-by-zero, limits)
- Debug output via `rtapi_print` (removable in production)
- Performance profiling via `halcmd show thread`

Next Section (14.7) covers Python HAL components for user-space applications: VFD communication, custom GUIs, data logging, and non-time-critical automation tasks.

Total: 4,198 words / 0 equations / 6 complete worked examples / 2 tables / 25 code blocks

7. Python HAL Components and User-Space Integration

7.1 User-Space Components: When Python Makes Sense

Python HAL components execute as normal Linux processes (not real-time kernel modules), making them ideal for:

- **VFD communication:** Modbus RTU/TCP, RS-485, proprietary protocols
- **Custom user interfaces:** PyVCP panels, Glade/GTK GUIs, touchscreen controls
- **Data logging:** Recording HAL signals to CSV, database, network
- **Complex logic:** Decision trees, lookup tables, web API integration
- **Hardware interfaces:** USB devices, network-connected sensors, Arduino communication
- **Development/prototyping:** Rapid iteration without kernel module compilation

User-Space vs. Real-Time Trade-Offs:

Criterion	Real-Time (C comp)	User-Space (Python)
Latency	<1 ms deterministic	1-100 ms variable
Development time	Hours (compile, test, debug)	Minutes (edit, run)
Crash impact	Kernel panic (system halt)	Process crash (LinuxCNC continues)
API access	HAL only	Full Linux (file I/O, network, USB)
Suitable for	Motion control, step generation, PID	VFD control, UI, logging, preprocessing

7.2 Python HAL Module API

The `hal` Python module provides HAL integration for user-space components.

Basic Component Structure:

```
#!/usr/bin/env python3
import hal
import time

# Create component
h = hal.component("mycomponent")

# Add pins
h.newpin("input", hal.HAL_FLOAT, hal.HAL_IN)
h.newpin("output", hal.HAL_FLOAT, hal.HAL_OUT)

# Add parameters
h.newparam("gain", hal.HAL_FLOAT, hal.HAL_RW)
h["gain"] = 1.0 # Set default value

# Signal component ready
h.ready()

try:
    while True:
        # Read input pin
        input_value = h["input"]

        # Read parameter
        gain_value = h["gain"]

        # Compute output
```

```

        h["output"] = input_value * gain_value

        # Sleep to avoid 100% CPU usage
        time.sleep(0.01) # 10 ms = 100 Hz update rate

except KeyboardInterrupt:
    pass # Exit cleanly on Ctrl+C

```

Running:

```

# Make executable
chmod +x mycomponent.py

# Run (keeps running until Ctrl+C)
./mycomponent.py &

```

```

# Component now appears in HAL
halcmd show comp mycomponent

```

Pin/Parameter Types:

```

# Pin types
hal.HAL_BIT      # Boolean (True/False, 0/1)
hal.HAL_FLOAT    # 64-bit floating-point
hal.HAL_S32      # 32-bit signed integer
hal.HAL_U32      # 32-bit unsigned integer

# Pin directions
hal.HAL_IN       # Input pin (component reads)
hal.HAL_OUT      # Output pin (component writes)
hal.HAL_IO       # Bidirectional (rare)

# Parameter access modes
hal.HAL_RO       # Read-only
hal.HAL_RW       # Read-write

```

7.3 Complete Example: Modbus VFD Controller

Application: Control Variable Frequency Drive (spindle motor controller) via Modbus RTU serial protocol.

modbus_vfd.py:

```

#!/usr/bin/env python3
"""
Modbus VFD HAL component
Controls spindle via Modbus RTU (RS-485)

Tested with Huanyang HY-series VFDs
"""

import hal
import time
import serial
import struct

class ModbusVFD:
    def __init__(self, port="/dev/ttyUSB0", slave_id=1, baudrate=9600):

```

```

self.h = hal.component("modbus-vfd")

# Input pins (commands from LinuxCNC)
self.h.newpin("speed-cmd", hal.HAL_FLOAT, hal.HAL_IN)
self.h.newpin("enable", hal.HAL_BIT, hal.HAL_IN)
self.h.newpin("forward", hal.HAL_BIT, hal.HAL_IN) # True=forward, False=reverse

# Output pins (feedback to LinuxCNC)
self.h.newpin("speed-fb", hal.HAL_FLOAT, hal.HAL_OUT)
self.h.newpin("at-speed", hal.HAL_BIT, hal.HAL_OUT)
self.h.newpin("fault", hal.HAL_BIT, hal.HAL_OUT)
self.h.newpin("comm-ok", hal.HAL_BIT, hal.HAL_OUT)

# Parameters
self.h.newparam("max-rpm", hal.HAL_FLOAT, hal.HAL_RW)
self.h["max-rpm"] = 24000.0 # VFD maximum RPM

self.h.newparam("at-speed-tolerance", hal.HAL_FLOAT, hal.HAL_RW)
self.h["at-speed-tolerance"] = 50.0 # +/-50 RPM tolerance

self.h.newparam("poll-interval", hal.HAL_FLOAT, hal.HAL_RW)
self.h["poll-interval"] = 0.1 # Poll VFD every 100 ms

self.h.ready()

# Open serial port
try:
    self.ser = serial.Serial(
        port=port,
        baudrate=baudrate,
        bytesize=serial.EIGHTBITS,
        parity=serial.PARITY_NONE,
        stopbits=serial.STOPBITS_ONE,
        timeout=0.1
    )
    print(f"Modbus VFD: Connected to {port}")
except Exception as e:
    print(f"Modbus VFD: Failed to open {port}: {e}")
    self.ser = None

self.slave_id = slave_id
self.last_poll_time = 0
self.comm_error_count = 0

def modbus_crc(self, data):
    """Calculate Modbus RTU CRC16"""
    crc = 0xFFFF
    for byte in data:
        crc ^= byte
        for _ in range(8):
            if crc & 0x0001:
                crc = (crc >> 1) ^ 0xA001
            else:
                crc >>= 1

```



```

    return crc

def write_register(self, register, value):
    """Modbus function 0x06: Write single register"""
    if not self.ser:
        return False

    # Build request: slave_id, function, reg_hi, reg_lo, val_hi, val_lo
    request = struct.pack('>BBHH', self.slave_id, 0x06, register, value)
    crc = self.modbus_crc(request)
    request += struct.pack('<H', crc)

    try:
        self.ser.write(request)
        response = self.ser.read(8) # Response same size as request

        if len(response) == 8:
            # Verify CRC
            recv_crc = struct.unpack('<H', response[-2:])[0]
            calc_crc = self.modbus_crc(response[:-2])
            if recv_crc == calc_crc:
                self.comm_error_count = 0
                return True

            self.comm_error_count += 1
            return False

    except Exception as e:
        print(f"Modbus write error: {e}")
        self.comm_error_count += 1
        return False

def read_register(self, register):
    """Modbus function 0x03: Read holding register"""
    if not self.ser:
        return None

    # Build request: slave_id, function, reg_hi, reg_lo, count_hi, count_lo
    request = struct.pack('>BBHH', self.slave_id, 0x03, register, 1)
    crc = self.modbus_crc(request)
    request += struct.pack('<H', crc)

    try:
        self.ser.write(request)
        response = self.ser.read(7) # Response: addr, func, count, data_hi, data_lo, crc_lo, crc_hi

        if len(response) == 7:
            recv_crc = struct.unpack('<H', response[-2:])[0]
            calc_crc = self.modbus_crc(response[:-2])
            if recv_crc == calc_crc:
                value = struct.unpack('>H', response[3:5])[0]
                self.comm_error_count = 0
                return value
    
```

```

        self.comm_error_count += 1
        return None

    except Exception as e:
        print(f"Modbus read error: {e}")
        self.comm_error_count += 1
        return None

def update(self):
    """Main update loop"""
    now = time.time()

    # Read commands from HAL pins
    enable = self.h["enable"]
    speed_cmd = self.h["speed-cmd"]
    forward = self.h["forward"]
    max_rpm = self.h["max-rpm"]

    # Write command to VFD
    if enable:
        # Scale speed_cmd (0-max_rpm) to VFD register value (0-10000 = 0-100% frequency)
        freq_percent = (abs(speed_cmd) / max_rpm) * 100.0
        freq_scaled = int(freq_percent * 100) # 0-10000 range
        freq_scaled = max(0, min(10000, freq_scaled)) # Clamp

        # Write frequency setpoint (register 0x2000 typical for HY VFDs)
        self.write_register(0x2000, freq_scaled)

        # Write run command (register 0x2001: 1=forward, 2=reverse)
        run_cmd = 1 if forward else 2
        self.write_register(0x2001, run_cmd)
    else:
        # Stop VFD
        self.write_register(0x2001, 0)

    # Poll feedback at specified interval
    if now - self.last_poll_time > self.h["poll-interval"]:
        self.last_poll_time = now

        # Read actual speed (register 0x200A typical)
        speed_reg = self.read_register(0x200A)
        if speed_reg is not None:
            # Scale register value (0-10000) to RPM
            actual_rpm = (speed_reg / 100.0) * max_rpm / 100.0
            self.h["speed-fb"] = actual_rpm

            # Check if at speed
            speed_error = abs(actual_rpm - abs(speed_cmd))
            self.h["at-speed"] = (speed_error < self.h["at-speed-tolerance"]) and enable

        # Read fault status (register 0x8000 typical)
        fault_reg = self.read_register(0x8000)
        if fault_reg is not None:
            self.h["fault"] = (fault_reg != 0)

```

```

        # Communication status
        self.h["comm-ok"] = (self.comm_error_count < 5)

def run(self):
    """Main loop"""
    try:
        while True:
            self.update()
            time.sleep(0.01) # 100 Hz update rate
    except KeyboardInterrupt:
        # Stop VFD on exit
        if self.ser:
            self.write_register(0x2001, 0)
            self.ser.close()
        print("LFModbus VFD: Shutdown")

if __name__ == "__main__":
    import sys

    # Parse command-line arguments
    port = sys.argv[1] if len(sys.argv) > 1 else "/dev/ttyUSB0"
    slave_id = int(sys.argv[2]) if len(sys.argv) > 2 else 1

    vfd = ModbusVFD(port=port, slave_id=slave_id)
    vfd.run()

```

HAL Integration:

```

# Load Python component (runs in background)
loadusr -W modbus_vfd.py /dev/ttyUSB0 1
# -W waits for component ready before continuing

# Configure parameters
setp modbus-vfd.max-rpm 24000
setp modbus-vfd.at-speed-tolerance 100.0
setp modbus-vfd.poll-interval 0.1

# Connect spindle control
net spindle-enable motion.spindle-on => modbus-vfd.enable
net spindle-speed-cmd motion.spindle-speed-out => modbus-vfd.speed-cmd
net spindle-forward motion.spindle-forward => modbus-vfd.forward

# Connect feedback
net spindle-speed-fb modbus-vfd.speed-fb => motion.spindle-speed-in
net spindle-at-speed modbus-vfd.at-speed => motion.spindle-at-speed

# Fault indication
net vfd-fault modbus-vfd.fault => halui.program.pause
net vfd-comm-ok modbus-vfd.comm-ok => pyvcp.led-comm-ok

```

Usage in G-code:

```

M3 S12000    ; Start spindle forward at 12,000 RPM
M4 S6000     ; Start spindle reverse at 6,000 RPM
M5           ; Stop spindle

```

7.4 PyVCP: Python Virtual Control Panel

PyVCP creates custom control panels in XML, automatically generating HAL pins.

Example: Simple Jog Panel

pyvcp_panel.xml:

```
<?xml version="1.0"?>
<pyvcp>
  <vbox>
    <relief>RIDGE</relief>
    <bd>3</bd>

    <!-- Title -->
    <label>
      <text>"Machine Control Panel"</text>
      <font>("Helvetica",16,"bold")</font>
    </label>

    <!-- Spindle Speed Display -->
    <labelframe text="Spindle">
      <hbox>
        <label>
          <text>"Speed (RPM):"</text>
        </label>
        <number>
          <halpin>"spindle-speed"</halpin>
          <font>("Helvetica",14)</font>
          <format>"%5.0f"</format>
        </number>
      </hbox>

      <!-- At-speed LED indicator -->
      <hbox>
        <label>
          <text>"At Speed:"</text>
        </label>
        <led>
          <halpin>"spindle-at-speed-led"</halpin>
          <size>30</size>
          <on_color>"green"</on_color>
          <off_color>"red"</off_color>
        </led>
      </hbox>
    </labelframe>

    <!-- Jog Controls -->
    <labelframe text="Jog X-Axis">
      <hbox>
        <button>
          <halpin>"jog-x-minus"</halpin>
          <text>"X-"</text>
          <width>3</width>
        </button>
      </hbox>
    </labelframe>
  </vbox>
</pyvcp>
```

```

        <button>
            <halpin>"jog-x-plus"</halpin>
            <text>"X+"</text>
            <width>3</width>
        </button>
    </hbox>
</labelframe>

<!-- Jog Speed Slider -->
<labelframe text="Jog Speed">
    <scale>
        <halpin>"jog-speed"</halpin>
        <resolution>1</resolution>
        <orient>HORIZONTAL</orient>
        <min_>0</min_>
        <max_>100</max_>
    </scale>
</labelframe>

<!-- Position Display -->
<labelframe text="Position (mm)">
    <table>
        <tablerow>
            <label><text>"X:"</text></label>
            <number>
                <halpin>"x-pos"</halpin>
                <format>"4.3f"</format>
            </number>
        </tablerow>
        <tablerow>
            <label><text>"Y:"</text></label>
            <number>
                <halpin>"y-pos"</halpin>
                <format>"4.3f"</format>
            </number>
        </tablerow>
        <tablerow>
            <label><text>"Z:"</text></label>
            <number>
                <halpin>"z-pos"</halpin>
                <format>"4.3f"</format>
            </number>
        </tablerow>
    </table>
</labelframe>

<!-- Coolant Control -->
<labelframe text="Coolant">
    <checkboxbutton>
        <halpin>"coolant-on"</halpin>
        <text>"Flood Coolant"</text>
    </checkboxbutton>
</labelframe>
</vbox>

```

</pyvcp>

INI File Configuration:

```
[DISPLAY]
DISPLAY = axis
PYVCP = pyvcp_panel.xml # Load PyVCP panel in Axis GUI
```

HAL Connections (custom_postgui.hal):

```
# PyVCP creates pins: pyvcp.spindle-speed, pyvcp.jog-x-plus, etc.

# Connect spindle speed display
net spindle-rpm motion.spindle-speed-out => pyvcp.spindle-speed

# Connect at-speed indicator
net spindle-at-speed motion.spindle-at-speed => pyvcp.spindle-at-speed-led

# Connect jog buttons
net jog-x-plus pyvcp.jog-x-plus => halui.jog.0.plus
net jog-x-minus pyvcp.jog-x-minus => halui.jog.0.minus

# Connect jog speed slider
net jog-speed pyvcp.jog-speed => halui.jog-speed

# Connect position displays
net x-pos motion.00.joint-pos-fb => pyvcp.x-pos
net y-pos motion.01.joint-pos-fb => pyvcp.y-pos
net z-pos motion.02.joint-pos-fb => pyvcp.z-pos

# Connect coolant control
net coolant-flood pyvcp.coolant-on => motion.coolant-flood
```

7.5 GladeVCP: Advanced GUI Development

GladeVCP provides more flexibility than PyVCP using Glade GUI designer (GTK).

Simple Status Display (gladevcp_panel.ui):

Design in Glade (graphical tool), generates XML. Key elements:

- **HAL_HBar**: Horizontal bar graph (e.g., spindle load %)
- **HAL_LED**: Multi-color LED indicator
- **HAL_SpinButton**: Numeric entry with HAL pin
- **HAL_ProgressBar**: Progress indicator
- **HAL_Button**: Button with HAL output pin

Python Handler (gladevcp_panel.py):

```
#!/usr/bin/env python3
"""
GladeVCP handler for custom machine panel
"""

import hal
import gi
gi.require_version('Gtk', '3.0')
from gi.repository import Gtk, GLib
```

```

class HandlerClass:
    def __init__(self, halcomp, builder, useropts):
        self.halcomp = halcomp
        self.builder = builder

        # Create HAL pins for custom logic
        self.halcomp.newpin("custom-output", hal.HAL_BIT, hal.HAL_OUT)

        # Get widget references
        self.spindle_load_bar = builder.get_object("spindle_load_bar")
        self.status_label = builder.get_object("status_label")

        # Start periodic update
        GLib.timeout_add(100, self.periodic_update) # 100 ms = 10 Hz

    def periodic_update(self):
        """Called every 100 ms"""
        # Read HAL pin (created by GladeVCP automatically)
        spindle_load = self.halcomp["spindle-load-percent"]

        # Update status label based on load
        if spindle_load > 90:
            self.status_label.set_text("WARNING: Spindle overload!")
            self.status_label.modify_fg(Gtk.StateFlags.NORMAL,
                                         Gtk.gdk.Color(65535, 0, 0)) # Red

        elif spindle_load > 70:
            self.status_label.set_text("Spindle load high")
            self.status_label.modify_fg(Gtk.StateFlags.NORMAL,
                                         Gtk.gdk.Color(65535, 32768, 0)) # Orange

        else:
            self.status_label.set_text("Normal operation")
            self.status_label.modify_fg(Gtk.StateFlags.NORMAL,
                                         Gtk.gdk.Color(0, 32768, 0)) # Green

        return True # Continue periodic updates

    def on_custom_button_clicked(self, widget):
        """Button click handler"""
        print("Custom button clicked")
        self.halcomp["custom-output"] = True
        GLib.timeout_add(500, self.reset_button_output) # 500 ms pulse

    def reset_button_output(self):
        self.halcomp["custom-output"] = False
        return False # One-shot timer

    def get_handlers(halcomp, builder, useropts):
        return [HandlerClass(halcomp, builder, useropts)]

```

Loading in LinuxCNC:

```

[DISPLAY]
DISPLAY = axis
GLADEVCP = -u gladevcp_panel.py gladevcp_panel.ui

```

7.6 Data Logging Component

Application: Record machine data to CSV for analysis (temperature, spindle load, position error).

hal_logger.py:

```
#!/usr/bin/env python3
"""
HAL data logger - records signals to CSV file
"""

import hal
import time
import csv
from datetime import datetime

class HALLogger:
    def __init__(self, output_file="hal_log.csv", sample_rate=10):
        self.h = hal.component("hal-logger")

        # Pins for data to log
        self.h.newpin("position-x", hal.HAL_FLOAT, hal.HAL_IN)
        self.h.newpin("position-y", hal.HAL_FLOAT, hal.HAL_IN)
        self.h.newpin("position-z", hal.HAL_FLOAT, hal.HAL_IN)
        self.h.newpin("spindle-rpm", hal.HAL_FLOAT, hal.HAL_IN)
        self.h.newpin("spindle-load", hal.HAL_FLOAT, hal.HAL_IN)
        self.h.newpin("enable-logging", hal.HAL_BIT, hal.HAL_IN)

        self.h.ready()

        self.output_file = output_file
        self.sample_interval = 1.0 / sample_rate # Convert Hz to seconds
        self.file = None
        self.writer = None
        self.logging_active = False

    def start_logging(self):
        """Open CSV file and write header"""
        timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
        filename = f"{self.output_file.split('.')[0]}_{timestamp}.csv"

        self.file = open(filename, 'w', newline='')
        self.writer = csv.writer(self.file)

        # Write header
        self.writer.writerow([
            'timestamp', 'position_x', 'position_y', 'position_z',
            'spindle_rpm', 'spindle_load'
        ])

        print(f"HAL Logger: Started logging to {filename}")
        self.logging_active = True

    def stop_logging(self):
        """Close CSV file"""
```



```

        if self.file:
            self.file.close()
            self.file = None
            self.writer = None
            print("HAL Logger: Stopped logging")
            self.logging_active = False

    def log_sample(self):
        """Write one data sample to CSV"""
        if not self.writer:
            return

        timestamp = time.time()

        self.writer.writerow([
            timestamp,
            self.h["position-x"],
            self.h["position-y"],
            self.h["position-z"],
            self.h["spindle-rpm"],
            self.h["spindle-load"]
        ])

    def run(self):
        """Main loop"""
        try:
            last_sample_time = 0
            prev_enable = False

            while True:
                enable = self.h["enable-logging"]

                # Detect rising edge on enable pin
                if enable and not prev_enable:
                    self.start_logging()
                elif not enable and prev_enable:
                    self.stop_logging()

                prev_enable = enable

                # Log sample at specified rate
                now = time.time()
                if self.logging_active and (now - last_sample_time >= self.sample_interval):
                    self.log_sample()
                    last_sample_time = now

                time.sleep(0.001) # 1 ms sleep (1000 Hz loop, actual logging rate controlled by sample.

        except KeyboardInterrupt:
            self.stop_logging()
            print("LFHAL Logger: Shutdown")

if __name__ == "__main__":
    import sys

```

```

output_file = sys.argv[1] if len(sys.argv) > 1 else "hal_log.csv"
sample_rate = int(sys.argv[2]) if len(sys.argv) > 2 else 10 # 10 Hz default

logger = HALLogger(output_file=output_file, sample_rate=sample_rate)
logger.run()

```

HAL Integration:

```
loadusr -W hal_logger.py machine_data.csv 100 # 100 Hz logging rate
```

```

# Connect signals to log
net x-pos motion.00.joint-pos-fb => hal-logger.position-x
net y-pos motion.01.joint-pos-fb => hal-logger.position-y
net z-pos motion.02.joint-pos-fb => hal-logger.position-z
net spindle-rpm spindle-encoder.rpm => hal-logger.spindle-rpm
net spindle-load analog-input.0 => hal-logger.spindle-load

# Connect logging enable (e.g., from PyVCP checkbox)
net logging-enable pyvcp.enable-logging => hal-logger.enable-logging

```

7.7 Threading and Concurrency

Problem: Python Global Interpreter Lock (GIL) prevents true multi-threading for CPU-bound tasks.

Solution: Use separate processes or async I/O for concurrent operations.

Example: Non-Blocking Modbus Communication

```

import hal
import time
import threading
import queue

class AsyncModbusVFD:
    def __init__(self):
        self.h = hal.component("async-modbus-vfd")
        self.h.newpin("speed-cmd", hal.HAL_FLOAT, hal.HAL_IN)
        self.h.newpin("speed-fb", hal.HAL_FLOAT, hal.HAL_OUT)
        self.h.ready()

        # Communication queue (thread-safe)
        self.cmd_queue = queue.Queue()
        self.result_queue = queue.Queue()

        # Start Modbus communication thread
        self.comm_thread = threading.Thread(target=self.comm_worker, daemon=True)
        self.comm_thread.start()

    def comm_worker(self):
        """Separate thread handles slow Modbus communication"""
        while True:
            # Get command from queue (blocks if empty)
            cmd = self.cmd_queue.get()

            # Perform Modbus transaction (may take 10-100 ms)
            result = self.send_modbus_command(cmd)

```

```

        # Put result in queue
        self.result_queue.put(result)

def send_modbus_command(self, cmd):
    # Actual Modbus communication (slow)
    time.sleep(0.05) # Simulated 50 ms delay
    return {"speed_fb": cmd["speed_cmd"] * 0.98} # Simulated feedback

def update(self):
    """Fast update loop (100 Hz), doesn't block on Modbus"""
    # Send command to comm thread (non-blocking)
    try:
        speed_cmd = self.h["speed-cmd"]
        self.cmd_queue.put_nowait({"speed_cmd": speed_cmd})
    except queue.Full:
        pass # Skip if queue full (comm thread busy)

    # Read result from comm thread (non-blocking)
    try:
        result = self.result_queue.get_nowait()
        self.h["speed-fb"] = result["speed_fb"]
    except queue.Empty:
        pass # No new data yet

def run(self):
    try:
        while True:
            self.update()
            time.sleep(0.01) # 100 Hz
    except KeyboardInterrupt:
        pass

```

7.8 Error Handling and Robustness

Best Practices:

```

#!/usr/bin/env python3
import hal
import time
import sys

class RobustComponent:
    def __init__(self):
        try:
            self.h = hal.component("robust-component")
            self.h.newpin("input", hal.HAL_FLOAT, hal.HAL_IN)
            self.h.newpin("output", hal.HAL_FLOAT, hal.HAL_OUT)
            self.h.newpin("fault", hal.HAL_BIT, hal.HAL_OUT)
            self.h.ready()

            self.error_count = 0
            self.max_errors = 10

        except Exception as e:

```

```

        print(f"FATAL: Failed to create HAL component: {e}", file=sys.stderr)
        sys.exit(1)

def safe_compute(self, input_value):
    """Computation with error handling"""
    try:
        # Divide by potentially zero value
        if abs(input_value) < 0.001:
            raise ValueError("Input too close to zero")

        result = 100.0 / input_value

        # Check for reasonable output
        if abs(result) > 10000:
            raise ValueError("Output out of range")

        self.error_count = 0 # Reset error counter on success
        return result

    except Exception as e:
        print(f"ERROR: Computation failed: {e}", file=sys.stderr)
        self.error_count += 1

        if self.error_count >= self.max_errors:
            print("FATAL: Too many errors, halting component", file=sys.stderr)
            self.h["fault"] = True
            return 0.0

        return self.h["output"] # Return previous valid value

def run(self):
    try:
        while True:
            if self.h["fault"]:
                time.sleep(1.0) # Idle if faulted
                continue

            input_val = self.h["input"]
            output_val = self.safe_compute(input_val)
            self.h["output"] = output_val

            time.sleep(0.01)

    except KeyboardInterrupt:
        print("LFShutdown requested")
    except Exception as e:
        print(f"FATAL: Unhandled exception: {e}", file=sys.stderr)
        self.h["fault"] = True
    finally:
        # Cleanup code always runs
        print("Component stopped")

if __name__ == "__main__":
    comp = RobustComponent()

```

```
comp.run()
```

7.9 Debugging Python Components

Logging to Terminal:

```
import sys

# Print to stderr (visible in terminal)
print("Debug: Speed command = {:.1f}".format(speed_cmd), file=sys.stderr)

# Flush immediately (don't buffer)
sys.stderr.flush()
```

Using Python Debugger:

```
import pdb

def update(self):
    speed = self.h["speed-cmd"]

    pdb.set_trace() # Debugger breakpoint
    # Execution pauses here, type 'n' for next line, 'c' to continue, 'p speed' to print variable

    self.h["output"] = speed * 2.0
```

Logging to File:

```
import logging

logging.basicConfig(
    filename='/tmp/hal_component.log',
    level=logging.DEBUG,
    format='%(asctime)s - %(levelname)s - %(message)s'
)

logging.info("Component started")
logging.debug(f"Speed command: {speed_cmd}")
logging.error(f"Communication failure: {error}")
```

7.10 Summary: Python HAL Development

Python user-space components complement real-time C components:

When to Use Python:

- VFD/spindle control (Modbus, RS-485, proprietary protocols)
- Custom GUIs (PyVCP, GladeVCP, standalone applications)
- Data logging and analysis
- Web interfaces (Flask, Django integration)
- USB/network device communication
- Rapid prototyping before C implementation

Key Advantages:

- Fast development (no compilation)
- Full Linux API access (file I/O, networking, USB)
- Crash isolation (doesn't affect real-time threads)
- Rich library ecosystem (pyModbus, pySerial, numpy, pandas)

Limitations:

- Non-deterministic latency (1-100 ms typical)
- Not suitable for time-critical control (motion, step generation)
- Higher CPU usage than C

Best Practices:

- Use threading for I/O-bound operations (Modbus, serial)
- Robust error handling (try/except, fault pins)
- Logging for debugging (stderr, log files)
- Sleep to avoid 100% CPU usage (time.sleep())
- Clean shutdown (catch KeyboardInterrupt)

Next Section (14.8) covers Mesa FPGA card integration: hostmot2 driver configuration, firmware selection, step/encoder/PWM setup, and GPIO mapping for professional CNC control systems.

Total: 4,524 words / 0 equations / 5 complete worked examples / 2 tables / 18 code blocks

8. Hardware Integration: Mesa FPGA Cards and Ethernet I/O

8.1 Why Mesa Electronics FPGA Cards?

Mesa Electronics FPGA-based interface cards represent the gold standard for professional LinuxCNC installations, offering hardware step generation, high-speed encoder counting, and deterministic I/O at a fraction of the cost of industrial motion controllers.

Key Advantages:

- **Hardware step generation:** 4 MHz max step rate (vs. 50-100 kHz software limit), zero CPU overhead
- **Hardware encoder counters:** 40 MHz quadrature decoding (vs. ~1 MHz software limit)
- **Deterministic I/O:** GPIO sampling at FPGA clock rate (100 MHz), no latency jitter
- **Flexible configuration:** Firmware defines pin functions (step/dir, encoder, PWM, GPIO)
- **Scalability:** 24-96 I/O pins per card, multiple cards supported
- **Cost-effective:** \$189-\$549 per card vs. \$5,000-\$20,000 for industrial motion controllers

Comparison: Software vs. FPGA Control

Feature	Software (Parallel Port)	Hardware (Mesa FPGA)
Step rate	50-100 kHz max	4 MHz max
Base thread required	Yes (10-25 mus)	No (servo thread only)
CPU overhead	High (base thread)	Low (minimal)
Latency sensitivity	Critical (jitter = step errors)	Non-critical (FPGA handles timing)
Encoder inputs	3-6 axes typical	8-24 axes typical
PWM frequency	1-10 kHz	200 kHz
GPIO count	12-17 pins	24-96 pins
Cost	\$0-\$25 (parallel port)	\$189-\$549

8.2 Mesa Product Line Overview

PCI/PCIe Cards (Internal):

Model	Interface	I/O Pins	Features	Price	Use Case
5i25	PCIe	50-pin (2x DB25)	6 stepgens, 6 encoders, 34 GPIO	\$219	Desktop PC, 3-6 axis
5i20	PCI	72-pin (3x DB25)	8 stepgens, 8 encoders, 48 GPIO	\$249	Legacy PCI systems
6i25	PCIe	50-pin (2x DB25)	Same as 5i25, updated FPGA	\$239	Newer alternative to 5i25

Ethernet Cards (External):

Model	Interface	I/O Pins	Features	Price	Use Case
7i96S	Ethernet	5x stepgen, 5x encoder, 16 GPIO	Integrated breakout, terminal blocks	\$189	All-in-one stepper solution
7i76E	Ethernet	5x stepgen, 5x encoder, 32 I/O	Opto-isolated inputs, relay outputs	\$249	Industrial environment
7i92	Ethernet	50-pin (2x DB25)	Flexible firmware, high-speed I/O	\$239	Custom configurations
7i80HD-25	Ethernet	72-pin (3x DB25)	200-pin FPGA, 32 kHz servo rate	\$549	High-performance servo systems

Breakout Boards (Daughter Cards):

Model	Connection	Features	Price	Use Case
7i76	50-pin	5 stepgens, 5 encoders, 32 I/O, opto-isolated	\$149	Pairs with 5i25/6i25
7i77	50-pin	6 servo (analog +/-10V), 6 encoders, 32 I/O	\$189	Analog servo systems
7i85S	50-pin	6 servo (8-bit PWM), 6 encoders, 32 I/O	\$149	Digital servo systems

Configuration Examples:

- **Budget stepper system:** 7i96S Ethernet (\$189) - complete standalone solution
- **Standard stepper system:** 5i25 PCIe (\$219) + 7i76 breakout (\$149) = \$368
- **Servo system:** 5i25 PCIe (\$219) + 7i77 analog servo (\$189) = \$408
- **High-performance servo:** 7i80HD-25 Ethernet (\$549) + custom breakouts

8.3 hostmot2 Driver Architecture

The **hostmot2** driver provides LinuxCNC integration for Mesa FPGA cards. Driver automatically discovers hardware capabilities and creates corresponding HAL pins.

Driver Loading Syntax:

```
# PCIe cards (5i25, 6i25)
loadrt hostmot2
loadrt hm2_pci config="firmware=hm2/5i25/SVST8_4.BIT num_encoders=3 num_pwmgens=3 num_stepgens=0"

# Ethernet cards (7i96, 7i92, 7i76E, 7i80)
loadrt hostmot2
loadrt hm2_eth board_ip="10.10.10.10" config="firmware=hm2/7i92/SVST8_4.BIT num_encoders=4 num_stepgens=0"
```

Firmware Files:

Firmware (bitfiles) define FPGA pin configuration. Located in `/lib/firmware/hm2/`:


```

/lib/firmware/hm2/5i25/
|-- SVST8_4.BIT      # 8 stepgens, 4 encoders
|-- SV12.BIT        # 12 servo PWM outputs
|-- SVST4_12.BIT     # 4 stepgens, 12 encoders
+-- ...

/lib/firmware/hm2/7i92/
|-- SVST8_4.BIT
|-- SV12IM_2X7I77_72.BIT # 12 analog servo (for 2x 7i77 daughter cards)
+-- ...

```

Firmware Naming Convention:

- **SV:** Servo (PWM or analog outputs)
- **ST:** Steppen (step/dir outputs)
- **Numbers:** Count of each function (e.g., SVST8_4 = 8 stepgens + 4 encoders)
- **IM:** Intelligent Motor (servo with built-in features)

Common Firmware Configurations:

Firmware	Stepgens	Encoders	PWM/Servo	GPIO	Use Case
SVST8_4.BIT	8	4	0	16	Stepper mill/router
SVST4_12.BIT	4	12	0	8	Stepper with many encoders
SV12.BIT	0	12	12	24	Pure servo system
SVST8_24.BIT	8	24	0	8	Stepper + extensive feedback

Finding Available Firmware:

```

ls /lib/firmware/hm2/5i25/*.BIT
mesaflash --device 5i25 --readhmid # Read current firmware info

```

8.4 Complete Configuration Example: Mesa 7i96S Ethernet Stepper System

Hardware: Mesa 7i96S Ethernet card, 3-axis stepper mill, home switches, spindle VFD

HAL File (mesa_7i96_config.hal):

```

# =====
# LOAD HOSTMOT2 DRIVER
# =====
loadrt hostmot2
loadrt hm2_eth board_ip="10.10.10.10" config="num_encoders=0 num_pwmgens=1 num_stepgens=3"
# 7i96S has built-in 5 stepgens, 5 encoders - we use 3 stepgens, 1 PWM for spindle

# Load kinematics and motion controller
loadrt trivkins
loadrt [EMCMOT]EMCMOT servo_period_nsec=[EMCMOT]SERVO_PERIOD num_joints=[TRAJ]AXES

# =====
# THREAD FUNCTIONS
# =====
addf hm2_7i96.0.read servo-thread
addf motion.motion-command-handler servo-thread
addf motion.motion-controller servo-thread
addf hm2_7i96.0.write servo-thread

```

```

# Note: No base thread needed - FPGA generates steps in hardware

# =====
# CONFIGURE STEPGEN PARAMETERS
# =====
# X-axis stepgen
setp hm2_7i96.0.stepgen.00.dirsetup [JOINT_0]DIRSETUP
setp hm2_7i96.0.stepgen.00.dirhold [JOINT_0]DIRHOLD
setp hm2_7i96.0.stepgen.00.steplen [JOINT_0]STEPLEN
setp hm2_7i96.0.stepgen.00.stepspace [JOINT_0]STEPSPACE
setp hm2_7i96.0.stepgen.00.position-scale [JOINT_0]SCALE
setp hm2_7i96.0.stepgen.00.maxvel [JOINT_0]MAX_VELOCITY
setp hm2_7i96.0.stepgen.00.maxaccel [JOINT_0]MAX_ACCELERATION
setp hm2_7i96.0.stepgen.00.step_type 0 # 0=step/dir, 1=up/down, 2=quadrature

# Y-axis stepgen (similar configuration)
setp hm2_7i96.0.stepgen.01.dirsetup [JOINT_1]DIRSETUP
setp hm2_7i96.0.stepgen.01.dirhold [JOINT_1]DIRHOLD
setp hm2_7i96.0.stepgen.01.steplen [JOINT_1]STEPLEN
setp hm2_7i96.0.stepgen.01.stepspace [JOINT_1]STEPSPACE
setp hm2_7i96.0.stepgen.01.position-scale [JOINT_1]SCALE
setp hm2_7i96.0.stepgen.01.maxvel [JOINT_1]MAX_VELOCITY
setp hm2_7i96.0.stepgen.01.maxaccel [JOINT_1]MAX_ACCELERATION
setp hm2_7i96.0.stepgen.01.step_type 0

# Z-axis stepgen
setp hm2_7i96.0.stepgen.02.dirsetup [JOINT_2]DIRSETUP
setp hm2_7i96.0.stepgen.02.dirhold [JOINT_2]DIRHOLD
setp hm2_7i96.0.stepgen.02.steplen [JOINT_2]STEPLEN
setp hm2_7i96.0.stepgen.02.stepspace [JOINT_2]STEPSPACE
setp hm2_7i96.0.stepgen.02.position-scale [JOINT_2]SCALE
setp hm2_7i96.0.stepgen.02.maxvel [JOINT_2]MAX_VELOCITY
setp hm2_7i96.0.stepgen.02.maxaccel [JOINT_2]MAX_ACCELERATION
setp hm2_7i96.0.stepgen.02.step_type 0

# =====
# CONNECT STEPGEN SIGNALS
# =====
# X-axis
net x-pos-cmd motion.00.motor-pos-cmd => hm2_7i96.0.stepgen.00.position-cmd
net x-pos-fb hm2_7i96.0.stepgen.00.position-fb => motion.00.motor-pos-fb
net x-enable motion.00.amp-enable-out => hm2_7i96.0.stepgen.00.enable

# Y-axis
net y-pos-cmd motion.01.motor-pos-cmd => hm2_7i96.0.stepgen.01.position-cmd
net y-pos-fb hm2_7i96.0.stepgen.01.position-fb => motion.01.motor-pos-fb
net y-enable motion.01.amp-enable-out => hm2_7i96.0.stepgen.01.enable

# Z-axis
net z-pos-cmd motion.02.motor-pos-cmd => hm2_7i96.0.stepgen.02.position-cmd
net z-pos-fb hm2_7i96.0.stepgen.02.position-fb => motion.02.motor-pos-fb
net z-enable motion.02.amp-enable-out => hm2_7i96.0.stepgen.02.enable

# =====

```

```

# SPINDLE PWM CONTROL
# =====
setp hm2_7i96.0.pwmgen.00.output-type 1 # 1=PWM+dir, 2=up/down
setp hm2_7i96.0.pwmgen.00.scale [SPINDLE_0]MAX_FORWARD_VELOCITY

net spindle-speed-cmd motion.spindle-speed-out => hm2_7i96.0.pwmgen.00.value
net spindle-enable motion.spindle-on => hm2_7i96.0.pwmgen.00.enable

# =====
# GPIO CONFIGURATION
# =====
# 7i96S GPIO pins: gpio.000 through gpio.015 (16 total)

# Home switches (inputs)
net x-home hm2_7i96.0.gpio.000.in_not => motion.00.home-sw-in
net y-home hm2_7i96.0.gpio.001.in_not => motion.01.home-sw-in
net z-home hm2_7i96.0.gpio.002.in_not => motion.02.home-sw-in

# Limit switches (inputs, normally closed switches)
net x-limit-min hm2_7i96.0.gpio.003.in_not => motion.00.neg-lim-sw-in
net x-limit-max hm2_7i96.0.gpio.004.in_not => motion.00.pos-lim-sw-in
net y-limit-min hm2_7i96.0.gpio.005.in_not => motion.01.neg-lim-sw-in
net y-limit-max hm2_7i96.0.gpio.006.in_not => motion.01.pos-lim-sw-in
net z-limit-min hm2_7i96.0.gpio.007.in_not => motion.02.neg-lim-sw-in
net z-limit-max hm2_7i96.0.gpio.008.in_not => motion.02.pos-lim-sw-in

# E-stop chain (input)
net estop-ext hm2_7i96.0.gpio.009.in_not => iocontrol.0.emc-enable-in

# Coolant and spindle outputs
setp hm2_7i96.0.gpio.010.is_output 1
net coolant-flood motion.coolant-flood => hm2_7i96.0.gpio.010.out

setp hm2_7i96.0.gpio.011.is_output 1
net coolant-mist motion.coolant-mist => hm2_7i96.0.gpio.011.out

# Tool change indicator (output LED)
setp hm2_7i96.0.gpio.012.is_output 1
net tool-change-active motion.tool-change => hm2_7i96.0.gpio.012.out

```

Network Configuration:

```

# Set static IP on PC Ethernet port
sudo ip addr add 10.10.10.1/24 dev eth0

# Verify connectivity
ping 10.10.10.10 # Mesa card default IP (configurable via jumpers or mesaflash)

# Read Mesa card identification
mesaflash --device 7i96 --addr 10.10.10.10 --readhmid

```

INI File Additions:

```

[EMCMOT]
EMCMOT = motmod
SERVO_PERIOD = 1000000 # 1 ms (no base thread needed)

```

```

[JOINT_0]
TYPE = LINEAR
SCALE = 800 # 200 steps/rev x 4 microsteps / 1 mm/rev
DIRSETUP = 5000 # 5 mus
DIRHOLD = 5000
STEPLEN = 2000 # 2 mus
STEPSPACE = 2000
MAX_VELOCITY = 50.0
MAX_ACCELERATION = 500.0

```

8.5 Mesa 5i25 + 7i76 Configuration (PCIe + Breakout)

Hardware: 5i25 PCIe card + 7i76 breakout board, 5-axis stepper system with spindle encoder

HAL File:

```

# =====
# LOAD HOSTMOT2 DRIVER (PCIe)
# =====
loadrt hostmot2
loadrt hm2_pci config="firmware=hm2/5i25/7i76x2.BIT num_encoders=1 num_pwmgens=1 num_stepgens=5 sserial.
# 7i76x2.BIT supports 2x 7i76 daughter cards (we use 1)
# sserial_port_0=00000000 configures smart serial port 0 for 7i76 mode

# Load kinematics and motion
loadrt trivkins coordinates=XYZAB # 5-axis: X Y Z A B
loadrt [EMCMOT]EMCMOT servo_period_nsec=[EMCMOT]SERVO_PERIOD num_joints=5

# =====
# THREAD FUNCTIONS
# =====
addf hm2_5i25.0.read servo-thread
addf motion.motion-command-handler servo-thread
addf motion.motion-controller servo-thread
addf hm2_5i25.0.write servo-thread
addf hm2_5i25.0.pet_watchdog servo-thread # FPGA watchdog timer

# =====
# 7i76 FIELD I/O CONFIGURATION
# =====
# 7i76 provides 32 opto-isolated inputs and relay outputs
# Accessed via hm2_5i25.0.7i76.0.0.input-XX and output-XX

# Digital inputs (24V opto-isolated)
net x-home-sw hm2_5i25.0.7i76.0.0.input-00 => motion.00.home-sw-in
net y-home-sw hm2_5i25.0.7i76.0.0.input-01 => motion.01.home-sw-in
net z-home-sw hm2_5i25.0.7i76.0.0.input-02 => motion.02.home-sw-in
net a-home-sw hm2_5i25.0.7i76.0.0.input-03 => motion.03.home-sw-in
net b-home-sw hm2_5i25.0.7i76.0.0.input-04 => motion.04.home-sw-in

# Relay outputs (24V, 2A per channel)
net coolant-flood motion.coolant-flood => hm2_5i25.0.7i76.0.0.output-00
net spindle-enable motion.spindle-on => hm2_5i25.0.7i76.0.0.output-01

```

```
# =====
# SPINDLE ENCODER FEEDBACK
# =====
# 7i76 has dedicated spindle encoder input
setp hm2_5i25.0.encoder.00.scale 1024 # 1024 pulses per rev (PPR)
setp hm2_5i25.0.encoder.00.counter-mode 0 # Quadrature mode

net spindle-position hm2_5i25.0.encoder.00.position => motion.spindle-revs
net spindle-velocity hm2_5i25.0.encoder.00.velocity => motion.spindle-speed-in
net spindle-index-enable hm2_5i25.0.encoder.00.index-enable <=> motion.spindle-index-enable

# =====
# STEPGEN CONFIGURATION (5 axes)
# =====
# Configure all 5 axes (X Y Z A B)
# (Similar to previous example, repeated for axes 0-4)

# X-axis (stepgen.00)
setp hm2_5i25.0.stepgen.00.dirsetup [JOINT_0]DIRSETUP
setp hm2_5i25.0.stepgen.00.dirhold [JOINT_0]DIRHOLD
setp hm2_5i25.0.stepgen.00.steplen [JOINT_0]STEPLEN
setp hm2_5i25.0.stepgen.00.stepspace [JOINT_0]STEPSPACE
setp hm2_5i25.0.stepgen.00.position-scale [JOINT_0]SCALE
setp hm2_5i25.0.stepgen.00.maxvel [JOINT_0]STEPVEL
setp hm2_5i25.0.stepgen.00.maxaccel [JOINT_0]STEPACCEL
setp hm2_5i25.0.stepgen.00.step_type 0

net x-pos-cmd motion.00.motor-pos-cmd => hm2_5i25.0.stepgen.00.position-cmd
net x-pos-fb hm2_5i25.0.stepgen.00.position-fb => motion.00.motor-pos-fb
net x-enable motion.00.amp-enable-out => hm2_5i25.0.stepgen.00.enable

# (Repeat for Y, Z, A, B axes with stepgen.01 through stepgen.04)
```

7i76 Wiring:

7i76 Breakout Board:

- TB2: Stepgen outputs (step/dir pairs for 5 axes)
- TB3: Field power input (24V DC for opto-isolated I/O)
- TB4: Digital inputs 0-15 (opto-isolated, NPN/PNP configurable)
- TB5: Digital inputs 16-31
- TB6: Relay outputs 0-15 (24V, 2A per channel)
- P1: 50-pin ribbon cable to 5i25 card

8.6 Mesa 7i77 Analog Servo Configuration

Hardware: 5i25 PCIe + 7i77 breakout, 6-axis analog servo system (+/-10V analog drives)

Key Features: - 6x analog servo outputs (+/-10V, 16-bit DAC, ~153 μ V resolution) - 6x encoder inputs (differential RS-422, up to 5 MHz) - 32x opto-isolated digital I/O

HAL File (Excerpt):

```
# =====
# LOAD FIRMWARE FOR 7i77 ANALOG SERVO
# =====
loadrt hostmot2
loadrt hm2_pci config="firmware=hm2/5i25/7i77x1.BIT num_encoders=6 sserial_port_0=00000000"
```

```

# PID components for closed-loop control
loadrt pid num_chan=6
addf pid.0.do-pid-calcs servo-thread
addf pid.1.do-pid-calcs servo-thread
addf pid.2.do-pid-calcs servo-thread
addf pid.3.do-pid-calcs servo-thread
addf pid.4.do-pid-calcs servo-thread
addf pid.5.do-pid-calcs servo-thread

# =====
# X-AXIS SERVO CONFIGURATION
# =====
# Encoder input
setp hm2_5i25.0.encoder.00.scale 2000.0 # 2000 counts/mm
setp hm2_5i25.0.encoder.00.filter 1 # Enable digital filter

# Analog output (+/-10V)
setp hm2_5i25.0.7i77.0.0.analogout0-scalemax 10.0 # +10V at scale 1.0
setp hm2_5i25.0.7i77.0.0.analogout0-minlim -10.0 # Minimum -10V
setp hm2_5i25.0.7i77.0.0.analogout0-maxlim 10.0 # Maximum +10V

# PID tuning
setp pid.0.Pgain [JOINT_0]P
setp pid.0.Igain [JOINT_0]I
setp pid.0.Dgain [JOINT_0]D
setp pid.0.FF0 [JOINT_0]FF0
setp pid.0.FF1 [JOINT_0]FF1
setp pid.0.FF2 [JOINT_0]FF2
setp pid.0.maxoutput 10.0 # +/-10V limit

# Signal connections
net x-pos-cmd motion.00.motor-pos-cmd => pid.0.command
net x-pos-fb hm2_5i25.0.encoder.00.position => pid.0.feedback motion.00.motor-pos-fb
net x-output pid.0.output => hm2_5i25.0.7i77.0.0.analogout0
net x-enable motion.00.amp-enable-out => pid.0.enable hm2_5i25.0.7i77.0.0.analogena0

# (Repeat for remaining 5 axes)

```

Analog Output Scaling:

7i77 analog outputs are 16-bit signed (-32768 to +32767 maps to +/-10V):

$$V_{out} = \text{HAL_value} \times \text{scalemax}$$

For HAL value = 1.0 and scalemax = 10.0:

$$V_{out} = 1.0 \times 10.0 = +10V$$

Resolution:

$$\text{Resolution} = \frac{20V \text{ range}}{65536 \text{ steps}} = 305 \text{ muV/step}$$

With oversampling and filtering, effective resolution ~16 bits = 153 muV.

8.7 GPIO and Special Functions

GPIO Pin Configuration:

```
# Set pin as output
setp hm2_7i96.0.gpio.010.is_output 1

# Set pin as input (default, explicit setting optional)
setp hm2_7i96.0.gpio.000.is_output 0

# Invert input (active-low logic)
net x-home-active hm2_7i96.0.gpio.000.in_not => motion.00.home-sw-in

# Direct output (active-high)
net coolant-on motion.coolant-flood => hm2_7i96.0.gpio.010.out
```

Watchdog Timer:

Mesa cards include hardware watchdog—if not petted periodically, FPGA disables all outputs (safety feature):

```
addf hm2_7i96.0.pet_watchdog servo-thread # Pet watchdog every servo cycle

# Watchdog timeout typically 10-20 ms
# If servo thread stops (crash), watchdog trips, motors stop
```

DPLL (Digital Phase-Locked Loop):

For spindle synchronization (threading, rigid tapping):

```
# Enable DPLL for precise spindle position tracking
setp hm2_7i96.0.dpll.01.timer-us -100 # -100 mus offset compensation

# Connect spindle encoder to DPLL input
net spindle-pos hm2_7i96.0.encoder.00.position => motion.spindle-revs
```

8.8 Firmware Flashing and Updates

Reading Current Firmware:

```
mesaflash --device 7i96 --addr 10.10.10.10 --readhmid
# Output:
# Board: 7i96
# FPGA: Xilinx XC6SLX9
# Pins: 96
# Firmware: SVST8_4
# ...
```

Flashing New Firmware:

```
# Download firmware from Mesa website or LinuxCNC forum

# Flash via Ethernet
mesaflash --device 7i96 --addr 10.10.10.10 --write 7i96_SVST8_24.bit

# Flash PCIe card
mesaflash --device 5i25 --write 5i25_7i76x2.bit

# Power cycle required after flashing
```

Reverting to Factory Firmware:

```
# Flash default firmware (stores in EEPROM, survives power cycle)
mesaflash --device 7i96 --addr 10.10.10.10 --write 7i96_default.bit --fix-boot-block
```

8.9 Troubleshooting Mesa Cards

Problem: Card Not Detected

```
# PCIe cards
lspci | grep Mesa
# Should show: "04:00.0 FPGA: Mesa Electronics 5i25"

# If not visible:
# - Check PCIe slot seating
# - Try different PCIe slot
# - Check BIOS PCIe settings (enable legacy interrupts)

# Ethernet cards
ping 10.10.10.10
# If no response:
# - Check Ethernet cable (use Cat5e or better)
# - Verify IP configuration (static IP on PC)
# - Check card power LED (green = powered, red = bootload mode)
# - Try factory reset (jumper W5 on 7i96, power cycle)
```

Problem: LinuxCNC Fails to Load hostmot2

```
# Check kernel log
dmesg | grep -i hm2

# Common errors:
# "hm2: no hm2 devices found"
# -> Card not detected, check lspci/ping

# "hm2: firmware not found: hm2/7i96/SVST8_4.BIT"
# -> Missing firmware file, install linuxcnc-firmware-dev package

# "hm2: board mismatch, expected 7i96, got 7i92"
# -> Wrong firmware file for card type
```

Problem: Stepgen Not Generating Pulses

```
# Check enable signal
halcmd show pin hm2_7i96.0.stepgen.00.enable
# Should show: TRUE when machine enabled

# Check position command
halcmd show pin hm2_7i96.0.stepgen.00.position-cmd
# Should change when jogging

# Check stepgen parameters
halcmd show param hm2_7i96.0.stepgen.00
# Verify maxvel, maxaccel, position-scale set correctly

# Monitor step output with oscilloscope or logic analyzer
# Look for step pulses on step/dir pins
```

Problem: Following Error on Servo System


```

# Check encoder feedback
halcmd show pin hm2_5i25.0.encoder.00.position
# Should change when motor moves

# Check encoder scale
halcmd getp hm2_5i25.0.encoder.00.scale
# Verify correct counts per position unit

# Check PID output
halcmd show pin pid.0.output
# Should be non-zero when error exists

# Check analog output
halcmd show pin hm2_5i25.0.7i77.0.0.analogout0
# Should show +/-10V range proportional to PID output

```

8.10 Performance Optimization

Thread Rate Selection:

```

# Standard performance (1 kHz servo thread)
[EMCMOT]
SERVO_PERIOD = 1000000 # 1 ms

# High performance (2 kHz servo thread, requires 7i80HD or 7i92)
[EMCMOT]
SERVO_PERIOD = 500000 # 0.5 ms

# Maximum performance (4-8 kHz, 7i80HD only)
[EMCMOT]
SERVO_PERIOD = 250000 # 0.25 ms (requires careful tuning)

```

Step Rate Configuration:

```

# Conservative (robust, low EMI)
setp hm2_7i96.0.stepgen.00.steplen 5000 # 5 mus
setp hm2_7i96.0.stepgen.00.stepspace 5000

# Standard (200 kHz max)
setp hm2_7i96.0.stepgen.00.steplen 2000 # 2 mus
setp hm2_7i96.0.stepgen.00.stepspace 2000

# Aggressive (500 kHz max, requires quality drivers)
setp hm2_7i96.0.stepgen.00.steplen 1000 # 1 mus
setp hm2_7i96.0.stepgen.00.stepspace 1000

# Extreme (1 MHz+, careful cable routing required)
setp hm2_7i96.0.stepgen.00.steplen 500 # 0.5 mus
setp hm2_7i96.0.stepgen.00.stepspace 500

```

8.11 Summary: Mesa Hardware Integration

Mesa FPGA cards transform LinuxCNC from hobbyist-grade to industrial-quality control:

Key Advantages:

- **Eliminates base thread:** FPGA handles step generation, reduces CPU load, removes latency sensitivity
- **Higher step rates:** 4 MHz vs. 50-100 kHz software limit (40-80x improvement)
- **More I/O:** 24-96 pins vs. 12-17 parallel port pins
- **Hardware features:** Watchdog, DPLL, smart serial, high-speed encoder counting
- **Scalability:** Multiple cards, flexible firmware, daughter card expansion

Product Selection:

- **Budget stepper:** 7i96S Ethernet (\$189) - all-in-one solution
- **Standard stepper:** 5i25 + 7i76 (\$368) - industrial I/O
- **Servo system:** 5i25 + 7i77 (\$408) - analog servo control
- **High-performance:** 7i80HD-25 (\$549) - 32 kHz update rate

Configuration Workflow:

1. Select card and firmware based on machine requirements
2. Load hostmot2 driver with correct firmware file
3. Configure stepgen/encoder/PWM parameters
4. Map GPIO to limit switches, coolant, spindle control
5. Test and verify I/O with halcmd and halmeter
6. Tune PID if using closed-loop servo control

Next Section (14.9) explores advanced HAL techniques: electronic gearing, custom kinematics, tool length probing, spindle synchronization, and complex automation sequences.

Total: 4,883 words / 2 equations / 6 complete configuration examples / 8 tables / 22 code blocks

9. Advanced HAL Techniques

9.1 Electronic Gearing and Axis Slaving

Electronic Gearing synchronizes one axis to another at a fixed or variable ratio, enabling tandem axis control, gantry machines, and master-slave configurations.

Application: Gantry Mill with Dual Y-Axis Motors

Problem: Gantry with motors on left and right sides (Y1, Y2) must move synchronously to prevent racking.

Solution 1: Trivial Gantrykins (Built-In)

```
# Use gantrykins instead of trivkins
loadrt gantrykins coordinates=XYZY # Y appears twice = dual motor Y-axis

# Joint mapping:
# Joint 0 = X-axis
# Joint 1 = Y1 (left motor)
# Joint 2 = Y2 (right motor)
# Joint 3 = Z-axis

# Configure both Y stepgens identically
setp hm2_7i96.0.stepgen.01.position-scale 800
setp hm2_7i96.0.stepgen.02.position-scale 800

# Connect joints
net x-pos-cmd motion.00.motor-pos-cmd => hm2_7i96.0.stepgen.00.position-cmd
net y1-pos-cmd motion.01.motor-pos-cmd => hm2_7i96.0.stepgen.01.position-cmd
net y2-pos-cmd motion.02.motor-pos-cmd => hm2_7i96.0.stepgen.02.position-cmd # Synchronized Y2
net z-pos-cmd motion.03.motor-pos-cmd => hm2_7i96.0.stepgen.03.position-cmd

# Homing: Y-axis homes both joints simultaneously
# G-code: G0 Y100 moves both Y1 and Y2 to 100 mm
```

Solution 2: Manual Electronic Gearing

```
# Use trivkins (4 independent axes)
loadrt trivkins

# Scale component for ratio adjustment
loadrt scale count=1
addf scale.0 servo-thread

# Y2 follows Y1 with adjustable ratio
setp scale.0.gain 1.0 # 1:1 ratio (adjust if Y2 needs compensation)

net y1-cmd motion.01.motor-pos-cmd => scale.0.in
net y2-cmd scale.0.out => hm2_7i96.0.stepgen.02.position-cmd

# Y1 controlled normally
net y1-pos-cmd motion.01.motor-pos-cmd => hm2_7i96.0.stepgen.01.position-cmd
```

Variable Ratio Gearing:

```
# Example: Rotary axis (A) drives tangential knife at variable ratio
loadrt mult2 count=1
addf mult2.0 servo-thread
```

```
# Ratio = f(cutting radius)
net cutting-radius ui.radius-input => mult2.0.in0
net a-velocity motion.03.joint-vel-cmd => mult2.0.in1
net knife-speed mult2.0.out => knife-motor.speed-cmd

# Tangential velocity = angular velocity x radius
# knife-speed = A-axis angular velocity (deg/s) x cutting radius (mm)
```

9.2 Spindle Synchronization and Rigid Tapping

Rigid Tapping requires precise spindle position tracking for coordinated Z-axis motion during threading.

Requirements:

1. Spindle encoder with index pulse (Z channel)
2. DPLL (Digital Phase-Locked Loop) for position interpolation
3. Motion controller configured for spindle sync

Configuration:

```
# =====
# SPINDLE ENCODER SETUP
# =====
setp hm2_7i96.0.encoder.04.scale 1024 # 1024 PPR encoder
setp hm2_7i96.0.encoder.04.counter-mode 0 # Quadrature mode
setp hm2_7i96.0.encoder.04.filter 1 # Enable input filter

# Connect spindle encoder to motion
net spindle-position hm2_7i96.0.encoder.04.position => motion.spindle-revs
net spindle-velocity hm2_7i96.0.encoder.04.velocity => motion.spindle-speed-in
net spindle-index-enable hm2_7i96.0.encoder.04.index-enable <=> motion.spindle-index-enable

# =====
# DPLL CONFIGURATION (Interpolation)
# =====
# DPLL provides sub-count position resolution for smooth motion
setp hm2_7i96.0.dpll.01.timer-us -100 # Timing compensation (tune empirically)

# Connect DPLL to spindle encoder
net spindle-pos-dpll hm2_7i96.0.dpll.01.phase-position => motion.spindle-phase-position
```

INI File:

```
[TRAJ]
SPINDLE_0 = 0

[SPINDLE_0]
MAX_FORWARD_VELOCITY = 3000 # Maximum spindle RPM
ENCODER_SCALE = 1024 # Encoder PPR
```

G-code Usage:

```
M3 S500 ; Start spindle at 500 RPM
G4 P2 ; Wait 2 seconds for spindle to reach speed
G33.1 Z-25 K1.5 ; Rigid tap to 25 mm depth, 1.5 mm pitch
G33.1 Z0 ; Retract (reverse tap)
M5 ; Stop spindle
```

Threading (G33):

```
G33 Z-50 K2.0      ; Thread 50 mm depth, 2.0 mm pitch
; Z-axis velocity synchronized to spindle rotation
; Feed rate = spindle RPM x pitch (mm/rev)
```

9.3 Custom Kinematics: Delta Robot Example

Delta Robot: 3 vertical arms control XYZ position of end effector platform via inverse kinematics.

Kinematics Component (delta.comp):

```
component delta_kins "Delta robot inverse kinematics";
description ""
Delta parallel robot kinematics.
Three vertical linear joints control Cartesian XYZ position.
"";

// Cartesian coordinates (world space)
pin in float pos_x "Commanded X position";
pin in float pos_y "Commanded Y position";
pin in float pos_z "Commanded Z position";

// Joint positions (motor space, vertical travel)
pin out float joint_0 "Vertical position of arm 0";
pin out float joint_1 "Vertical position of arm 1";
pin out float joint_2 "Vertical position of arm 2";

// Geometric parameters
parameter rw float base_radius = 100.0 "Base triangle radius";
parameter rw float platform_radius = 50.0 "Platform triangle radius";
parameter rw float arm_length = 150.0 "Upper arm length";
parameter rw float rod_length = 300.0 "Parallel rod length";

function inverse_kins fp "Calculate joint positions from Cartesian";

license "GPL";

;;

#include <rtapi_math.h>

// Helper function: Calculate single arm joint position
static float calc_joint_pos(float x, float y, float z, float angle,
                           float base_r, float platform_r, float arm_len, float rod_len) {
    // Base pivot position
    float base_x = base_r * cos(angle);
    float base_y = base_r * sin(angle);

    // Platform attachment position
    float platform_x = x + platform_r * cos(angle);
    float platform_y = y + platform_r * sin(angle);

    // Vector from base to platform
    float dx = platform_x - base_x;
    float dy = platform_y - base_y;
    float dz = z;
```

```

    // Distance in XY plane
    float r = sqrt(dx*dx + dy*dy);

    // Total 3D distance
    float d = sqrt(r*r + dz*dz);

    // Check reachability
    if (d > (arm_len + rod_len) || d < fabs(arm_len - rod_len)) {
        return 0.0; // Unreachable, return safe position
    }

    // Law of cosines: solve for arm angle
    float cos_angle = (arm_len*arm_len + d*d - rod_len*rod_len) / (2.0 * arm_len * d);
    float arm_angle = acos(cos_angle);

    // Angle from horizontal to platform
    float platform_angle = atan2(dz, r);

    // Joint vertical position
    return z - arm_len * sin(arm_angle + platform_angle);
}

FUNCTION(inverse_kins) {
    // Arm angles: 0 deg, 120 deg, 240 deg (equilateral triangle)
    const float angles[3] = {0.0, 2.0*M_PI/3.0, 4.0*M_PI/3.0};

    float x = pos_x;
    float y = pos_y;
    float z = pos_z;

    // Calculate each joint position
    joint_0 = calc_joint_pos(x, y, z, angles[0],
                             base_radius, platform_radius, arm_length, rod_length);
    joint_1 = calc_joint_pos(x, y, z, angles[1],
                             base_radius, platform_radius, arm_length, rod_length);
    joint_2 = calc_joint_pos(x, y, z, angles[2],
                             base_radius, platform_radius, arm_length, rod_length);
}

```

HAL Integration:

```

# Load custom kinematics
loadrt delta_kins
addf delta-kins.inverse-kins servo-thread

# Configure geometry
setp delta-kins.base-radius 100.0
setp delta-kins.platform-radius 50.0
setp delta-kins.arm-length 150.0
setp delta-kins.rod-length 300.0

# Connect Cartesian commands to kinematics
net x-cmd motion.00.pos-cmd => delta-kins.pos-x
net y-cmd motion.01.pos-cmd => delta-kins.pos-y

```

```
net z-cmd motion.02.pos-cmd => delta-kins.pos-z
```

```
# Connect joint commands to motors
net joint-0-cmd delta-kins.joint-0 => pid.0.command
net joint-1-cmd delta-kins.joint-1 => pid.1.command
net joint-2-cmd delta-kins.joint-2 => pid.2.command
```

9.4 Tool Length Probing and Automatic Offsets

Application: Touch probe automatically measures tool length, updates tool table.

Probe HAL Setup:

```
# Tool probe input (normally open switch, closes when contact)
net probe-input hm2_7i96.0.gpio.015.in => motion.probe-input

# Probe LED indicator
setp hm2_7i96.0.gpio.014.is_output 1
net probe-active motion.probe-input => hm2_7i96.0.gpio.014.out
```

G-code Probing Subroutine (O sub):

```
O<tool_probe> sub ; Tool length measurement subroutine

; Assumes:
; - Z zero set on top of gauge block (known height)
; - Gauge block height in parameter #<gauge_height>
; - Rapid to position above gauge block

#<gauge_height> = 25.0 ; 25 mm gauge block

G91 ; Relative mode
G38.2 Z-50 F25 ; Probe toward gauge, 25 mm/min feed
G90 ; Absolute mode

; Calculate tool length offset
#<probe_z> = #5063 ; Z position when probe tripped (absolute)
#<tool_offset> = [#<probe_z> + #<gauge_height>] ; Offset from Z zero

; Update tool table (requires M66/M67 or Python component)
; For LinuxCNC 2.8+, use G10 L1:
G10 L1 P#<_current_tool> Z#<tool_offset> ; Set tool offset

G91 G0 Z5 ; Retract 5 mm
G90 G0 Z50 ; Move to safe height

O<tool_probe> endsub

Usage:

T1 M6 ; Load tool 1
G43 H1 ; Apply tool length offset
O<tool_probe> call ; Measure and update tool 1 offset
M0 ; Pause for verification

; Tool offset now automatically compensates for tool length
G0 Z0 ; Tool tip now at Z=0 (gauge block top surface)
```

9.5 Adaptive Feed Rate Control

Application: Reduce feed rate based on real-time spindle load, preventing tool breakage.

HAL Component (adaptive_feed.comp):

```
component adaptive_feed "Adjust feed rate based on spindle load";

pin in float spindle_load "Spindle load percentage (0-100)";
pin out float feed_scale "Feed override scale (0-1.0)";

parameter rw float max_load = 80.0 "Maximum allowable spindle load (%)";
parameter rw float min_scale = 0.1 "Minimum feed scale (10%)";

function _ fp;
license "GPL";

;;

FUNCTION(_) {
    if (spindle_load > max_load) {
        // Load too high: reduce feed rate proportionally
        float overload = (spindle_load - max_load) / 20.0; // 20% overload = 1.0
        feed_scale = 1.0 - overload;

        // Clamp to minimum
        if (feed_scale < min_scale) {
            feed_scale = min_scale;
        }
    } else {
        // Normal load: full feed rate
        feed_scale = 1.0;
    }
}
```

HAL Integration:

```
loadrt adaptive_feed
addf adaptive-feed servo-thread

# Configure thresholds
setp adaptive-feed.max-load 75.0 # Start reducing at 75% spindle load
setp adaptive-feed.min-scale 0.25 # Never go below 25% feed rate

# Connect spindle load sensor (analog input scaled to 0-100%)
net spindle-load-pct analog-input.0 => adaptive-feed.spindle-load

# Connect feed scale to motion controller
net feed-scale-adaptive adaptive-feed.feed-scale => motion.adaptive-feed

Result: Feed rate automatically reduces when spindle load exceeds 75%, preventing overload and tool damage.
```

9.6 Multi-Pass Cutting Logic

Application: Automatically execute multiple cutting passes with incremental depth.

Python HAL Component (multi_pass.py):


```

#!/usr/bin/env python3
"""
Multi-pass cutting controller
Executes G-code subroutine multiple times with incremental Z depth
"""

import hal
import time

class MultiPass:
    def __init__(self):
        self.h = hal.component("multi-pass")

        # Configuration
        self.h.newpin("start", hal.HAL_BIT, hal.HAL_IN)
        self.h.newpin("reset", hal.HAL_BIT, hal.HAL_IN)
        self.h.newpin("total-depth", hal.HAL_FLOAT, hal.HAL_IN)
        self.h.newpin("depth-per-pass", hal.HAL_FLOAT, hal.HAL_IN)

        # Status outputs
        self.h.newpin("current-pass", hal.HAL_S32, hal.HAL_OUT)
        self.h.newpin("total-passes", hal.HAL_S32, hal.HAL_OUT)
        self.h.newpin("current-depth", hal.HAL_FLOAT, hal.HAL_OUT)
        self.h.newpin("complete", hal.HAL_BIT, hal.HAL_OUT)

        self.h.ready()

        self.pass_num = 0
        self.active = False

    def update(self):
        if self.h["reset"]:
            self.pass_num = 0
            self.active = False
            self.h["complete"] = False

        if self.h["start"] and not self.active:
            # Calculate number of passes
            total_depth = abs(self.h["total-depth"])
            depth_per_pass = abs(self.h["depth-per-pass"])

            if depth_per_pass > 0:
                num_passes = int(total_depth / depth_per_pass)
                if total_depth % depth_per_pass > 0:
                    num_passes += 1 # Partial final pass

            self.h["total-passes"] = num_passes
            self.active = True

        if self.active:
            # Update current depth for pass
            depth_per_pass = abs(self.h["depth-per-pass"])
            self.h["current-depth"] = (self.pass_num + 1) * depth_per_pass

```

```

        # Clamp to total depth
        if self.h["current-depth"] > abs(self.h["total-depth"]):
            self.h["current-depth"] = abs(self.h["total-depth"])

        self.h["current-pass"] = self.pass_num + 1

        # Advance pass (triggered externally by G-code M66)
        # In practice, G-code would read current-depth and execute pass

    def run(self):
        try:
            while True:
                self.update()
                time.sleep(0.01)
        except KeyboardInterrupt:
            pass

if __name__ == "__main__":
    mp = MultiPass()
    mp.run()

```

G-code Integration:

```

; Multi-pass pocket milling

#<total_depth> = 10.0      ; 10 mm total depth
#<depth_per_pass> = 2.0    ; 2 mm per pass

; Set multi-pass parameters (writes to HAL pins)
M68 E0 Q#<total_depth>
M68 E1 Q#<depth_per_pass>
M68 E2 Q1 ; Start multi-pass sequence

; Loop for each pass
O100 while [#<multi_pass_complete> EQ 0]
    #<current_depth> = #5399 ; Read current-depth from HAL (example parameter)

    ; Execute cutting pass at current depth
    G0 X0 Y0
    G1 Z[-#<current_depth>] F100
    ; ... pocket toolpath at current Z depth ...
    G0 Z5 ; Retract

    ; Advance to next pass (increment counter in HAL component)
    M66 P0 L0 ; Wait for HAL input (trigger next pass)
O100 endwhile

```

9.7 Backlash Compensation

Problem: Mechanical backlash causes position error during direction reversal.

Solution: HAL backlash component offsets commanded position based on direction.

```

# Load backlash compensation
loadrt backlash count=3
addf backlash.0 servo-thread

```

```

addf backlash.1 servo-thread
addf backlash.2 servo-thread

# Configure X-axis backlash (0.05 mm measured)
setp backlash.0.backlash 0.05

# Insert between motion and stepgen
net x-cmd-raw motion.00.motor-pos-cmd => backlash.0.in
net x-cmd-compensated backlash.0.out => hm2_7i96.0.stepgen.00.position-cmd
net x-pos-fb hm2_7i96.0.stepgen.00.position-fb => motion.00.motor-pos-fb backlash.0.filt-in

# Backlash component adds 0.05 mm when direction changes from negative to positive
# Removes 0.05 mm when changing from positive to negative
# Net effect: compensates for mechanical slack

```

Alternative: INI-Based Backlash (Simpler but Less Precise):

```

[JOINT_0]
BACKLASH = 0.05 ; mm (applied by motion controller, not HAL)

```

9.8 Tool Changer Sequencing

Application: Automatic tool changer with rotary carousel and pneumatic arm.

State Machine Component (tool_changer.comp):

```

component tool_changer "Automatic tool changer controller";

pin in bit start "Start tool change sequence";
pin in s32 tool_pocket "Tool pocket number (1-8)";
pin in bit tool_clamped "Sensor: Tool clamped in spindle";
pin in bit carousel_aligned "Sensor: Carousel at correct position";

pin out bit unclamp "Solenoid: Release spindle collet";
pin out bit carousel_fwd "Motor: Rotate carousel forward";
pin out bit arm_extend "Solenoid: Extend arm to carousel";
pin out bit complete "Tool change complete";

parameter rw u32 timeout_ms = 5000 "Timeout per step (ms)";

function _ "Tool changer state machine";
license "GPL";

;;

#define STATE_IDLE 0
#define STATE_UNCLAMP 1
#define STATE_ROTATE 2
#define STATE_EXTEND 3
#define STATE_CLAMP 4
#define STATE_RETRACT 5
#define STATE_DONE 6

static int state = STATE_IDLE;
static int prev_tool_pocket = 0;
static unsigned long timer = 0;

```

```

FUNCTION(_) {
    static bool prev_start = 0;
    bool start_edge = start && !prev_start;
    prev_start = start;

    unsigned long timeout_cycles = timeout_ms; // Assume 1 kHz servo thread = 1 ms/cycle

    switch (state) {
        case STATE_IDLE:
            unclamp = 0;
            carousel_fwd = 0;
            arm_extend = 0;
            complete = 0;

            if (start_edge) {
                state = STATE_UNCLAMP;
                timer = 0;
            }
            break;

        case STATE_UNCLAMP:
            unclamp = 1;
            timer++;

            if (!tool_clamped || timer > timeout_cycles) {
                state = STATE_ROTATE;
                timer = 0;
            }
            break;

        case STATE_ROTATE:
            unclamp = 0;

            // Rotate carousel to correct pocket
            if (carousel_aligned) {
                carousel_fwd = 0;
                state = STATE_EXTEND;
                timer = 0;
            } else {
                carousel_fwd = 1;
                timer++;
                if (timer > timeout_cycles) {
                    // Timeout: abort
                    state = STATE_IDLE;
                    rtapi_print_msg(RTAPI_MSG_ERR, "Tool changer: Carousel timeoutLF");
                }
            }
            break;

        case STATE_EXTEND:
            arm_extend = 1;
            timer++;
    }
}

```

```

        if (timer > 1000) { // 1 second delay for arm extend
            state = STATE_CLAMP;
            timer = 0;
        }
        break;

    case STATE_CLAMP:
        unclamp = 0; // Clamp tool (solenoid de-energized)
        timer++;

        if (tool_clamped || timer > timeout_cycles) {
            state = STATE_RETRACT;
            timer = 0;
        }
        break;

    case STATE_RETRACT:
        arm_extend = 0;
        timer++;

        if (timer > 1000) { // 1 second delay for arm retract
            state = STATE_DONE;
        }
        break;

    case STATE_DONE:
        complete = 1;

        if (!start) { // Wait for start to go FALSE
            state = STATE_IDLE;
            prev_tool_pocket = tool_pocket;
        }
        break;
    }
}

```

HAL Integration:

```

loadrt tool_changer
addf tool-changer servo-thread

setp tool-changer.timeout-ms 5000

# Connect motion controller
net tool-change-start iocontrol.0.tool-change => tool-changer.start
net tool-pocket iocontrol.0.tool-prep-number => tool-changer.tool-pocket
net tool-change-done tool-changer.complete => iocontrol.0.tool-changed

# Connect sensors
net tool-clamped-sensor hm2_7i96.0.gpio.010.in => tool-changer.tool-clamped
net carousel-aligned-sensor hm2_7i96.0.gpio.011.in => tool-changer.carousel-aligned

# Connect outputs
setp hm2_7i96.0.gpio.012.is_output 1
setp hm2_7i96.0.gpio.013.is_output 1

```

```

setp hm2_7i96.0.gpio.014.is_output 1

net unclamp-out tool-changer.unclamp => hm2_7i96.0.gpio.012.out
net carousel-motor tool-changer.carousel-fwd => hm2_7i96.0.gpio.013.out
net arm-extend-out tool-changer.arm-extend => hm2_7i96.0.gpio.014.out

```

9.9 Plasma Torch Height Control (THC)

Application: Maintain constant arc voltage (proportional to torch-to-workpiece distance) during plasma cutting.

THC HAL Component (plasma_thc.comp):

```

component plasma_thc "Plasma torch height controller";

pin in float arc_voltage "Measured arc voltage (V)";
pin in float target_voltage "Target arc voltage (V)";
pin in bit arc_ok "Arc established and stable";
pin in bit torch_on "Torch firing";

pin out float z_offset "Z-axis position correction (mm)";

parameter rw float voltage_tolerance = 5.0 "Voltage deadband (V)";
parameter rw float correction_scale = 0.5 "mm per volt error";
parameter rw float max_correction = 5.0 "Maximum Z offset (mm)";

function _ fp;
license "GPL";

;;

FUNCTION(_) {
    if (!arc_ok || !torch_on) {
        // No correction when arc off
        z_offset = 0.0;
        return;
    }

    float voltage_error = target_voltage - arc_voltage;

    // Apply deadband
    if (fabs(voltage_error) < voltage_tolerance) {
        // Within tolerance, no correction
        return;
    }

    // Calculate correction (mm)
    float correction = voltage_error * correction_scale;

    // Update offset (integrating controller)
    z_offset += correction;

    // Clamp to limits
    if (z_offset > max_correction) z_offset = max_correction;
    if (z_offset < -max_correction) z_offset = -max_correction;
}

```

```
}
```

HAL Integration:

```
loadrt plasma_thc
addf plasma-thc servo-thread

# Load sum component to add offset to Z position
loadrt sum2 count=1
addf sum2.0 servo-thread

# Configure THC
setp plasma-thc.target-voltage 120.0 # 120V arc voltage target
setp plasma-thc.voltage-tolerance 5.0 # +/-5V deadband
setp plasma-thc.correction-scale 0.1 # 0.1 mm per volt error
setp plasma-thc.max-correction 3.0 # +/-3 mm max correction

# Connect arc voltage sensor
net arc-voltage analog-input.0 => plasma-thc.arc-voltage

# Connect torch status
net torch-on motion.digital-out-00 => plasma-thc.torch-on
net arc-ok hm2_7i96.0.gpio.015.in => plasma-thc.arc-ok

# Insert THC offset into Z-axis command
net z-cmd-raw motion.02.motor-pos-cmd => sum2.0.in0
net z-offset-thc plasma-thc.z-offset => sum2.0.in1
net z-cmd-compensated sum2.0.out => hm2_7i96.0.stepgen.02.position-cmd
```

9.10 Summary: Advanced HAL Techniques

Advanced HAL techniques unlock LinuxCNC's full potential for specialized applications:

Key Techniques:

1. **Electronic gearing:** Gantry machines, master-slave axes, synchronized motion
2. **Spindle synchronization:** Rigid tapping, threading, position-synchronized operations
3. **Custom kinematics:** Non-Cartesian robots (delta, SCARA, cable-driven)
4. **Tool probing:** Automatic tool length measurement, workpiece probing
5. **Adaptive control:** Real-time feed rate adjustment based on process feedback
6. **State machines:** Complex sequencing (tool changers, multi-pass operations)
7. **Process control:** THC, laser power modulation, EDM gap control

Design Principles:

- **Modularity:** Break complex logic into discrete components
- **Robustness:** Handle edge cases, timeouts, and error conditions
- **Observability:** Expose status pins for monitoring and debugging
- **Configurability:** Use parameters for tuning without code changes

Next Section (14.10) covers diagnostic tools: halcmd, halmeter, halscope, and systematic troubleshooting procedures for HAL configuration and runtime issues.

Total: 4,231 words / 0 equations / 9 complete worked examples / 0 tables / 18 code blocks